GEK-84000

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## Safety Symbol Legend

**WARNING** — Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in personal injury or death.

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**NOTE** — Commands attention to an essential operating or maintenance procedure, condition, or statement that must be highlighted.

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# CHAPTER 1

# SYSTEM DESCRIPTION

## 1-1. BASIC DRIVE CONTROL

The DC Link Commutated (DCLC) Induction Motor Drive (IMD) is a static, adjustable frequency drive system which controls an induction motor. Depending on size of the load capacitors, $C_1$ but typically from 50-70% to 100% speed, the load inverter is load commutated by leading KVAR load. From 0% to 50-70% speed the load inverter is forced commutated by shunting the DC link current into a shunt path in the DC link. For the current to transfer from the load inverter to the DC link shunt and thereby commutating the load inverter thyristors, the voltage that the DC shunt switches into must be less than the motor voltage reflected to the load side of the DC link at the instant of commutation. A simplified one-line diagram of a single channel IMD is given in Figure 1-1.

The transformer provides isolation from the ac system bus and provides the correct voltage at the terminals of the rectifier. Also, the internal impedance of the transformer limits the magnitude of any downstream bus faults.

The rectifier is a thyristor bridge whose gating is controlled to produce a variable dc voltage at its output. The output of the rectifier is fed through the dc link reactor, whose function is to smooth the current and keep it continuous over the operating range of the system.

The dc link reactor output is then fed into the DCLC and inverter bridge which produces variable frequency ac at the stator terminals of the motor. The inverter bridge , the DCLC, and the rectifier bridge use different power hardware and are controlled by microprocessor-based electronics. Although the bridges are labelled "rectifier" and "inverter" it is possible for their roles (when load has leading KVARS) and hence power flow to reverse. In this case the motor would be braked by pumping its energy back into the ac line. A more general nomenclature is to call the line side bridge the source converter, and to call the motor side bridge the load converter.

The filter provides the leading KVARS so that the drive may operate in the more efficient self-commutating mode (rather than forced comutated) when sufficient motor volts and speed are reached.

A more detailed diagram of a single channel IMD is shown in Figure 1-2. As shown in this figure, the electronic control receives the following signal inputs:

1. Attenuated line and load bus voltage signals.
2. Attenuated line and load current signals from current transformers.
3. Speed reference signal.
4. Process commands such as stop, start, etc.

The attenuated bus voltage signals are used by the control to synchronize thyristor firing with the line and load voltages, to provide voltage sensing across the thyristor cells, and to provide electronic over/under voltage detection. The attenuated current signals are

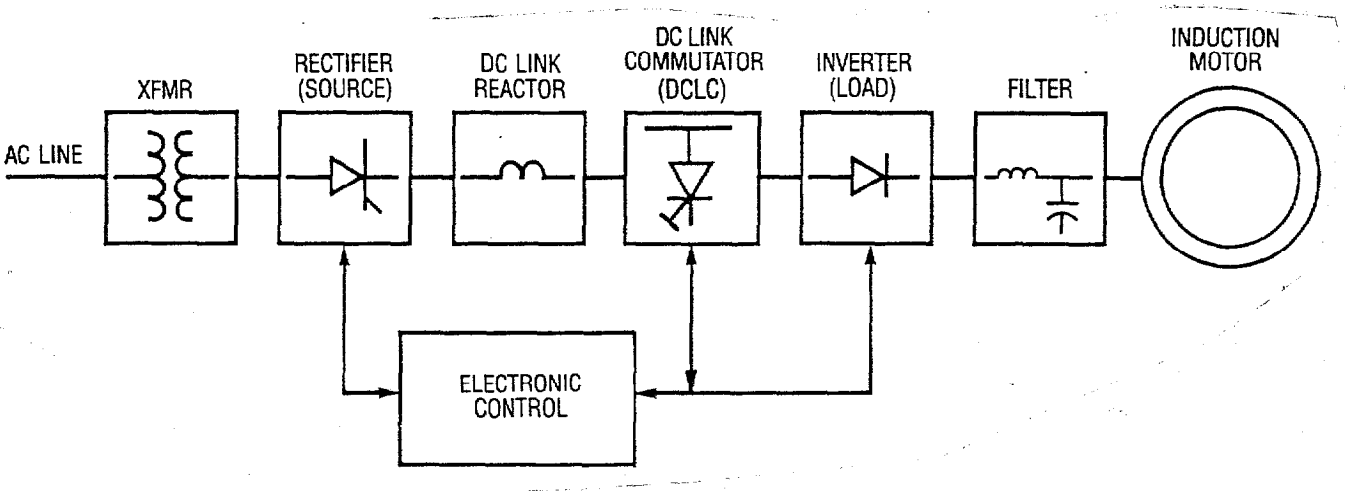



*Figure 1-1. Simplified IMD One-Line Diagram*

used for regulator current feedback, electronic overcurrent detection, and software-implemented fault detection.

The electronic control evaluates the process commands and internal status signals to determine whether the drive should be in a stopped, started, alarmed, or faulted condition. If in a started condition, the electronics provides gate signals to control the thyristor bridges. These signals are low level and are conditioned in the power bridge circuitry to provide the necessary isolation and power level.

The power bridges are of the six pulse, double way type and the thyristor bridge legs "fire" in the order that they are numbered in Figure 1-2 and as shown in Figure 1-3.

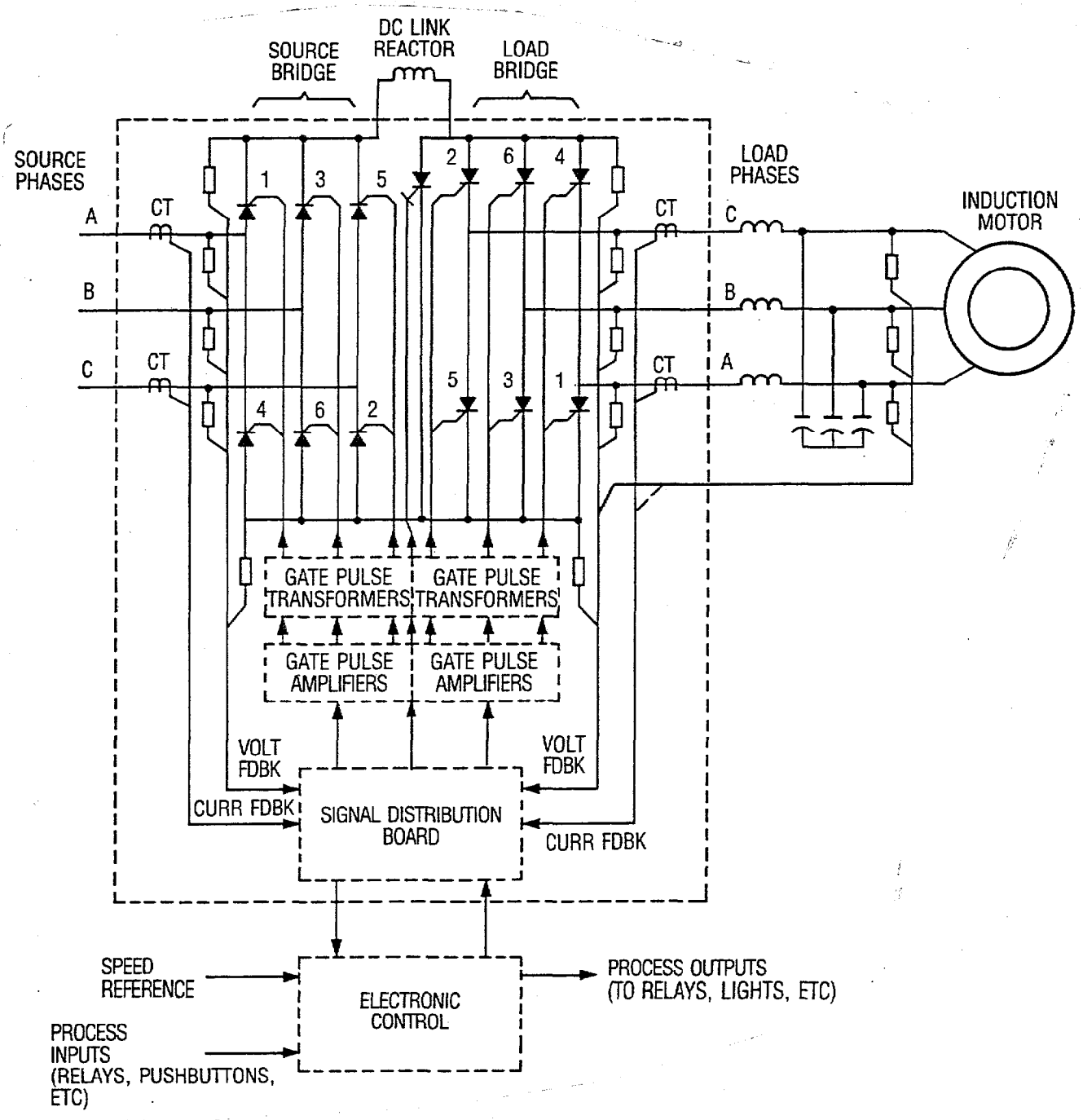



*Figure 1-2. Single Channel IMD Drive*

The source converter current is successfully transferred from one leg to the next by the AC line voltages. Similarly, the inverter current is successfully transferred by load's leading KVAR (75% SPD).

The process of switching bridge current from one leg to the next is illustrated in Figure 1-4. The principles illustrated here apply to a rectifier bridge as well as to an inverter bridge (when load has leading KVARS). This phase-controlled switching is accomplished by using the following two thyristor characteristics: (1) When the voltage across the thyristor is positive, it can be triggered into conduction, and (2) it will not permit current flow in the reverse direction - thus, in an alternating voltage circuit, thyristor conduction will cease and reverse voltage will begin to appear when the current becomes zero.

Current transfer must be completed before voltage cross-over, with a margin angle $\gamma$ . This angle must be long enough to allow the previously conducting leg thyristors to recover to their blocking state before reverse voltage is applied. This is why the fundamental component of current must lead the voltage, from an inverter/load viewpoint. From the rectifier/source viewpoint, the fundamental component of current will always lag the voltage; but for successful commutation, the angle $\gamma = 180° - \beta = 180° - \mu - \gamma$ must always be less than 180° ; a practical limit is 155°. Correspondingly, a practical minimum value for $\beta$ , for the inverter bridge, is 25°.

## 1-2. GENERAL OPERATIONAL THEORY

Startup modes and assumptions for GTO DC Link Commutation approach:

1. Commutation by phasing the source converter to inversion limit: (typically 0 to 15 Hz).
2. Initially ther is no motor flux, so at .25 PU IDC the OPEN_LOOP frequency is ramped from 1 to 5 Hz.
3. At 5 Hz, motor flux has been established - thus the load PLL (phase-lock-loop), speed regulator, flux regulator, and load angle regulator are activated.
4. Now in CLOSED-LOOP mode, continue to ramp up motor speed to 15 Hz using source converter commutation. 15 Hz is about the limit of source converter commutation capability. At this speed there should be sufficient motor voltage to allow shifting to the GTO DC Link Commutation mode.
5. GTO DC Link Commutation mode: (typically 15 Hz to 50-70% speed)
6. From 15 hz to about 50-70% PU speed, the load thyristers are commutated by turning on the GTO shunt. GTO commutation is maintained as long as the commanded load thyrister firing angle (alpha) is greater than (180 degrees - margin) i.e. about 165 degrees. For succesful transfer of current from the load thyrister to the GTO, there must be sufficient commutating VOLT-SEC and this is accomplished by limiting the max firing angle (alpha) to about 225 deg. This prevents the vector control from being able to independently control flux and torque as could be done if alpha was unrestricted. The DC link voltage wa-

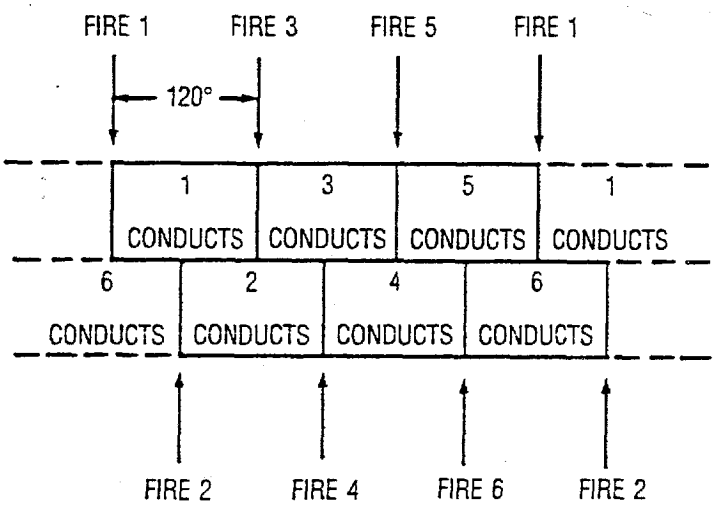



*Figure 1-3. Thyristor Bridge Leg Conduction Sequence*

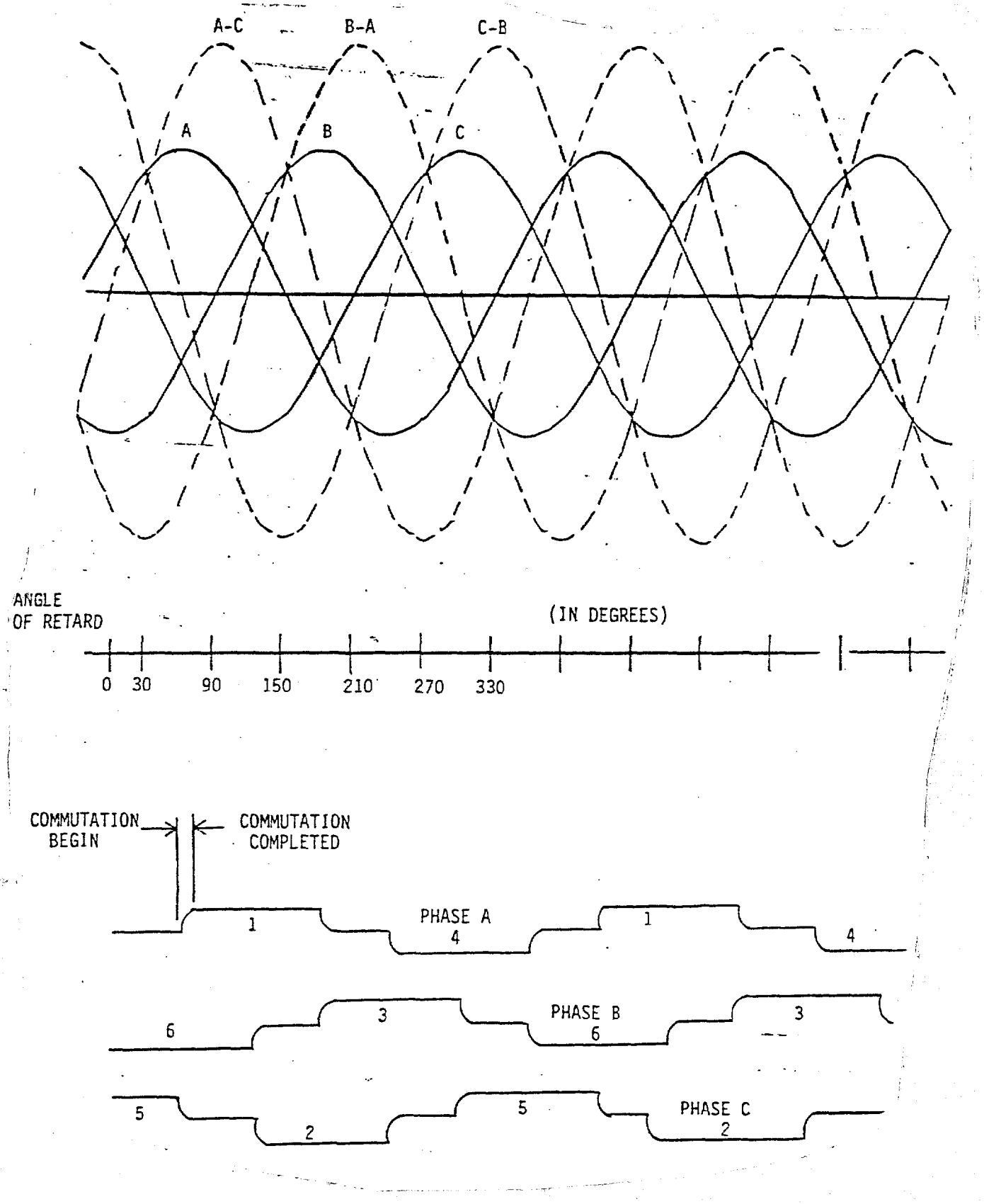



*Figure 1-4A. Source Voltage and Current*

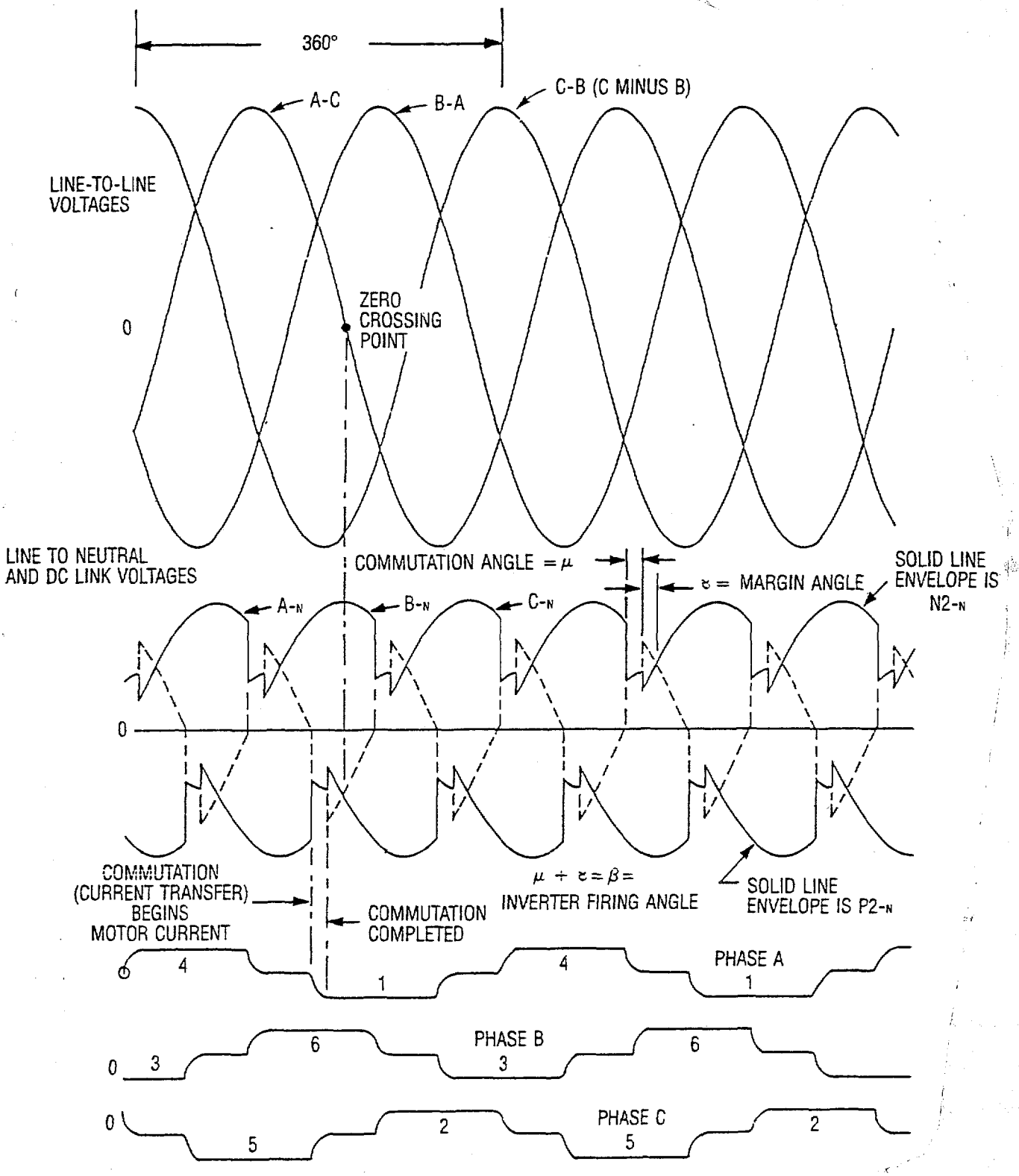



*Figure 1-4B. Load Converter Voltage and Current (Self-Commutation Mode)*

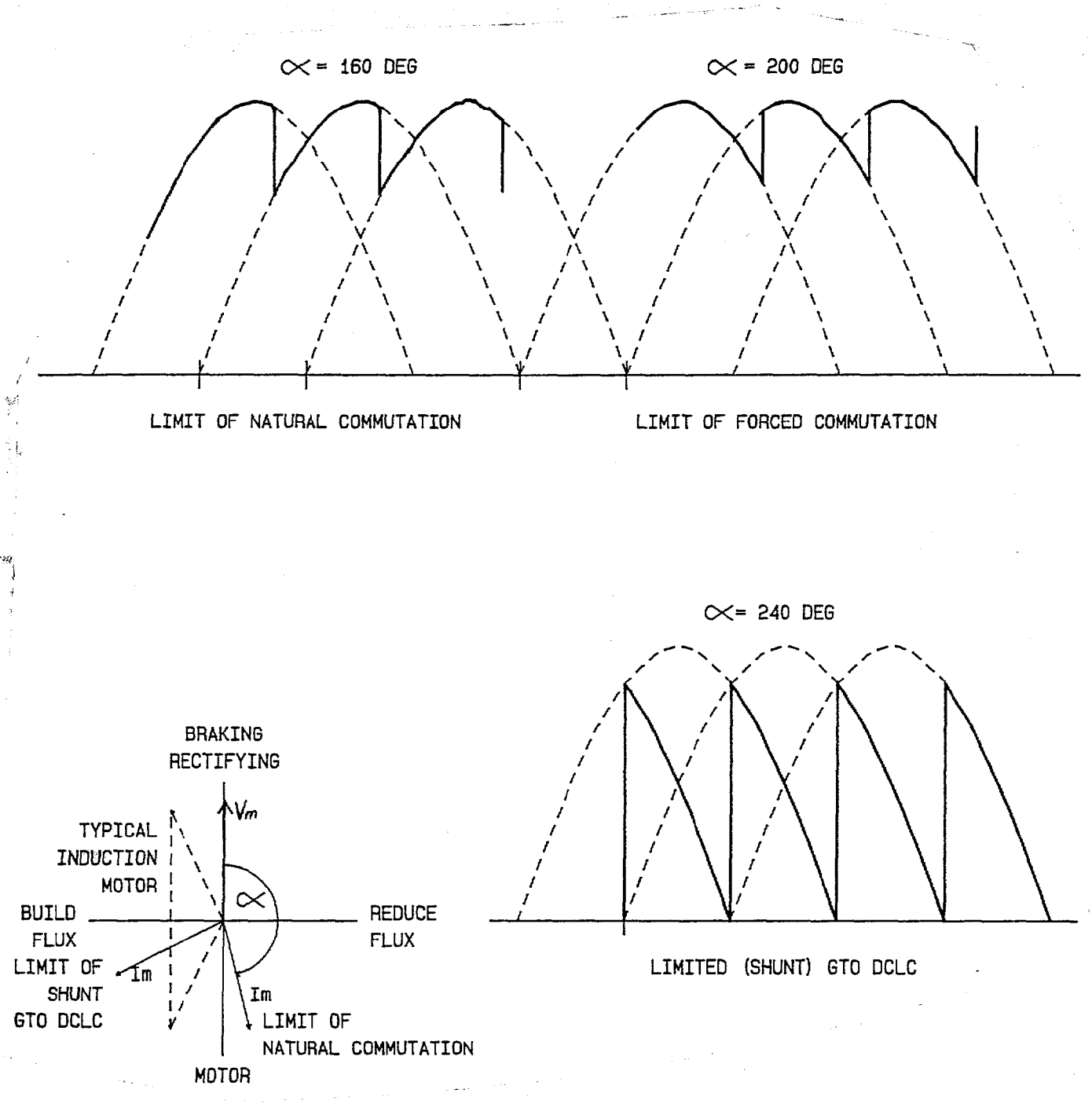

∝ = 160 DEG
∝ = 200 DEG
LIMIT OF NATURAL COMMUTATION
LIMIT OF FORCED COMMUTATION
∝ = 240 DEG
BRAKING
RECTIFYING
Vm
TYPICAL
INDUCTION
MOTOR
∝
BUILD
FLUX
REDUCE
FLUX
LIMIT OF
SHUNT
GTO DCLC
Im
Im
LIMIT OF
NATURAL COMMUTATION
MOTOR
LIMITED (SHUNT) GTO DCLC

*Figure 1-5.*

veshape is illustrated in Figure 1-2 for various inverter firing angles. From this figure the firing angle alpha is limited to 240 degrees. For regenerative braking, the firing angle range is alpha between 270 to 360 degrees, thus thisinverter approach does not provide regenerative braking at low speed. As a means of maintaining speed regulator control down to light load-low speed conditions, when the commanded firing angle exceeds the max angle limit, link current is commanded to current-lower-limit and the angle spillover is used to stretch the GTO conduction interval - reducing torque and flux and increasing harmonic content in the inverter output. This all serves to make the control task of satisfying the speed regulator while ensuring successful load thyrister commutation in this low speed region very difficult - particularly because this is also the region of resonance of the load capacitor and motor leakage inductance with the 5th and 7th harmonics of the fundamental freguency. To reduce this "self-excitation", the current in the 8-18 Hz region is "chopped" to reduce the 5th and 7th harmonics and 'chopped' in the 18-28 Hz region to reduce the 5th harmonic. This essentially limits sustained drive operation to the 50-100% speed range although the drive will allow acceleration through the 0-50% speed range.

7. Load Commutation Mode: (typically 50-100% speed)
8. For (0 degrees < commanded firing angle (alpha) < 165 deg) the load thyristers are commutated by the load capacitor. In fact, at high speed the load inverter must 'eat' some of the excess leading KVAR to prevent motor overvoltage. In this region the GTO DC Link forced Commuation mode does not need to operate. The relationship between motor, inverter, and load capacitor currents as a fuction of inverter firing angle is illustrated in Figure 1-3. At high speed motoring the firing angle (alpha) range is 90 deg < alpha < 180 deg. For transient high speed regeneration the alpha range is 0 deg < alpha < 90 deg. For low speed motoring operation the alpha range is 180 deg < alpha < 270 deg. These three cases are illustrated in vector diagrams in Figures 3a, 3b, 3c respectively.
9. Assumptions in Power Circuit Design:
10. Fan load, IQ = K1*(SPD**2), IX = K2, IDC = K3*SQRT((IX**2)+(IQ**2))
11. Nominal motor volts = rated voltage * actual speed / rated speed
12. Diverter will not operate for alpha < 165 deg, which implies that the diverter will not operate for speed > .5-.7 PU. Nonlinear reactors (Lf) with 20% unsaturated impedance, 1.5% saturated impedance, and saturation at 10% rated current will buffer the inverter from the load capacitor. A voltage spike is experinced on turning off the GTO and picking up current in the load inverter. This voltage is proportional to Vp = IDC * SQRT(2*Lf/Csnub_eqv) . This voltage spike does not occur when the GTO is not operating but the equipment will be sized such that semiconductors will not be overvoltaged if a control failure occurs and the GTO inadvertantly fires at high speed, high IDC provided that all series cells are intact. The forced commutation circuit will allow nominal operation with one shorted GTO but will not be rated for two concurrent faults such as one cell shorted plus the control failure discussed above.
13. The drive will not provide regeneration in the GTO DC Link Commutation mode of operation.
14. The drive will be able to accelerate through the resonance region of the load capacitor with the motor leakage inductance excited by harmonics of the fundamental frequency, but need not operate continously in this region.
15. No Synchronization-and-Transfer-to-Bus feature.
16. Does have Start-on-the-Fly feature.
17. The GTO DC Link forced Commutation design uses either three or four series 4500 volt GTOs - depending on the voltage and horsepower requirements.
18. Nominal DI/DT in the GTO string is controlled by the load inductors, Lf. However, there could be parasitic capacitor discharge associated with the DC link inductor and its cabling The GTO leg is designed for DI/DT protection for a max of .01 microfarad parasitic capacitance.
19. Shunt-Series GTO DCLC:
20. For drives needing to use the 10-50% speed range during normal operation, it is possible to get successful commutation with less Volt-Secs by using a GTO string in the DC link between the shunt GTO string and the inverter bridge (hereby called Series GTO). The series GTO firing pattern is essentially the opposite of the Shunt GTO i.e. when turn-on the Shunt GTO then turn-off the Series GTO and vis-versa.

This allows the drive to essentailly operate over the entire speed range (10-100%) although the DCLC and harmonic chopping are still used in the 10-70% range.

## 1-3. AC INDUCTION MOTOR

AC motor torque is produced by a force or pull between two magnetic fields in the air gap. One being produced by current flowing in the conductors of the stator or stationary member and the other by current flowing in the conductors of the rotor or rotating member.

In the controlled inverter system, only the stator circuit is accessible therefore both magnetic fields in the air gap must be controlled by controlling the stator parameters (or stator current, volts, and frequency).

Torque is proportional to the "torque producing component" of stator current, $I_S \sin \Theta$ and motor flux,$\psi$

$$T_{AC} = K_4 I_S \sin \Theta (\psi) = K_5 I_S \sin \Theta (I_S \cos \Theta) = K_6 I_S^{\ 2} \sin \Theta \cos \Theta$$

AC motor flux is produced by the flux producing component of stator current $I_S \cos \Theta$. The motor angle $\Theta$ is described as the angle between the stator current and the motor flux produced in the air gap by the stator current. See Figure 1-8.

In the induction motor, constant torque can be maintained by controlling the torque producing component and the flux producing component of $I_S$ which is done by regulating constant stator current and motor angle $\Theta$.

When the motor takes on additional load, or torque is required to change, the speed drops back and both $\Theta$ and motor slip increase which, in turn increases the rotor current and draws corresponding torque producing stator current to maintain load.

Figure 1-9 shows the speed-torque characteristic of a typical induction motor. With a constant voltage applied and a frequency equal to B applied, the motor would accelerate to speed A where the torque curve equals the load torque at that speed. As the motor accelerates, the developed torque equals the characteristic shown. As long as the applied frequency, voltage and load remain constant, the drive will run at speed A and a slip equal to B-A. Under these conditions of constant volts/cycle and constant slip-stator current, motor angle $\Theta$ and flux are constant. To regenerate power back into the line, the motor speed exceeds the applied frequency B, the torque becomes negative, and the motor acts as a generator.

Constant torque is controlled by holding constant stator current and motor angle. Motor angle is increased (Figure 1-8) to reduce flux (and torque) for constant HP operation. Stator current and slip also change during this region.

In addition to torque control, motor speed can be controlled by being able to go from motoring to regenerating as required to hold the set speed. This can be done without changing speed, by changing the input frequency. See Figure 1-9. Assume the motor is running in the set speed mode of operation in the motoring quadrant at speed A. The load changes to require a regenerating torque equal to E to maintain speed A. The control would change the input frequency to D (motor angle $\Theta$ and slip change polarities).

To control motor torque and speed, the inverter current and stator voltage, signals are fed back to the control. These components are used to calculate flux, torque, capacitor current and motor current. The converter circuit is controlled to regulate current. The inverter circuit is controlled to regulate motor speed and volts.

At high load on the motor, the apparent power factor "seen" by the power source increases. This is because the source converter firing advances ( is reduced) to obtain more current. The harmonics in the current and the resultant harmonics in the voltage, due to commutation "notching", will decrease.

The fundamental control strategy is: To increase motor current in response to a load torque increase, the D-C link voltage, on the source side, is increased, thereby raising motor current so as to keep motor speed constant.

## 1-4. SERIES TWELVE-PULSE OPERATION (OPTION)

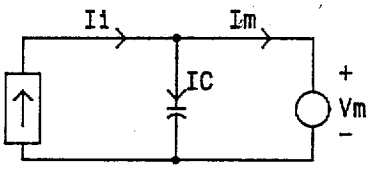
Series Twelve-pulse operation consists of two identical, separate, six-pulse bridges operating from a common source at the same current with the voltage and firing reference angles shifted 30 degrees by an incoming isolation transformer. But the two drives are combined into one by connecting the "+" bus of one to the "N" bus of the other .

The two bridges deliver equal power and the transformer design makes the source-side converter voltage equal, but also 30 degrees apart between the two channels. It is necessary that source-side and load-side voltages be phase-shifted. The transformer windings are shifted to reduce the harmonic distortion on the power system and to raise the harmonic frequencies, and to increase the power flow.

Normally the source master voltages (WYE) LEAD the source slave voltages (DELTA) .

## 1-5. BYPASS (ACROSS THE LINE) OPERATION - OPTION

On drives with a bypass mode, the motor can be connected directly to the line - "bypassing" the IMD




PER PHASE EQUIV. CIRCUIT

Ii = Ic + Im

Im = MOTOR CURRENT
Ic = LOAD CAPACITOR CURRENT
Ii = INVERTER LOAD CURRENT
Vm = MOTOR = CAPACITOR VOLTAGE

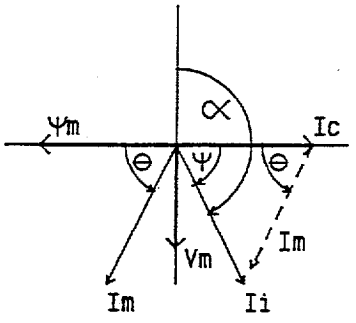



FIGURE 1-3A

$90° < \alpha \leq 180°$
HIGH SPEED MOTORING
LOAD CAPACITOR COMMUTATION

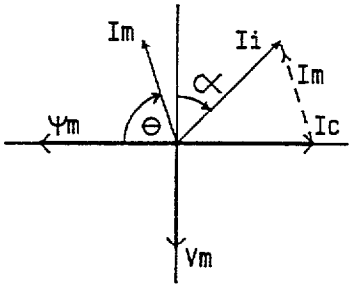



FIGURE 1-3B

$0° < \alpha \leq 90°$
TRANSIENT HIGH SPEED REGENERATION
LOAD CAPCAITOR COMMUTATED

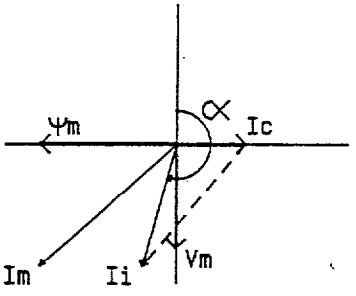



FIGURE 1-3C

$180° < \alpha \leq 240°$
LOW SPEED MOTORING
DC LINK COMMUTATION

*Figure 1-6.*

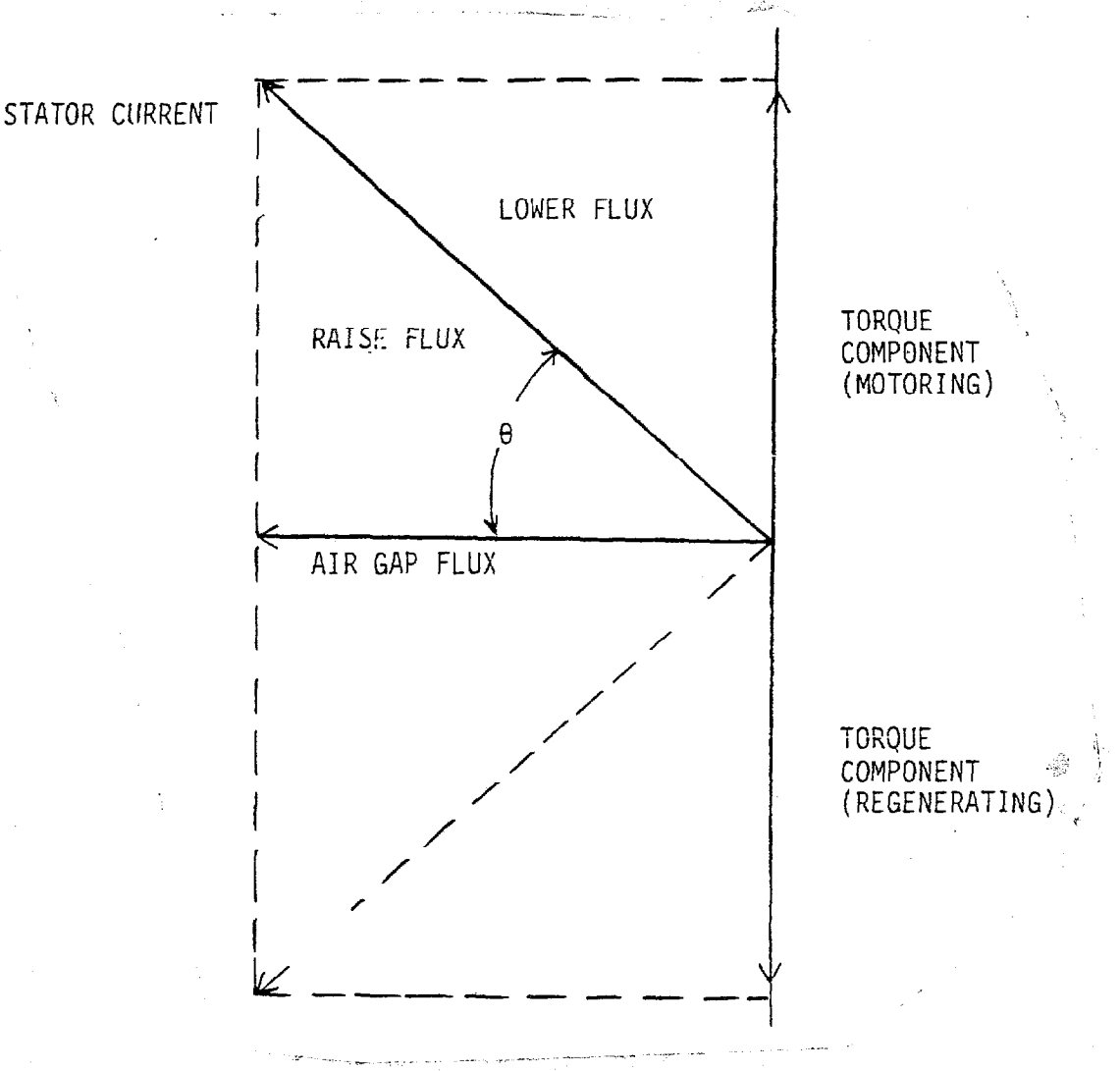



*Figure 1-8. Motor Angle*

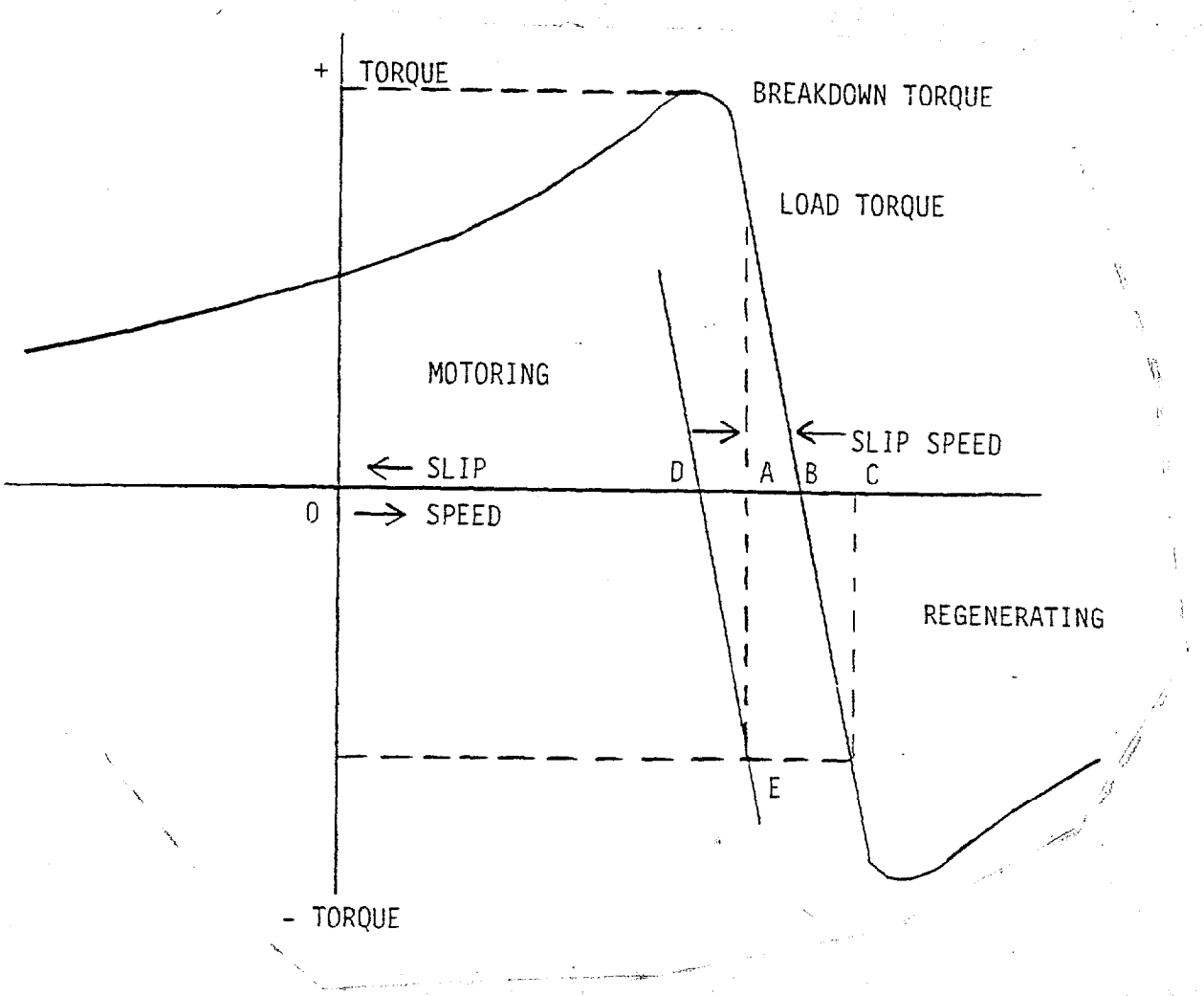



*Figure 1-9. Speed Torque Characteristics of AC Induction Motor*

drive. This feature gives "back-up" capability to the system, i.e. the motor can still be run if the drive is faulted.

# CHAPTER 2

## 2-1. EQUIPMENT OPTIONS

Three types of bridges may be utilized for the power conversion equipment. One type uses three thyristors in series and will be designated the medium voltage bridge (MVB). Another type utilizes one or two thyristors in series and will be designated the SILCOMATIC® Plus power conversion bridge. The third uses three or four gate-turnoff thyristors (GTO) in series and is designated as a GTO leg.

## 2-2. 2KV MEDIUM VOLTAGE BRIDGE

The arrangement of the power converter SCR stack assemblies is shown in Figure M2-1. This figure and the elementary diagram show the major power components.

The SCRs are stacked horizontally; three in series, between four heat sinks. The 18 SCRs between the P-bus (P1, P2) and the AC lines, source and load, are on the left side in the bridge compartment; also CT1, CT3, FU1-4 and FU-6.

The N-bus SCRs are on the right side of the bridge compartment; also CT4, CT6, FU-5, and FU7-10.

The blower compartment contains the blowers, source and load AC and DC filters, MOVs, CT1A, CT3A, CT4A, and CT6A. The hinged base in this compartment contains the blower starters, the 70V gating power supply, the thyristor fan-out PWB XTFH, and bottom blower selection relay.

## 2-3. THYRISTOR MODULES, MEDIUM VOLTAGE BRIDGE (MVB)

Each power converter leg module consists of three thyristors arranged in series in a heatsink assembly.

The master leg module is mounted in an insulated box with associated printed wiring boards in front as shown in Figure M-2.

An R-C snubber network is also contained in the leg module box, above the cells.

Each cell stack assembly consists of three cells sandwiched between heatsinks as shown in Figure M-3. The stack assemblies are oriented horizontally in the module as shown. Air flow over the heatsinks is from front to back of the bridge enclosure and the cell modules.

The cells are selected, matched, and assembled at the factory under the correct amount of pressure. For these reasons, it is recommended that the entire stack be replaced if a cell failure occurs. The defective stack may then be sent to the factory for repair.

Each cell in the module is paralleled by a voltage-equalizing resistor and by a neon lamp and dropping resistor network which indicates when voltage is present across the cell. These lights are visible through the front doors of the equipment and can be used to detect a shorted cell. The cell light network is mounted on a small card (NTDA) located above each cell on the front of the thyristor modules. The location of these cards is as shown in Figure M-2. The NTDA card also contains thyristor gate interface circuitry which is illustrated in Figure M-4. Thyristor gating signals are routed from the gating transformers through the NTDA cards to the thyristor gates.

## 2-4. GATE DRIVER TRANSFORMERS

The cell gates are driven by current transformers mounted on the front of the module as shown in Figure M-2. The current transformers for each module are wired so that all their primary windings are in series. The primary winding is simply a conductor through the center hole of the transformer. Current pulses are applied in this conductor by the gate driver circuitry. This circuit insures that all thyristors in the string are gated simultaneously and it further provides voltage isolation between the thyristor gates and the electronic gate pulse generation circuitry.

## 2-5. GATE DRIVER CARD (HPTK)

Gate pulses of proper duration and timing are developed in an electronics module located in a control enclosure at the left end of the MVB. These signals do not possess the necessary drive and isolation to interface directly with the gate driver current transformers. This interface is provided by the HPTK gate driver card located in the thyristor module in each leg. The HPTK receives the low level signals and, after conditioning, passes them through a pulse transformer to provide isolation. The pulse transformer output drives a pulse amplifier circuit which is powered by a 70 VDC gating power supply located in the fan and filter compartment of the MVB. The pulse amplifier output drives all gate pulse transformers in the leg as described above. A red LED on the board glows to indicate when gate pulses are present.

A second function of the HPTK card is to detect an undervoltage condition of the 70 VDC supply. In the event that an undervoltage is detected, the HPTK outputs a logic signal to the electronics module indicating the fault. The electronic control then takes appropriate action to shut down the drive. Since there is an undervoltage detector on each gating card, wiring defects in the gating power will also be detected.

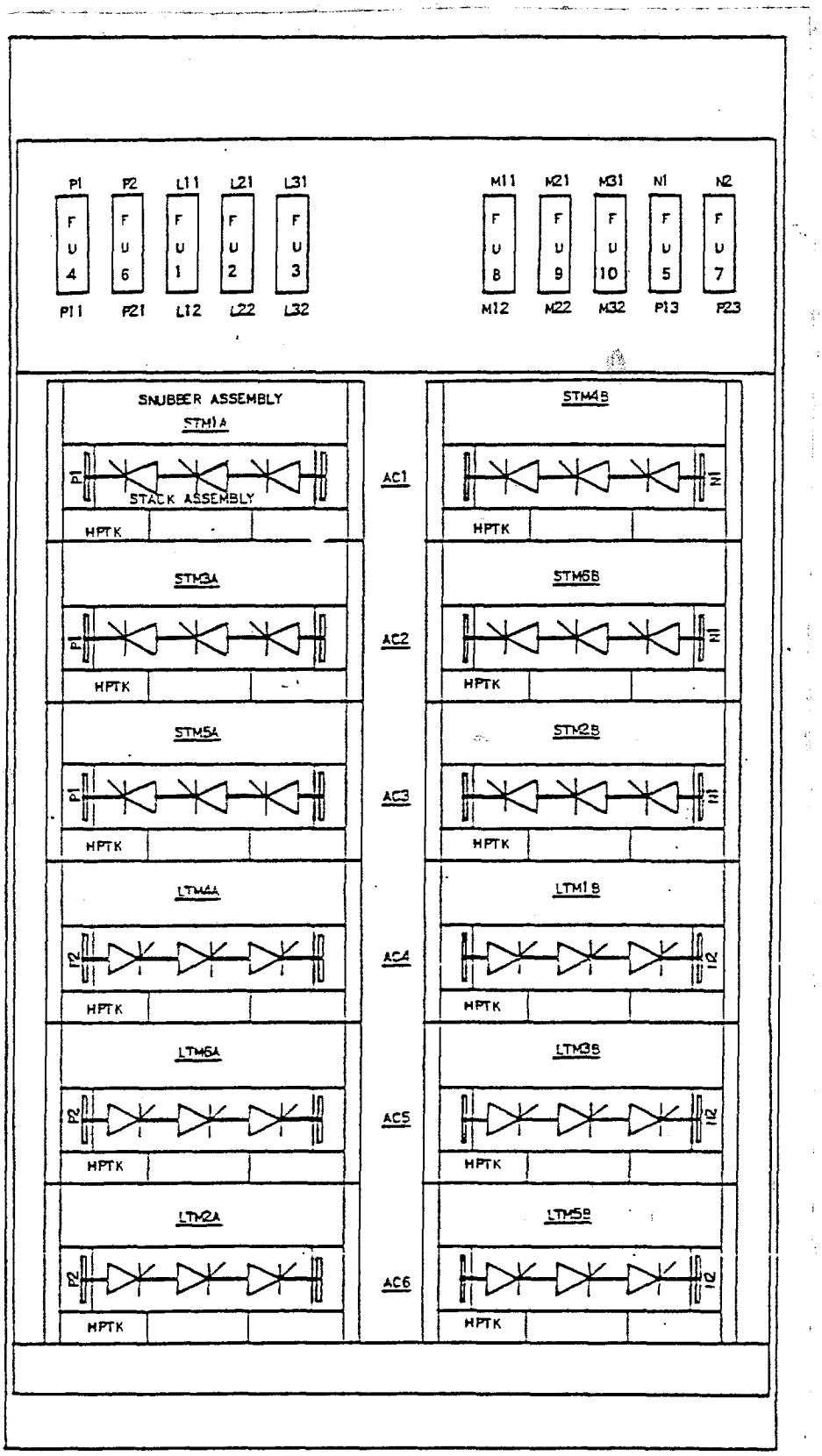



*Figure M2-1. Medium Voltage Bridge Assembly*

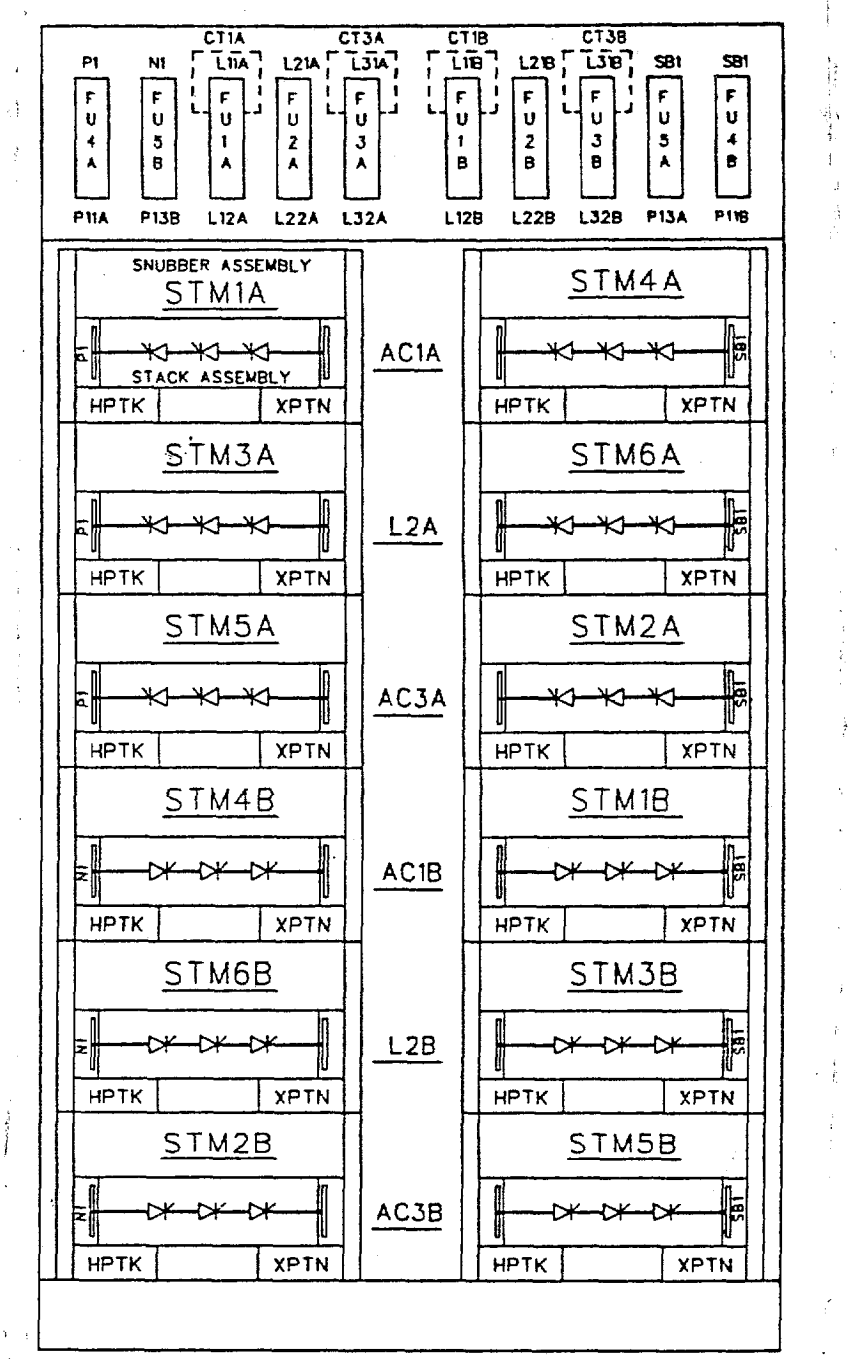



*Figure M2-2. Medium Voltage Bridge Assembly*

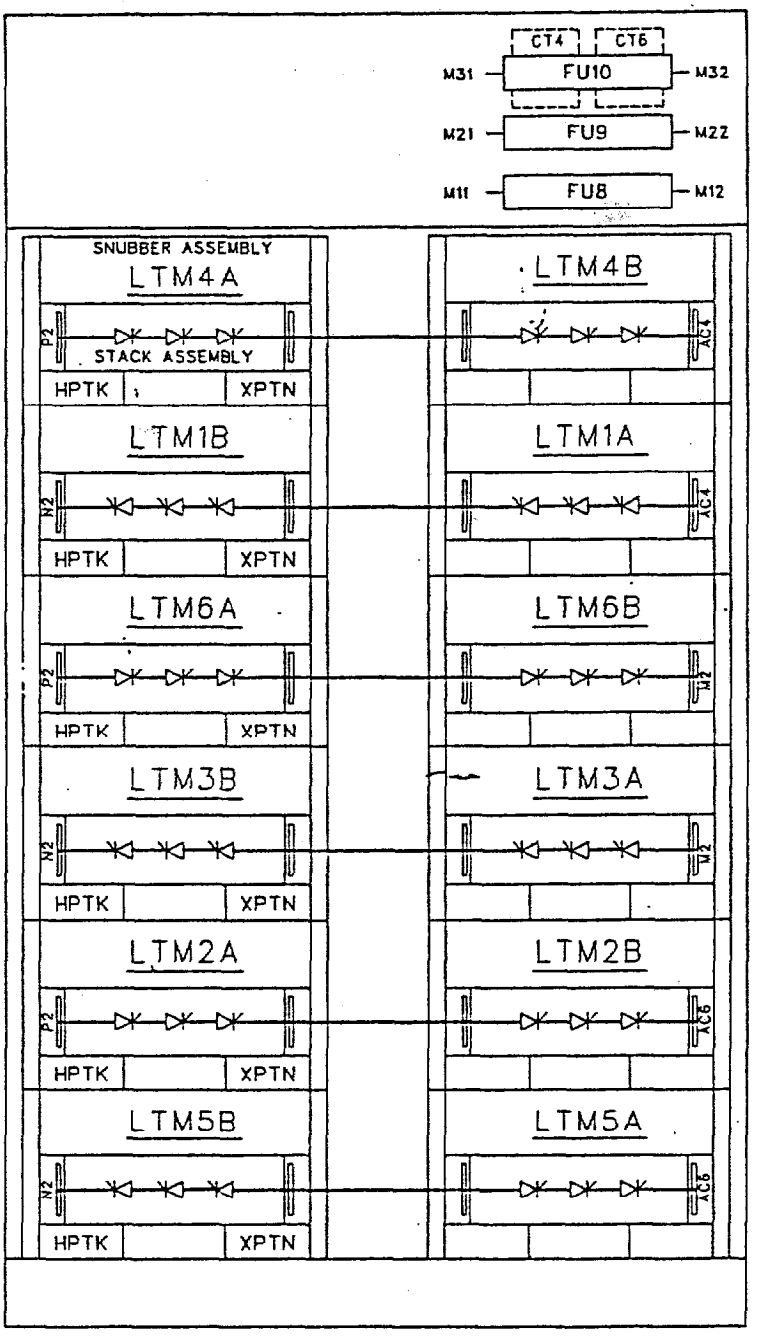



*Figure M4-2. Medium Voltage Bridge Assembly, 4KV*

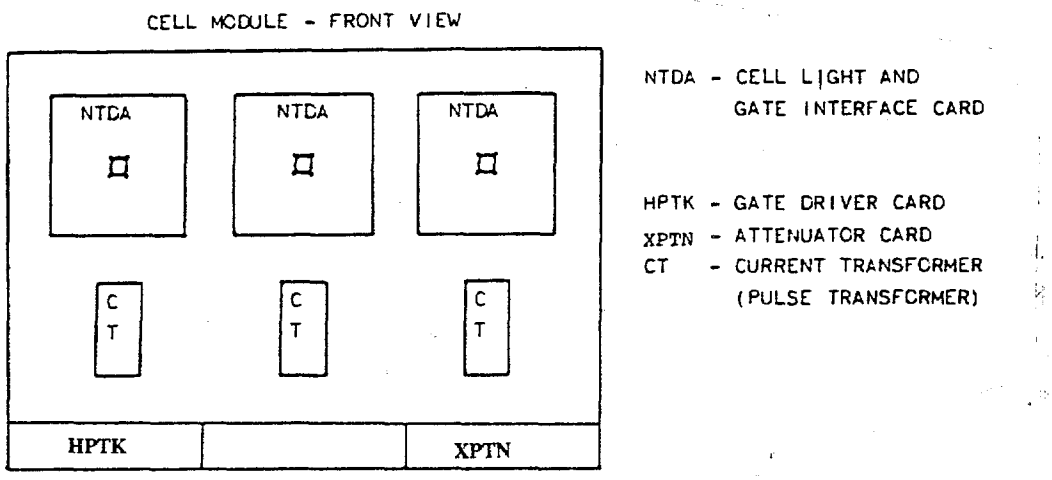



*Figure M-2. MVB Thyristor Module*

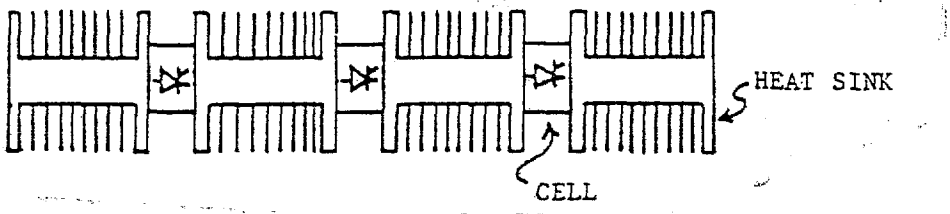



*Figure M-3. MVB Cell Stack Assembly*

A block diagram of the HPTK card is given in Figure M-5. The pulse driver delivers a current pulse simultaneously to all thyristors in the leg module.

## 2-6. VOLTAGE FEEDBACK ATTENUATOR CARD (XPTN)

In order to properly control the gating signals to the MVB, the electronics in the control enclosure must have available analogs of all AC and DC bus voltages. The XPTN attenuator cards in the thyristor modules provide bus-to-ground voltage signals which are combined in the electronics module to produce bus-to-bus signals for each converter bridge.

Figure M-6 illustrates the overall attenuator connections for developing bus voltage feedback signals for an entire bridge. Note that only five bus-to-ground signals are required. Consult MVB elementary DS5201 for complete details of attenuator connections.

Five signals are derived from the attenuator units as illustrated in Figrue M-6:

In Figure M-6, a series string of thyristors is represented by a single SCR symbol.

Note that the line-to-ground attenuator in leg module 4 is not used.

As explained in the paragraph discussing power converters, the voltage signals are used to control firing of the SCRs and to provide various protective functions.

The attenuation is variable via a Berg Jumper (BJ) for various system voltages: 1.75KV-STO; 2KV-LO; 4KV-HI.

## 2-7. SIGNAL DISTRIBUTION BOARD (XTFH)

All signal interchange between the drive electronic control and the medium voltage bridge is made through the XTFH signal distribution board located in the fan/filter compartment. A small ribbon cable from each HPTK gating card connects to the XTFH board. These ribbons each contain gating control signals, attenuated bus voltage signals, and 70 volt power

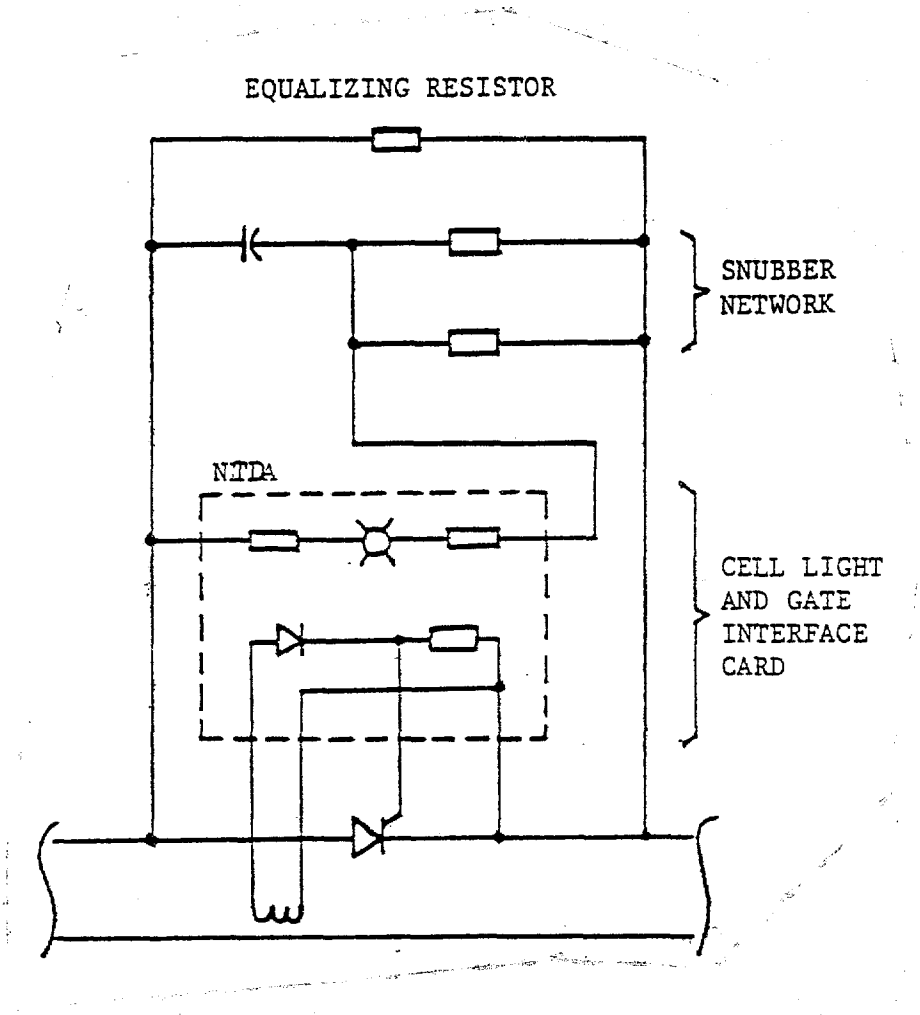



*Figure M-4. Cell with Gating, Snubber, and Voltage Equalizing circuits*

supply undervoltage signals. The undervoltage signals from each bridge are combined in the XTFH card to produce a composite undervoltage signal for each bridge. The gating signals originate in the electronics module in the control compartment, while the remaining signals originate in the medium voltage bridge and must terminate in the electronics module. Signal interchange to the electronics module is made via two large ribbons which connect the XTFH board and cards in the source and load electronics.

The XTFH contains the burden resistor circuits for the source and motor line current transformers. The auxiliary ct leads plug into the XTFH board and two small ribbons route these signals to the source and load electronics in the control cabinet.

## 2-8. PWB ADJUSTMENTS, TEST RINGS, LIGHTS, AND SWITCHES (MVB)

1. HPTK

TP1 - Gate pulse signals from SEM. Leg 1 same as SIG1, leg 2 same as SIG2, etc. 0 to 28VDC.

TP2 - 70V Gating power. Less than 10 volts DC indicates power undervoltage. Measure from PC1, PC2, or PC3 (Stabs on card).

TP3 - Gate Pulse signal common (reference for TP1)

CR -Light is on while gate pulses present.

2. NTDA

NE - Neon light. Normally illuminated. Light goes out if cell shorts.

TP1 - Cell cathode. Voltage across cell can be measured between adjacent test points.

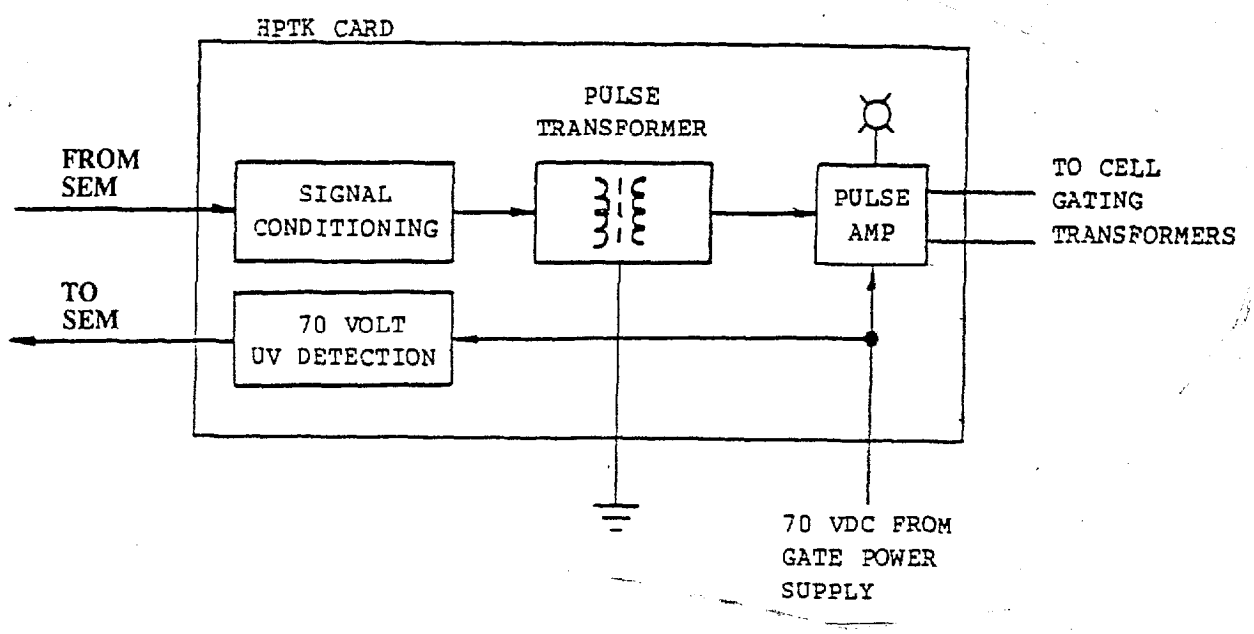



*Figure M-5. HPTK Block Diagram*

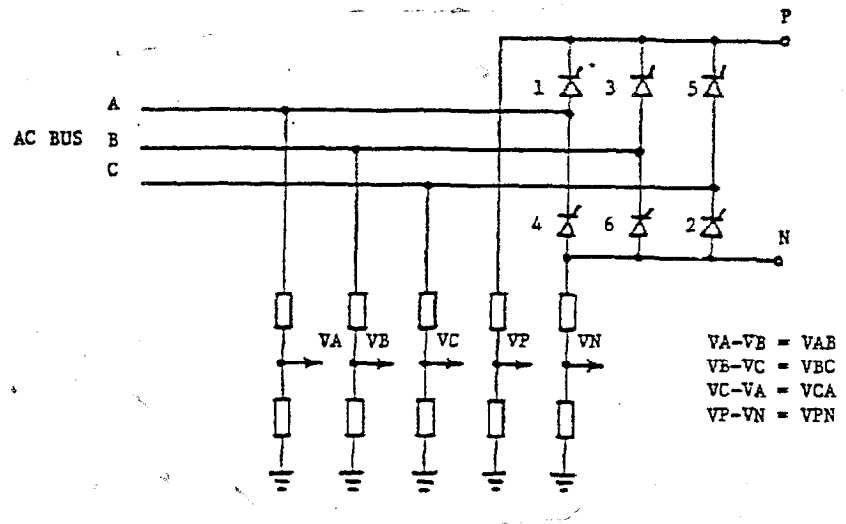



*Figure M-6. Bridge Attenuator Circuits*

# CHAPTER 3

# CONTROL ELECTRONICS PANEL

## 3-1. INTRODUCTION

The control electronics panel is made-up mainly by various modules interconnected thru ribbon cables and harnessed wiring. The main modules are:

| | |
|---|---|
| DS3800IME | Main Control electronics |
| DS3820IOM | Inputs/Outputs (I/O) |
| DS3820SIM | Auxiliary interface electronics (When more than the normal signal are needed) |
| DS3820LDM | Display Module |

The modules are typically configured as in Figure 3-1. Following are brief descriptions of the functions of the module. For more specific information the corresponding modules elementary and documentation should be consulted.

## 3-2. ELECTRONICS MODULE (IME)

The control electronics module consists of cards in 2-rows x 6-slots. Its primary purpose is to control the firing of the power converters and field exciter module SCRS for driving a motor. It also provides various protective functions for the motor and bridges. Interface between the IME to the customer is thru the IOM and SIM.

See Figure 3-2A for the physical layout of the module. Connection is made to the electronics module (SEM) through the plugs located at the top, and also through card front connectors.

From a functional standpoint, the electronics module processes signals as represented in Figure 3-2B.

The signal interfaces for the SEM are as follows: Attenuator signals - Attenuated bus to ground signals from each converter.

SIM module - See SIM section

Line/load current feedback - Signals from line/load bus current transformer burden resistors.

Power supply - The module requires +5, ±15, and +28 volts which are provided by a base-mounted power supply.

Thyristor gate signals - These are the electronic signals which drive the gate pulse circuits in the medium voltage bridge.

The hardware of the electronics module is represented from a block diagram standpoint in Figure 3-2C.

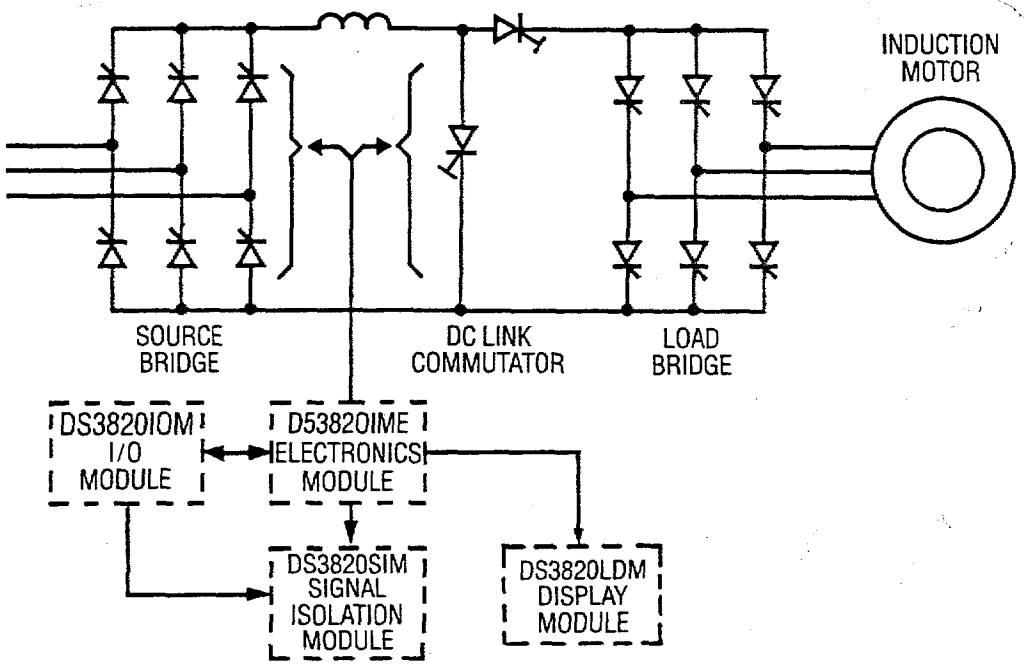



*Figure 3-1. Control Electronic Panel*

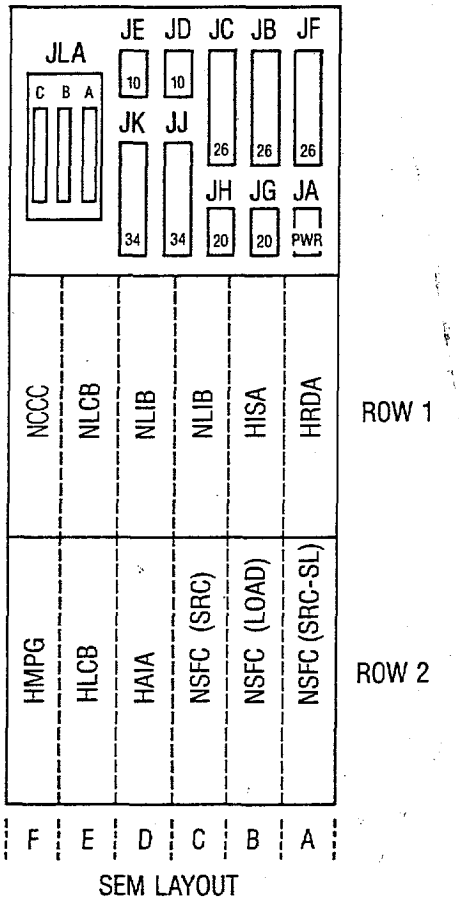



*Figure 3-2A. Small Electronics Module (SEM) Card and Connector Layout*

There is one microcomputer controlling the firing of both of the power converters and the field exciter.p

The microcomputer establishes a local bus (L-bus) structure on which the input/output for it is interfaced. In addition, it communicates to an 8k byte read/write (RAM) memory and a 2k byte electrically alterable memory (EAROM). This memory combines features of read/write memory (RAM) and read only memory (ROM) in that it is electrically alterable (although slow) and at the same time will not lose its contents on power loss. The primary use of this EAROM memory is to store parameters which must be tailored to each different system. These parameters are field adjustable for drive tuneup.

The local bus has a timer and interrupt card which provides the necessary hardware for control of interrupt priority and timing for the microcomputer.

There are input/output ports on the local bus which interface static logic signals to/from the microcomputer. As an example, thyristor gating commands are written out to one of these ports (on each side) and sent to an analog card to produce the gating signals to the converter bridge.

The electronic control contains a number of analog cards which process the voltage feedback, current feedback, meter signals, etc.

Analog to digital converter cards interface the speed reference, voltage feedback, current feedback signals, etc. to the microcomputer bus on each side.

I/O cards interface process command and status signals to the microcomputer. These inputs are combined logically by a sequencing program in the microcomputer to control the relays in the IOM. These outputs are used to control system devices such as panel lights, contactors, etc.

## 3-3. CONTROL OPERATIONAL DESCRIPTION

**3-3.1. Basic Principles.** Refer to the Control System block diagram in Figure 3-3.1A. The speed refer-

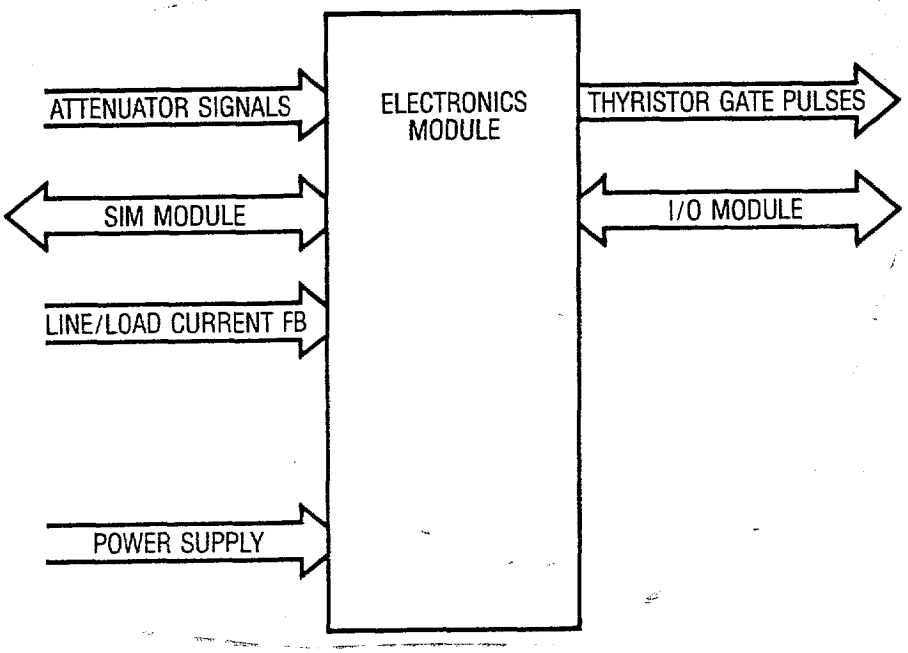



*Figure 3-2B. SEM Input/Output*

ence from the process controller is compared to a speed feedback derived from the integrated motor voltage to develop a speed error input to the speed regulator. The output of the speed regulator is a torque command. The speed regulator program resides in the load side processor.

The torque command signal is applied to both the source and load side control. As illustrated in Figures 1-6 and 1-7, motor torque is a function of voltage, current, and the angle between them. Therefore, torque can be controlled by adjusting current magnitude at a fixed angle or maintaining a constant current and varying the displacement angle. The current is controlled from the source side , whereas the angle is varied from the load side . At any instant in time, the torque is actively controlled by only one of these means.

The torque command to the source side control is applied to a maximum and minimum current limiter. The minimum current level is set to maintain continuous current in the DC link.

The current level control also affects the load side angle control. When the voltage spillover is not activated, the angle control is in one of its limits. If current is adjusted to control voltage, the angle control gain is decreased and displacement angle O is increased, which reduces developed torque to that required by the load.

The current command is compared with current feedback, $I_{DC}$, and the error is applied to the current regulator. The current regulator controls the firing of the thyristors in the source side converter (rectifier). Thus, the source side control adjusts the DC link voltage as required (within its limits) to produce the motor current needed to drive the load.

The load side control is designed to adjust firing angle so as to maintain sufficient flux, by controlling the angle between voltage and current .

Referring to Figure 1-4, it is shown that the angle of advance,B, is the sum of the commutation angle,U, and the margin angle,$\gamma$. For maximum efficiency, it is desired to keep the margin angle as small as possible without encountering commutation failure. The objective is to make the displacement between voltage and current as small as possible while maintaining torque and flux.

When the drive requires regeneration, the load-side thyristors are fired full advance (point "X" in Figure 3-3.1B), and the source side controls current by reversing the DC voltage to match the rectified motor voltage.

**3-3.2. Phase Locked Loop Operation.** Since the inverter thyristors are fired in synchronism with the integrated motor stator voltages, a phase locked loop

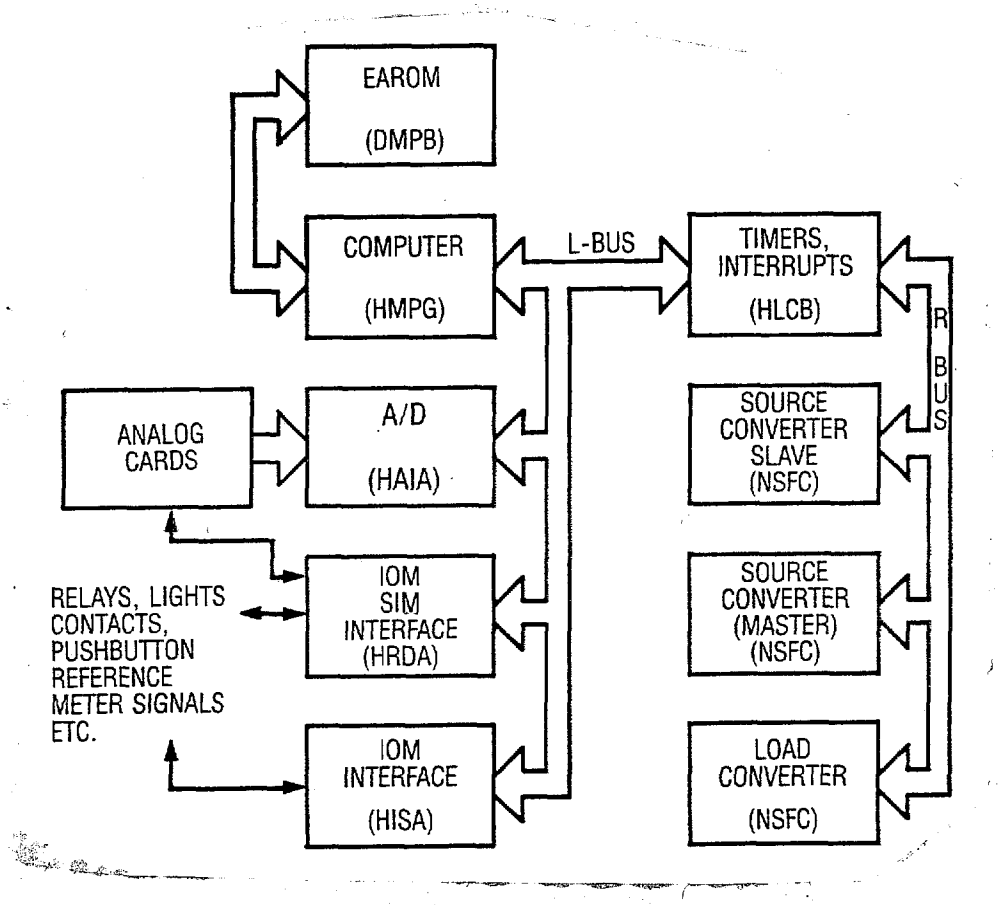



*Figure 3-2C. Electronics Module Block Diagram*

(PLL) counter is used to maintain correspondence between the integrated voltage wave zero crossings (Figure 3-3.1B) and the firing pulses.

When the PLL counter output is precisely in synchronism with the "flux" wave zero crossing, the loop is said to be "locked". If the counter output is lagging (slower than) the flux crossings, the PLL controlloer acts to increase the counter frequency (by decreasing PLLC LOAD). Similarly, if the counter frequency is leading (faster than) the flux crossings, the controller acts to decrease frequency (by increasing PLLC LOAD).

Although the output of a phase locked loop, when "locked", will have the same frequency as the reference signal, there will be a phase difference between the two due to the integrating action of the loop low-pass filters; thus, the name phase locked loop.

## 3-4. ELECTRONICS MODULE (IME) PRINTED WIRING BOARD (PWB)

Descriptions Following is a list of cards used in the electronics module. The function and condensed theory of operation of these cards follows the list. It is not the intent to describe in great detail the inner workings of each card, but to relate their function in the overall system. For more detail, refer to the appropriate DS3800 drawing, Section 2, for operating notes and Section 4 for elementary diagram. It is also generally helpful to refer to the card symbols in the SEM elementary diagram.

**3-4.1. ALCI Processor extender board = DS3800HLCB .** The HLCB board extends the functions provided by HMPG to include those required for an LCI system.

The following functions are provided.

1. Address decoding for on-board functions. NSFC, and NSFF.
2. Interrupt controller. - Source crossover
   - load crossover
   - source angle control
   - field excitation control
3. 5 bit purposes ionput port used for:
   - ground fault
   - I/O module input ribbon cable continuity

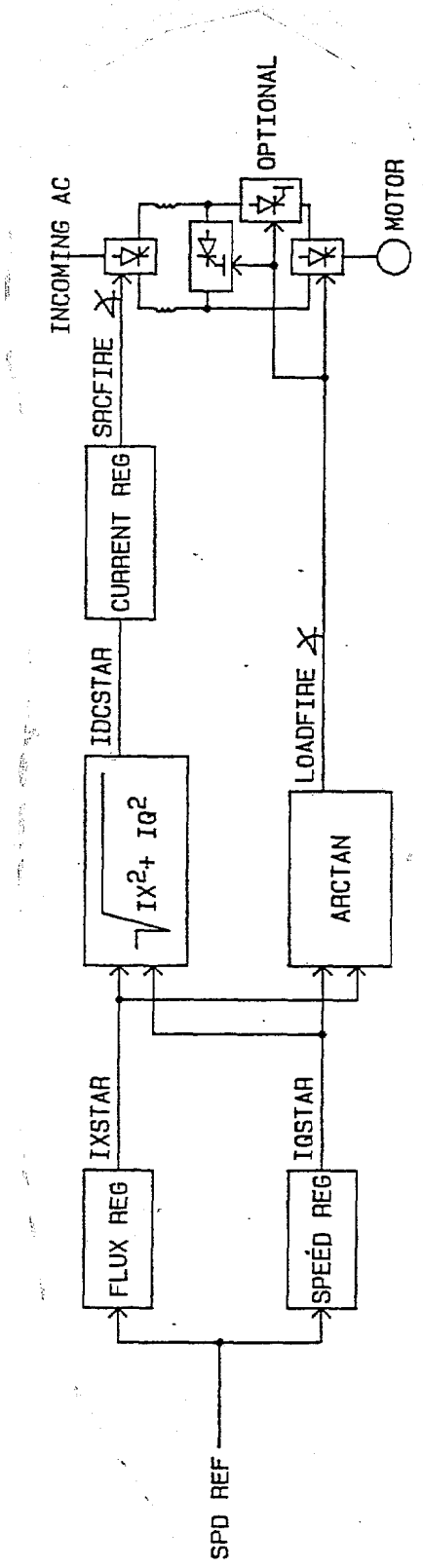



*Figure 3-3.1A. DCLC-IMD System Digital Regulator Block Diagram*

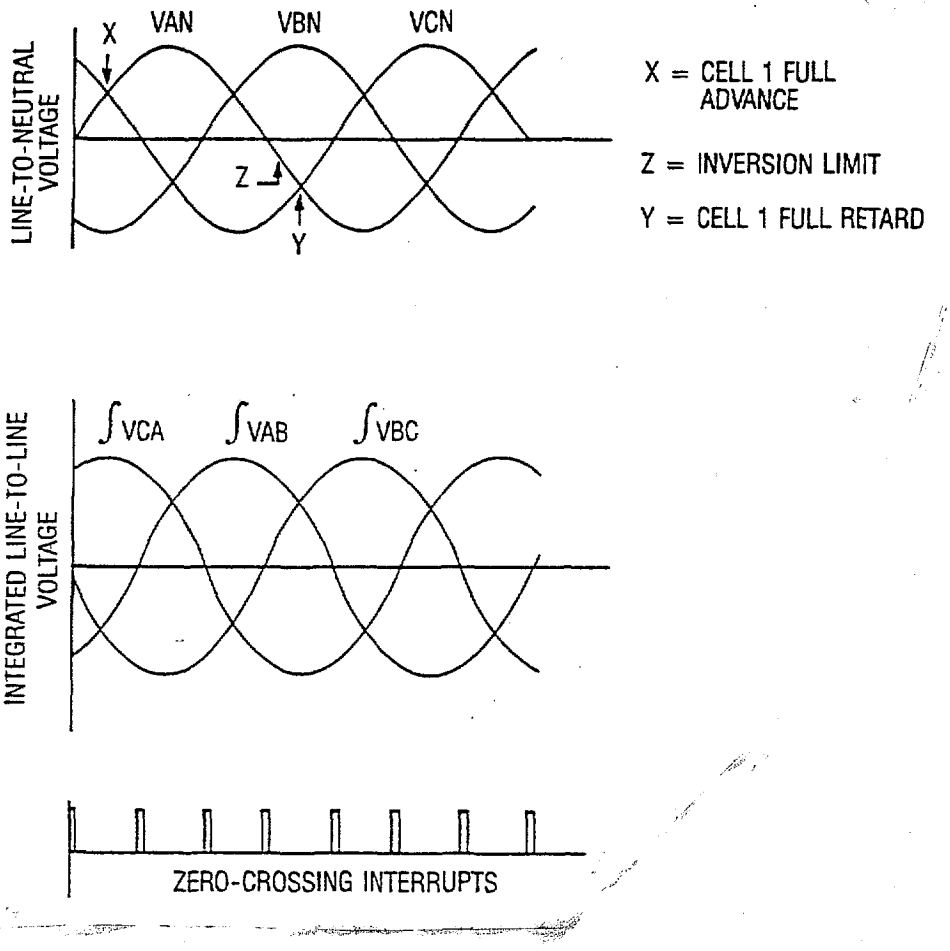



*Figure 3-3.1B. Flux Wave Zero-Crossings*

- I/O module signal cable continuity
- door module ribbon cable continuity

4. Zero crossing detection. Circuity is included to derive source and load crossover interrupts from square wave information originating on the MLIB boards. A single clock pulse interrupt is obtained at each crossover.
5. Two programmable timers provide the functions of - PLL clock (source and load)
   - TIME counter (source and load)
   - SPEED calculation
   - CLOCK for serial communications port
6. RS-232 compatible port. Only the transmitt function is implemented at the port is intended for driving a printer locaated in the door module. The control line, DSR (data set ready), is also monitored to determine the status of the remote printer.
7. Digital to analog converters. three 8-bit D/A converters are suplied. D/A 1 and D/A 2 are 8 bit mon-committed outputs which are available for transmission to the signal isolation module (SIM) if supplied. The 12 bit D/A is committed to supplying the torque reference between 6 pulse converter controls when used in a 12 pulse redundant system. System level documentation will define its use (if any) in other systems.

Berg jumpers

None used.

LED's

| Nomenclature | Function |
|---|---|
| IMOK | Software controlled as a function or selftest. On normally, will be off if selftest is not passed. |

Testpoints

| Nomenclature | Function |
|---|---|
| ACOM | Analog common. |
| DA1 | Digital to analog output 1. Nor- |

| | |
|---|---|
| | mally uncommitted. |
| DA2 | Digital to analog output. Normally uncommitted. |
| DA3 | Digital to analog output 3. Normally used for horsepower output. |
| DA4 | Digital to analog output 4. Normally used for torque command to other channel (if supplied). |
| OTESTIN0 | ? |
| OTESTIN1 | ? |
| SFT1 | Fxecution flag for source crossover and angle control routine. |
| SFT2 | Execution flag for load crossover and angle control routine. |
| SFT3 | Execution flag for field angle control interrupt. |
| SFT4 | Execution flag for background program. |
| VA1 | Phase A signal for source power. |
| VA2 | Phase B signal for motor voltage. |
| DCOM | Digital common. |

**3-4.2. Microprocessor - HMPG/DMPB.** The DS3800HMPG is a single board computer with the following features.

1. 80286 microprocessor
2. Programmable communications interface
3. Programmable interval timer
4. On-board RAM memory (16K bytes)
5. On-board PROM memory (64K bytes)
6. Mapped I/O
7. Bus hog timer
8. Stall timer

The configuration chosen gives a powerful 16 bit processor.

Connectors.

A 50 pin ribbon connector is included on the DMPG to interface the EAROMmemory (DMPB).

A 25 pin male connector is present to serve as an RS232C communications port. (See Tables 3-1A and 3-1B) A "null modem" is supplied to allow a standard data terminal interface cable to be used with either communications port.

**3-4.3. Electrically-alterable . ROM = DS3800DMPB**

This is a non-volatile memory board containing 8k bytes of memory used to store drive tuneup parameters which may be customized on a per drive basis. Physically, the board is configured as a daughter board and mounted on the HMPG.

Constants stored in the board's memory are normally changed by first enabling board writes with switch "WREN" and then supplying the data through the drive monitor.

In the Drive Systems documentation procedures, DMPB and the two PROMS in sockets U2 and U3 comprise module DS3815PLHA.

DS3815PLHA and HMPG are then combined to form module, DS3815PLGA. (See Table 3-2)

**3-4.4. Analog/Digital Interface Card DS3800HAIA**

The HAIA board accepts 16 analog input signals, multiplexes them to one line and coverts to a two byte digital word. The A/D provides 11 bits of magnitude resolution. Also provided are three, 8-bit digital input ports, three 8-bit output ports and one 4-bit digital port. Two boards are used, one for source microprocessor and one for the load. (See Table 3-3)

**3-4.5. DC LINK COMMUTATOR CONTROL CARD - DS3800NCCC, 1F.** This card sends out the gating signal (from the microprocessor control) for the shunt and series (if used) GTO to the XTFZ card (which fans it out via light pipes). It also receives the status (OK) of each of the GTO's. This information is available for use by the control program via the L-Bus.

Other functions include:

- Source slave bridge's current conditioning.
- A fault trip comparator of source slave current vs load current.
- A shorted motor filter capacitor detector.
- Motor current calculation circuitry which consists of the filter capacitor current and subtracting it from the load current.
- synchronizing function circuitry which includes PT match comparator (motor volts vs source (bypass) volts) and motor current as read from the bypass or motor isolator.
- DC link overvoltage detection

Test points and lights include:

| | | |
|---|---|---|
| TP1 | OIA2 | SRC slave phase A current |
| TP2 | OIB2 | SRC slave phase B current |
| TP3 | OIC2 | SRC slave phase C current |
| TP4 | IFB2 | SRC slave bridge current 5V=1PU |
| TP5 | IERR | (SRC slave current)-(load current) |
| TP6 | CIAB | Capacitor phase A-B current |
| TP7 | CICA | Capacitor phase C-A current |
| TP8 | MIAB | Motor phase A-B current |
| TP9 | MIBC | Motor phase B-C current |
| TP10 | MICA | Motor phase C-A current |
| TP11 | MIFB | Motor current 5V=1PU |
| TP12 | PTER | Motor PT(B) - Source PT(A) |
| TP13 | GSH | Shunt GTO firing pulse |
| TP14 | GSR | Series GTO firing pulse |
| TP15 | Test | |
| LED | IMOK | On board op-amps not saturated |

Daughter Board

*Table 3-1A. Microprocessor Board Features - RS232C Connector Signals*

| Pin# | Name | Signal | Direction | Function |
|---|---|---|---|---|
| | | **RS232C CONNECTOR SIGNALS** | | |
| JB-2 | BA | TXD | output | transmit data |
| -3 | BB | RXD | input | receive data |
| -4 | CA | RTS | output | request to send |
| -5 | CB | CTS | input | clear to send |
| -6 | CL | DSR | input | data set only |
| -7 | AB | DCOM | input | signal ground |
| -9 | CD | DTR | output | data terminal ready |

*Table 3-1B. Microprocessor - Berg Jumpers, Testpoints and LED's*

| Nomen. | Function | Position |
|---|---|---|
| | **BERG JUMPERS** | |
| J1 | Selects type of PROM | 8/16 |
| J2 | Selects size of PROM | 16k |
| J3 | Selects bootup method (on/off bd.) | DGN(not) |
| J4 | Selects 8 or 16 address segments on L bus | 8 |
| J5 | ground for RS232 | GND |
| | **TESTPOINTS** | |
| RST | Reset jumper to common to reset processor | |
| DCOM | DCOM – Digital common | |
| I/S | Inhibit stall – Jumper to common to activate | |
| TNMI | Test nonmaskable interrupt – Jumper to common to activate | |
| | **LED'S** | |
| BHOG | Bushog – On when non-existent address is called by software | |
| TST | Software controlled and defined – On during initialization self test | |

*Table 3-2. Electrically-Alterable ROM - LED'S and Switches*

**(DMPB)**

| Nomen. | Function |
|---|---|
| | **BERG JUMPERS** |
| None Supplied | |
| | **LED'S** |
| IMOK | Software controlled as a function of self diagnostics — On normally, will be off if selftest is not passed |
| WREN | *Write enable. On when write capabiltiy to board is* enabled with switch, "WREN" |
| | **SWITCHES** |
| WREN | Activates the board to allow alteration of the drive tuneup parameters. |

| | |
|---|---|
| TP1/R1 | Adjustment of current unbalance (IERR) trip point |
| TP2/R2 | Adjustment of PT match point |
| TP3/R3 | Adjustment of shorted capacitor detection point |
| R4 | *Gain adjustment of contactor motor current for* metering |
| TP4 | TP4 is motor current 5V = 1PU |
| TP5/R5 | Adjustment of DC link overvoltage detection point |

**3-4.6. 4KV MEDIUM VOLTAGE BRIDGE-COMPACTED.** The arrangement of the power converter SCR stack assemblies is shown in Figure M4-1. This figure and the elementary diagram sheets show the major power components.

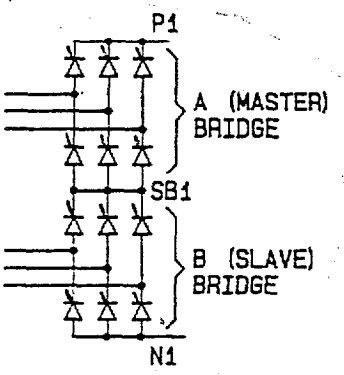
Thyristors (SCRs) are stacked horizontally, three in series, between four heat sinks. The AC buses are on the right side (consecutive stack pairs). The DC buses are on the left side (every other stack). The fuses (AC and DC filters) are located at the top of the component. The primary CTs are located behind the fuses on the bus work.

The blower compartment contains blowers, AC and DC filters, MOVs, secondary CTs and the terminal boards. Also, the hinged base in this compartment contains blower starters the 70V gating power supply, and thyristor fan-out PWB XTFH.

**3-4.7. 2KV MEDIUM VOLTAGE BRIDGE-SERIES TWELVE PULSE.** The arrangement of the power converter SCR stack assemblies is shown in Figure M2-2. This figure and the elementary diagram show the major power components.

SCRs are stacked horizontally; three in series, between four heat sinks. The top three rows are the master bridge, while the bottom three rows are the slave stacks. The AC lines come into the center of the rows respectively. The DC buses are on the outside edges of the rows. The fuses (AC and DC filters) are located at the top of the compartment. The primary CTs are located behind the fuses on the bus work.

The blower compartment contains the blowers, AC and DC filters, MOVs, secondary CTs, and terminal boards. The hinged base in this compartment contains the blower starters, the 70V gating power supply, the thyristor fan-out PWB XTFH, and bottom blower selection relay.




**Typical 12-Pulse Series Bridge**

### 3-4.8. GTO POWER CONVERSION SECTION.

The arrangement of the GTO power converter stack assemblies is as shown in figure G4-1. This figure and the elementary diagram show the major power components.

The GTO is mounted with its power diode in a heat sink assembly. The GTO may be mounted in series or in "center tap" arrangement depending on the GTO ratings and characteristics. Like the SCR's, it is recommended that the entire stack be replaced if a failure occurs and that the stack be sent back to the factory for repair.

Each GTO stack is mounted behind a GTO driven module which provides gating power, the gating signals (on, off) and status indication. Gating signals come in and status signals go out on fiber optic "pipes" to the XTFZ which converts these to signals the SEM uses.

Mounted to the side and behind the stack is the snubber network for the GTO diode, and the NTDA which contains a LED that goes out when the GTO shorts.

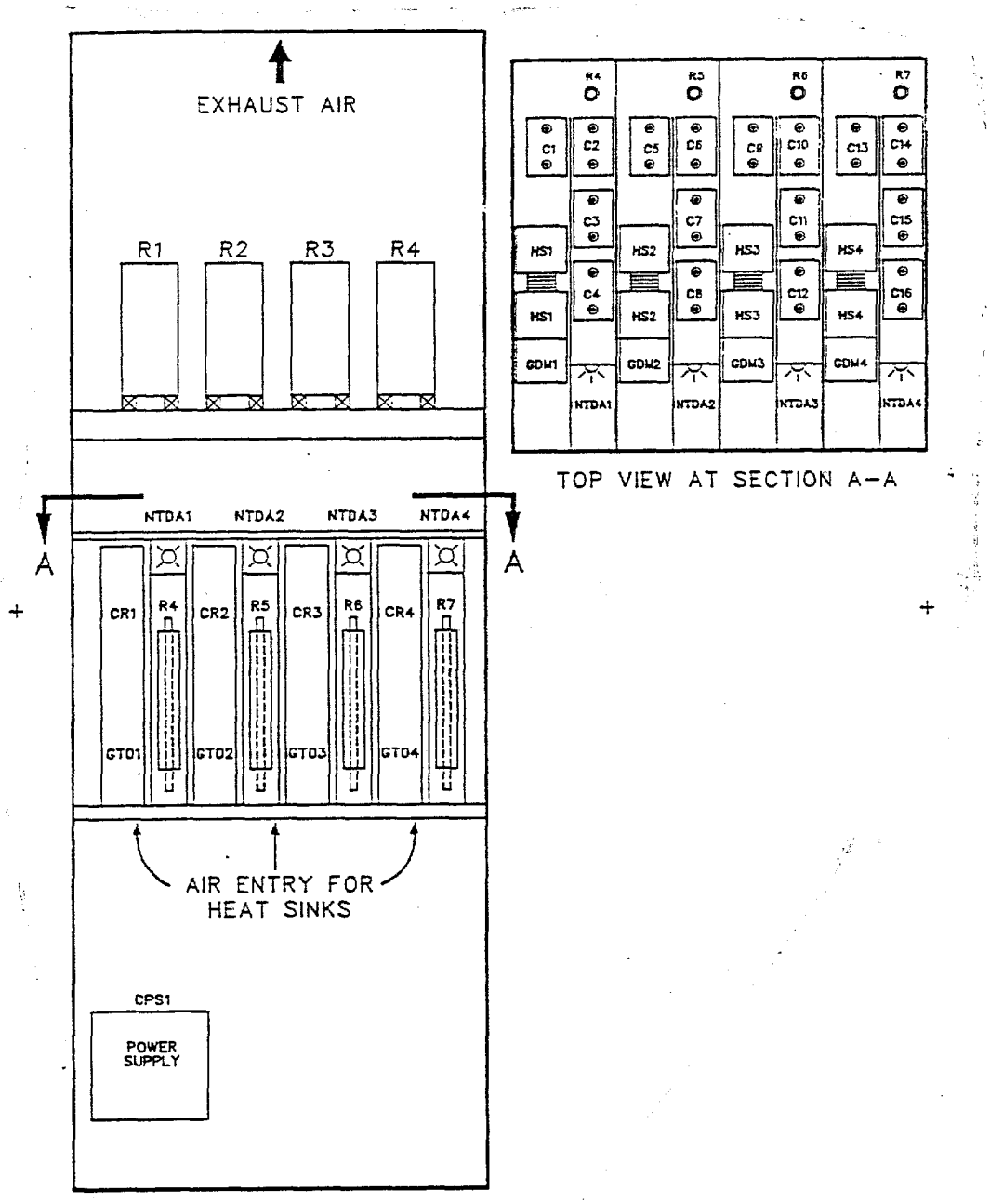



*Figure G4-1*

In the bottom of the compartment is located the GTO chopper power supply. It converts the 120 Vac, 60HZ into 120 Vac 400 HZ which is feed to the GT0 driven modules.

Mounted in the top of the compartment are some low ohm, high power resistors that are also used in the GTO's snubber network, also located above is the cooling blower.

**3-4.9. Flux Integrators - DS3800NLIB.** The NLIB contains various analog signal conditioning and protective functions. Two identical boards are used - one for the source side and one for the load. The following functions are provided:

1. Voltage integrators
2. Line/Motor current transformer receivers
3. Voltmeter driver
4. Overvoltage protection
5. Undervoltage sensing
6. Suppression
7. Variable reference comparator
8. Current meter driver

**3-4.10. Voltage integrators..** A signal proportional to each line-to-line voltage is obtained from the NSFC board. This signal is applied to an analog integrator to obtain a replica of motor flux according to the equation:

$$V = K * dPhi/dt$$
or
$$Phi = V * dt/K$$

Components of line currents are used to compensate for drops due to motor resistance and commutating impedance so that the flux represents the motor internal counter-emf or the so called voltage behind the commutating reactance. The analog sine wave corresponding to flux is then applied to a zero crossing comparator where the digitized output is then available to the HITA for phasing information. The analog flux signals are also read by the microprocessor based software for use in the control algorithms.

Line/motor current transformer receivers.

*Table 3-3. Analog/Digital Input Signals - Berg Jumpers and Testpoints*

(HAIA)

| Nomen. | Function | Position |
|---|---|---|
| | **Berg Jumpers** | |
| J1 | Selects either a fixed +7.5v input to channel 15 or a variable input thru pin12. The variable input is selected. | A |
| J2 | Selects either a fixed +7.5v input to channel 14 or a variable input thru pin12. The variable input is selected. | A |
| J3-J8 | Selects L bus address for board. | F |
| J9 | Selects filter for analog side of A/Dconverter. Filter is selected. | O |
| | **Testpoints** | |
| ACOM | Power supply analog common. | |
| FOG | Test input. Tie to common to latch internalcard data bus. | |
| AN | Analog signal at input to A/D conversion | |
| DB | Data buffer direction. High transmits to microprocessor and low receives. | |
| DV | "O" data valid signal | |
| BSY | "1" busy signal | |
| ES | "1" enable switch | |
| CLK | card clock - 1.152 mhz | |

Two signal conditioning differential receivers are included for line currents A and C. Line current B is calculated from A and C assuming a balanced load. Delta currents are also determined for use in the flux reconstruction.

**3-4.11. Voltmeter driver..** The magnitude of the line-to-line voltage is determined by a 3 phase absolute value network. The absolute value is then scaled by an operational amplifier for use as a line voltage signal.

**3-4.12. Overvoltage protection..** The absolute value of line/motor voltage obtained above is compared with an adjustable reference for a time determined by daughter board component, C37, then the overvoltage latch is set. The signal is wired to an I/O port for fault logging by the control software.

**3-4.13. Undervoltage sensing..** The absolute value of line voltage is compared with an adjustable reference. If the voltage is less than the reference for time delay set by daughter board component, C17, then signal "1LV" becomes true. This signal is used by the software system sequencing to determine low source volts condition. The function is not used on the load side.

**3-4.14. Suppression..** The absolute value sum of the line currents is taken and applied to a comparator. If the rectified current exceeds the reference threshold for 26ms, a fault latch is set. The time delay is included since the primary protection is in software which includes an on-line recovery. The function implemented here is strictly backup.

**3-4.15. Variable reference comparator..** "COMP" presently not used

**3-4.16. Current meter driver..** The absolute value of current is scaled and available from either source or load for meter driving. Scaling is 5v = 1pu/ (See Table 3-4)

## 3-5. GATING REFERENCE BOARD - DS3800NSFC

The NSFC's primary task is to convert information from the microprocessor based control to cell pair and timing for firing 6 thyristors configured into a 3 phase - double way bridge. In performing this task, the following functions are used:

1. R bus interface
2. Firing mask latch
3. Cell state sensors
4. Line-to-line voltage signal conditioners
5. Pulse train generation
6. Gate drivers
7. Attenuator buffers
8. D-C voltmeter driver

Quantity 4 NSFC's are used, assigned as follows:

1. Master source converter
2. Master load converter
3. Slave source converter
4. Slave load converter

R bus interface.Control logic is provided for decoding read and write strobes. Eight data lines (either upper or lower byte) are connected. All information exchange with host processor takes place through the R bus interface.

Firing mask latch.The particular cell pair to be fired is selected in software and communicated to the NSFC over the R bus. The mask information is temporarily stored in the primary firing mask latch. Also obtained from the software is time delay corresponding to the number of degrees the cell pair firing is to be delayed. This information is routed to a timer. At the end of the delay an interrupt is generated and on the load side the cell mask is clocked into the final latch. On the source side, the generated interrupt calls an "execute firing" program which (if all permissives are met) signals the NSFC to move the firing mask to the final latch.

**3-5.1. Gate drivers..** Six darlington transistors shift the gating information from logic level to analog levels sufficient to drive the remotely mounted gate pulse amplifiers. These drivers operate at 28 VDC levels.

Pulse train generation.The gate pulse reference driver signals consist of a "O" true pulse train nominal 120 degrees long during the conduction period. The reference pulse train is generated from a crystal clock and shaped for a duty cycle of 15 microseconds on and 30 microseconds off by a timer and control logic.

## 3-6. BUFFERS.

Attenuator buffers are provided to receive each attenuator string connected to phase voltages 1,2,3 and also the positive and negative bus. These buffered signals are then used for analog line-to-line voltage signals as well as cell state sensor inputs.

**3-6.1. Line-to-line voltage signal conditioners..** The buffered phase to neutral converter voltage signals are differentially combined to recreate line-to-line voltages. These are used after further processing (on NLIB) as information for the software control algorithms.

*Table 3-4. Flux Integrators - Jumpers, Adjustments, LEDs and Testpoints*

(NLIB)

| Nomen. | Function | Position |
|---|---|---|
| | **BERG JUMPERS** | |
| BJ1-BJ3 | Selects voltage integrator scaling for 60Hz or 120Hz | 60Hz or 120Hz (var) |
| | **Adjustments** | |
| FAB, FBC, FCA | Scale flux integrators which calculate flux. Scaling is 4.75 volts RMS corresponds to 1 pu. | |
| OV | Set overvoltage protection trip point. | |
| UV | Set undervoltage protection trip point. | |
| SUPP | Set trip point for overcurrent backup fault. | |
| COMP | Sets field loss level on source side and field overcurrent level of load side. | |
| VDC | Not used. | |
| | **LED's** | |
| OV | On when overvoltage fault latch set. | |
| SUPP | On when overcurrent fault latch set. | |
| IMOK | On if microprocessor based self test is passed and no saturation condition are present. | |
| | **Testpoints** | |
| OVR | Analog D-C voltage proportional to setpoint of over-voltage fault latch. | |
| UVR | Analog D-C voltage proportional to setpoint of source undervoltage sensor. | |
| SUPP | Analog D-C voltage proportional to setpoint of overcurrent fault latch. | |
| FAB,FAC FCA | Analog voltage proportional to motor/line flux reconstructed. See fig. 3-6 for typical wave forms. | |
| FBAR, FCBR, FACR | Analog voltage proportional to motor/line flux reconstructed for IX drop. See fig. 3-6 for typical wave form. | |
| OIA,OIB OIC | Analog waveform representative of inverted line current. See fig. 3-6 for typical waveforms. | |

**3-6.2. Cell state sensors.** . From a cursory check of the standard three phase double way bridge configuration, it can be seen that the voltage across each cell is a combination of one phase voltage and one D-C bus voltage. Since each of these voltages is available from an attenuator, the voltage across each cell or "on state" maybe uniquely determined. Circuitry is provided for reading the "state" of all cells at a particular instant and for making the information available to the microprocessor via the L bus.

**3-6.3. D-C Link volts.** . The positive and negative D-C link bus voltages are combined differentially to obtain a D-C link volts signal suitable for driving a remote analog meter with approximately 10 VDC maximum scale (See Table 3-5).

## 3-7. ANALOG LCI BOARD = DS3800NLCB

The NLCB provides various miscellaneous signal conditioning and test functions required for the structured LCI. These features include:

1. Speed feedback generation from motor voltage zero crossing signals with overspeed protection and indication

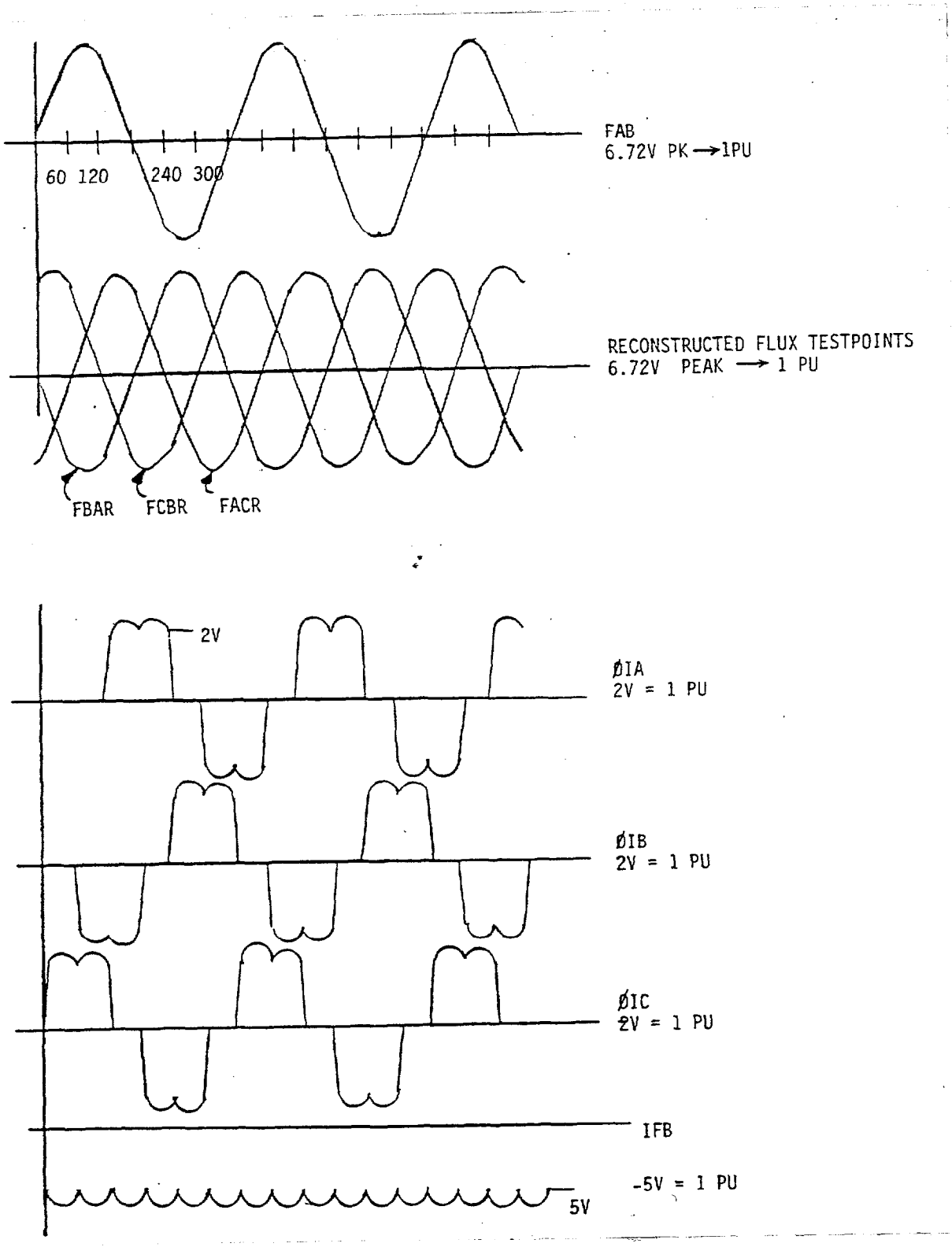



*Figure 3-6. Analog Typical Waveforms*

*Table 3-5. Gating Reference - Berg Jumpers, Adjustments, LED's and Testpoints (NSFC)*

| Nomen. | Function | Position |
|---|---|---|
| | **Berg jumpers** | |
| J1 | Determines if new mask should be clocked into final latch at termination of count down timer. | NINT(1d)<br>INT(src) |
| | **Adjustments** | |
| VDC | Sets remote meter full scale to corresponding D-C link volts. | |
| | **LED's** | |
| IMOK | Board hardware status indication. On normally. Off if self diagnostics determines a board hardware failure. | |
| | **Testpoints** | |
| SIG 1-SIG 6 | Gate pulse command. "O" true. Steady state length of 120 degrees relative to operating frequency. Each gate pulse is shifted 60 degrees relative to lower numbered ones. SIG numbers correspond to thyristor leg being gated. | |
| VDC | Output of D-C link voltage meter driver. | |
| VAC,VCB | Analog output proportional to motor/source line-to-line voltage. | |
| ACOM | Power supply analog common. | |

2. External reference signal conditioning
3. Test reference signal conditioning
4. Torque reference signal conditioning
5. Selftest features
6. 4-20 MA signal drivers
7. Ground fault detector
8. Power applied reset
9. Fault relay drivers
10. Horsepower meter driver
11. Continuity detection circuits
12. Test switch
13. Restart upon power loss

**3-7.1. Speed feedback:.** A pulse rate signal proportional to drive speed is obtained from the zero crossing detection circuitry on the HLCB. A pulse rate to analog converter then produces a corresponding D-C level proportional to speed suitable for driving a meter and protective functions.

The overspeed function is implemented by comparing an overspeed setting reference to the analog speed voltage attained above. A latch insures that transient conditions will be sealed. A red LED indicates the overspeed condition until reset.

A differential amplifier committed to accepting the remote speed reference is supplied. Berg jumpers are used to configure the circuit to accept a 0 to 10 volt signal or a 4 -20 MA signal. In either case, the output voltage is transmitted to channel of the HAIA analog to digital converter for reading by the system processor.

**3-7.2. Test speed reference..** A differential amplifier is included to condition a voltage source signal from a potentiometer located on the I/O module. This is also known as the local speed reference.

**3-7.3. Torque reference signal conditioning.** A differential amplifier for receiving the master torque reference from the other six pulse drive (in a 12 pulse system) is also supplied.

**3-7.4. Selftest features .** An array of static switches and voltage sources is included for use in testing the analog circuitry of this and other boards in the module during the module selftest mode. A sinewave oscillator is used to simulate line and motor voltage during "gate test mode". This is to allow the power converter electronics to be tested without the necessity of applying high voltage to the associated bridges.

**3-7.5. 4 - 20 MA signal drivers .** Two voltage to current converters provide capability of remote driving system information to customer interfaces. The signals selected for transmission are motor current and motor speed. In each case, 4 ma corresponds to 0.

**3-7.6. Ground fault detector .** The ground fault detector sums line-to-neutral voltages Va, Vb, and Vc. It operates on the principal that when all impedances from phase to neutral are balanced that the summation of the phase-to-neutral voltages will be zero. Any unbalanced impedance will cause an error to be detected and a ground fault noted by the system processor.

**3-7.7. Power applied reset .** A power applied reset is generated here and distributed throughout the electronics module. It is used to prevent false operations of relay drivers and fault latches during the process of powering up the module.

**3-7.8. Fault relay drivers .** Relay driver no. 1 is generally used as a method of tripping the source side drive switchgear from both hardware and software sensed fault conditions. It is inhibited from false operations during the power up phase by the power applied reset.

Relay driver no. 2 is generally used as a method of tripping the motor side isolator (if used) from both hardware and software sensed fault conditions. It is also inhibited from false operations during the power up phase.

**3-7.9. Horsepower meter driver .** An adjustable meter driver for powering a door mounted meter indicating horsepower is also included. The horsepower signal is calculated in software and converted to analog on the HLCB board.

**3-7.10. Continuity detection circuits .** Four circuits are included to monitor the continuity of various ribbon cables connected to the electronics module when it is integrated in to a system control panel. These signals are only monitored during the panel selftest. An appropriate diagnostic message is printed out if a cable is not in the correct location.

**3-7.11. Test switch .** The test switch on the board front edge is used to initiate the panel selftest mode. When the switch is operated, then the electronics module will begin to execute a selftest routine which extends to other components of the control panel. Upon completion of the selftest, the control will enter gate-test-mode using the three phase test oscillator as a source of synchronous waveforms.

**3-7.12. Restart upon power loss .** If the drive has been previously running and the control power is lost for a time (determined by setting of RTD potentiometer), then the drive will be automatically restarted when power returns by a pulse which appears at signal 1RSRT. Note this is an option and the I/O module must be properly configured. (See Table 3-7)

## 3-8. RELAY DRIVER BOARD = DS3800HRDA

The HRDA is relay driver board with 16 open-collector outputs. Output voltages can range from 5 to 50 volts. In the LCI system, it is applied at a nominal voltage of 28 volt D-C. The board may be connected to the computer R-bus for accessibility by the system control.

The relay driver board is used principally as an interface between the system software control and the power devices located in the I/O module. (See Table 3-8)

## 3-9. INPUT SIGNAL CONDITIONING = DS3800HISA

HISA is designed to work in conjunction with HRIA (in I/O module) to condition thirty high level (105VDC) signals to logic levels and allow access to them through the processor R bus. Each signal is separately isolated by a photocoupler and indicated by a bar LED. Two relay drivers are also included which are only utilized in the hardware selftest of the I/O module. (See Table 3-9)

## 3-10. DISPLAY MODULE (LDM)

The DS3820LDM is the display module for the structured LCI control panel. The display provides fault, alarm and diagnostic information. The display module comes in two basic layouts. Layout Option 1 provides the following features:

- 3 indicator lights (RUN, ALARM and FAULT)
- 2 push buttons (RESET AND DRIVE STATUS)
- 3 1/2 digit motor speed meter (RPM)
-24 column printer with real-time clock

and Layout Option 2 provides the following features:

- 4 push-to-test indicator lights (RUN, ALARM, FAULT and READY)
- 2 push buttons (RESET and DRIVE STATUS)
- 4 DB40 type meters (motor speed, motor current, motor voltage and horsepower)

All lights and push buttons are on a 28v circuit. The functions and color codes for each light are:

Red for the motor RUN light

Red for a system FAULT light

Amber for a system ALARM light

Green for the system READY light

*Table 3-7. Analog LCI Board - Berg Jumpers, Adjustments, Testpoints and LEDs*
(NLCB)

| Nomen. | Function | Position |
|---|---|---|
| | **BERG JUMPERS** | |
| LOAD | Selects remote speed reference for either current input or voltage input. Should be coordinated with GAIN. | variable |
| GAIN | Selects gain necessary for remote speed when used as current input or voltage input. | variable |
| | **Adjustments** | |
| SPD | Adjusts analog speed feedback circuit for 10 voltsat 60 Hz. | |
| SPM | Adjusts door mounted speed meter for proper fullscale reading | |
| OSP | Hardware overspeed reference. | |
| RTD | Adjusts allowable time for auto restart after loss of control power. | |
| | **Testpoints** | |
| SPD | Output of analog speed feedback calculation | |
| OSP | Reference to hardware overspeed detection circuit. Scaled same as speed i.e. 10 volts = 60Hz. | |
| IOA | Output of channel A current source. | |
| IOB | Output of channel B current source. | |
| REF | Output of remote speed reference. With a voltage input will be 0 to 10 volts. With a current source input, will be 2 to 10 volts. | |
| TREF | Output of local test reference. Varies from 0 to 8.3 volts. | |
| 12P | Output of torque reference obtained from other channel. | |
| NTL | Neutral. Summation of line to neutral voltages.Normally near 0 volts. | |
| RTD | Run time delay. Sets allowable time for loss of control power while still allowing an auto restart. Scaled 0 to 10 volts. | |
| PHA | Test oscillator, phase A. Shows a 55Hz. sine wave. | |
| PHB | Test oscillator, phase B. Shows a 55Hz. sine wave,lagging phase "A" by 120 degrees. | |
| | **LED's** | |
| OSP | Red LED indicating that the hardware overspeed latch is set. | |
| RSRT | Green LED which is pulsed when the auto restart circuit is activated. | |
| FT1 | Green LED indicating the status of fault relay driver No. 1. Will be on when the relay driver is energized. | |
| FT2 | Green LED indicating the status of fault relay driver No. 2. Will be on when the relay driver is energized. | |
| TEST | Green LED which luminous when the test switches closed and the system software indicates gate test mode. | |
| IMOK | Yellow indicating light will be on normally. An off condition indicates a possible hardware problem with the board. It can be turned off either locally or from the processor as a function of selftest. | |

*Table 3-8. Relay Driver - Testpoints and LED's*

(HRDA)

| Nomen. | Function |
|---|---|
| | **BERG JUMPER** |
| None supplied | |
| | **ADJUSTMENTS** |
| None supplied | |
| | **TESTPOINT** |
| TP1 | Reset. Connecting testpoint 1 to testpoint 2 resets only this board. This function isn't useful in the LCI system. |
| TP2 | DCOM |
| | **LED's** |
| O - F | A red LED for each of the 16 relay drivers indicates the energized status of each circuit. |
| ADR | Yellow LED. Turns on when the board is addressed; stays on if the board is addressed at least every 200 microseconds. |
| WPOK | Parity check. Not used in this system will always be on. |

The push buttons are connected to a 28v circuit and direct a 28v signal to the output when the button is depressed. The RESET push button is used to reset latched faults in either software or hardware (NLIB and NLCB latched faults - overspeed, suppression, overvoltage, etc.). The DRIVE STATUS push button; when depressed, provides information on the motor voltage, current, speed, etc. via the printer.

The digital meter is a DC voltage measurement panel type. The conversion factor for the meter is 2 volts correspond to 1999 cnts. The panel meter is powered by the 5 volt bus via the 26-pin ribbon cable.

The four DB40 type analog meters are 10 volt movement meters with ± 10% adjustment rheostats (back).

The printer included in the display module is a Digitec Model 6610 Panel mount printer. The 24 column printer inputs data and control signals via a serial RS232 connection and uses an impact print head mechanism to output the data. The printer also supplies a real-time clock that will be used to log the time of status information and system failures.

The LDM interfaces to the LCG thru a 26 pin ribbon cable JK.

## 3-11. SIGNAL ISOLATION MODULE (SIM)

The DS3820SIM is used to provide optional functions to an LCI control panel. In particular, it provides four impedance isolated signal outputs for use in a purchaser's external control system. The four signals may be selected as 0 to 10 volts or 4 to 20ma outputs by header connection jumpers at JM. Each circuit may be selected independantly. Inputs to the module (TEM) are received from the LCI control SEM via 26-pin ribbon cable (JM). The four inputs to the isolation circuits may be chosen by header connector jumpers at JB from the following list of signals: Source volts, source current, SPD ref., motor SPD, horsepower, DA1, DA2, exciter current, motor volts.

A DS3800NVAA provides the functions of voltage to current source transformation and isolation. A differential amplifier accepts the signal from the remote control SEM. The signal is isolated from the NVAA on-board ground by approximately 100 K ohms of resistance. The isolated signal is then applied as a reference to a current source with the gains weighted such that a 0 to 10 volt signal provides a delta output of 16 milliamperes. An additonal bias is provided ot give the 4 milliamperes at 0 volts input. A chopper isolated power supply is present on the NVAA board to power the output stages of the signal isolators. The

output of each of the four circuits is referenced to XCOM, the on-board power supply common. It is intended that the XCOM connection be tied to the customer's external circuit common. While it is not necessary for the signals used as inputs to the isolators be referenced to the customer's common or XCOM, the difference between ACOM and XCOM must not exceed 3 volts steady-state or transiently. The voltage developed across any external impedance in series with a current source must a not exceed 10 volts.

The output signals are also available as voltage followers (0 to 10v). Maximum output load is 1 ma.

An operational amplifier monitor (or saturation detector) is provided for monitoring the output stages of the current driver circuits an LED, "IMOK", is luminous as long as no fault condition exists. In additon, if FUSE 1 in the on-board power supply is blown, then the LED will be non-luminous. (See Table 4-0)

## 3-12. I/O MODULE (IOM)

The purpose fo the DS3820IOMA (I/O) module is to provide signal conditioning and connection for interpanel and purchaser wiring in a load commutated inverter (LCI) control panel. The following functions are provided.

1. Signal conditioning and isolation for interface wiring at the 105vdc level.
2. Isolated signal interlocks (form C) to remote locations as a function of programmed system requirements. Rated .25A, 115vac (Resistive).
3. Fused 105vdc power supply to provide a source of high level power for interface.
4. Power relay contro for interface with a motor-side isolator as well as interface with a line-side switchgear device. Rated 10A, 600vdc (Inductive).
5. Local operator's station for running the drive during maintenance modes. (Channel selector, local/remote, start/stop, local speed reference).
6. Termianl boards for terminating interconnecting wiring.
7. Diagnostic functions.

The IOMA module is designed to be intergrated into a system diagnostic test function residing in the system electronics module. Therefore, no local diagnostic messages are displayed. However, the necessary circuits aand feedbacks are provided to allow the system function resident in the electronics module to isolate failed components to the replaceable board level.

*Table 3-9. Input Signal Conditioning - Berg Jumpers, Adjustments, Testpoints and LED's*

(HISA)

| Nomen. | Function |
|---|---|
| | **BERG JUMPERS** |
| None Supplied | |
| | **ADJUSTMENTS** |
| None supplied | |
| **TESTPOINTS** | |
| None Supplied | |
| | **LED's** |
| I0O - I1D | Green bar LED's. Each indicates the status of the associated input, luminous when the input is energized. |
| IMOK | Yellow LED. On during normal operation. will be off only as the result of module selftest. |

*Table 4-0. Signal Isolation Module (SIM)*

| Nomen. | Function |
|---|---|
| | **TESTPOINTS** |
| TP1 | Voltage output of channel 1 relative to XCOM. |
| TP2 | Voltage output of channel 2 relative to XCOM. |
| TP3 | -2.5 VDC bias for V-I circuits. |
| TP4 | Voltage output of circuit 3 relative to XCOM. |
| TP5 | -2.5 VDC bias voltage for I-V circuit. |
| TP6 | XCOM, isolated supply ground. |
| TP7 | Voltage output of channel 4 relative to XCOM. |
| TP8 | Voltage output of 4-20ma to voltage converter. |
| TP9 | N15X, isolated supply. |
| TP10 | P15X, isolated supply. |
| TP11 | ACOM, analog power supply ground. |

# CHAPTER 4

# TUNE UP CONSTANT SETTINGS AND PUBLIC VARIABLES

## 4-1. TUNE UP CONSTANTS

EEPROM Memory is a non-volatile memory space used to hold critical drive constants that may be system dependent. The use of EEPROM Memory allows the drive constants to be changed in the field in order to fine tune the system performance. EEPROM Memory space resides on the DMPB card (HMPG daughter board) and is configured to contain 1024 bytes (1K x 8).

Default values for the drive constants are burned into the ROMs residing on the HMPG (the ROMs also contain the executable software to run the drive). These default values are intended to provide a reasonable starting base for tuning the drive. They may be loaded into the EEPROM Memory by turning the DMPB write enable switch "on", jumpering testpoint TINO on the HLCB pwb to DCOM and then executing a hardware SEM reset (Touch HMPG TP "reset" to DCOM). In order to change a value in the EEPROM memory, turn the DMPB write enable switch "on" and use the monitor commands: (MODIFY and SAVE).

After power up or hardware reset, the microprocessor on the HMPG makes a local copy (shadow) in RAM of the EEPROM Memory. During normal running, the microprocessor accesses the drive constants from this local RAM shadow.

Table 4-1. Description of Available Tune Up Settings

CRFIX1

CRFIX2

Units - pu dc link voltage command

Resolution - 0.01

Range - 0 to 0.5

Definition - Variables controlling the first rwo firings of the current regulator out of inversion limit when the drive is in source side forced commutation mode. The first two current regulator calculations are essentially open loop. Typical settings are 0.125 pu for the first firing and 0.095 pu for the second firing.

CRFIXQ = pu voltage command on first firing

CRFIX2 = pu voltage command on second firing

The related software variable definition is

IREG FIC ALPGA1,2 E = 16384 * CRFIC1,2

---

C R G A I N

Units - gain
Resolution - 1.0
Range - 10 to 1000

Definition - Current regulator gain. The relation to system parameters is:

CRGAIN = (682.24*Im*Wc*L)/Vll

Im = rms rated inverter current
Wc = desired crossover (200 rad)
L = commutating inductance
= reactor + 2*(trans comm reactance)
Vll = rated rms line-line source volts
(*2 V11 for series 12-pulse)

The related software variable definition is

IREG G2 E = 16*CRGAIN/(12*DRSFRQ)

---

C R L I M P

Units - pu impedance
Resolution - 0.01
Range - 0 to 0.25 pu

Definition - Variable controlling the source side retard limit as a function of current based on source impedance.

CRLIMP = per unit source commutating impedance

The related software variable definition is

LINE IMP E = 63.75 * CRLIMP

---

C R M A R G

Units - degrees
Resolution - 1.0
Range - 0 to 30

Definition - Variable controlling the fixed source side margin when inverting. Normal margin is 20 degrees.

CRMARG = desired fixed margin in degrees

The related software variable definition is

RTARD MARG SRC E = 255[1+COS(180-CRMARG)]

---

C R M A X

Units - pu dc link current
Resolution - 0.01
Range - 0.25 to 1.50

Definition - Variable controlling the maximum clamp level for dc link current command. Usually picked on inverter rating and drive loading requirements. Normal setting is 1.10 pu.

CRMAX = max pu dc link current

The related software variable definition is

ISTAR CLMP E = 1024 * CRMAX

---

C R M A X L C

Units - pu dc link current
Resolution - 0.01
Range - 0.25 to 1.50

Definition - Variable controlling the maximum clamp level for dc link current command when the drive is in the dc link commutation mode. Usually picked on dc link commutator rating and drive loading requirements. Normal setting is 1.6 pu.

CRMAXLC = max pu dc link current

The related software variable definition is

ISTAR CLMP1 E = 1024 * CRMAX

---

C R M I N

Units - pu dc link current
Resolution - 0.01
Range - 0.01 to 0.5

Definition - Variable controlling the minimum clamp level for dc link current command. Usually set to maintain continuous current. The link reactor is usually picked to maintain continuous current at 0.12 pu.

CRMIN = min pu dc link current

The related software variable definition is

CURLL SRC E = 1024 * CRMIN

---

C R T C
Units - seconds
Resolution - 0.001
Range - 0.001 to 0.5

Definition - Time constant used to stabilize the current regulator. Typical value is 0.03 seconds.

CRTC = time constant (seconds)

The related software variable definition is

IREG G1 E = 4 * CRGAIN * CRTC

---

C R V D C
Units - gain
Resolution - 0.01
Range - 0.8 to 1.2

Definition - Gain applied to the feedforward section of the current regulator. It is intended to compensate for different rated source line and motor voltages.

CRVDC = Vllm / Vlls

Vllm = rated rms line-to-line motor voltage

Vlls = rated rms line-to-line source voltage

(*2 V11 for series 12-pulse)

The related software variable definition is

VDC GAIN E = 24 * CRVDC

---

C R V D C L C
Units - gain
Resolution - 0.01
Range - 0.8 to 1.2

Definition - Gain applied to the feedforward section of the current regulator when the drive is in the dc link commutation mode. It is intended to compensate for different rated source line and motor voltages and for reduced link voltage due to link commutator "on" time.

CRVDCLC = 0.8 * Vllm / Vlls

Vllm = rated rms line-to-line motor voltage

Vlls = rated rms line-to-line source voltage

(* 2 V11 for series 12-pulse)

The related software variable definition is

DCLC VDC GAIN E = 24 * CRVDCLC

---

---

C T A D V L

| | |
|---|---|
| Units | - pu source Vdc |
| Resolution | - 0.01 |
| Range | - 0.5 to 1.00 |

Definition - Maximum output clamp on the current regulator output, source dc link volts command. Normal setting is 1 pu.

CTADVL = desired max pu dc link volts command

The related software variable definition is

ADVANC LINE E = 255 * CTADVL

---

C T G A I N

| | |
|---|---|
| Units | - gain |
| Resolution | - 1.0 |
| Range | - 5 to 100 |

Definition - Cross-tie regulator gain. The relation to system parameters is:

CTGAIN = 4 * Wc

Wc = desired crossover (17 rad)

The related software variable definition is

CRSST INT GAIN E = 64*CTGAIN/(2*DRFRQS)

---

C T L A N G L

| | |
|---|---|
| Units | - degrees |
| Resolution | - 1.0 |
| Range | - 90 to 270 |

Definition - Defines the retard threshold of the inverter firing angle at which point torque droop is activated. This function is only active when cross-tie is also active.

CTLARNGL = (inverter angle at rated) + 10 deg

where the angle (measured from alona=0) is the calculated inverter firing angle at rated load and speed.

The related software variable definition is

LD ANG CRSST CLMP E = (CTANGL-180) * 1024/360

---

C T M A R G
Units - pu source Vdc
Resolution - 0.01
Range - 0.0 to 0.5

Definition - Defines the minimum margin allowed between the actual current regulator output and the absolute maximum clamp on current regulator output, CTADVL. If current regulator output gets closer to the max clamp than is allowed by the margin CTMARG, then the crosstie becomes active to maintain the margin by reducing motor flux. Normal margin is 0.06 pu.

CTMARG = desired pu margin

The related software variable definition is

CROSS TIE MARG E = 255 * CTMARG

---

C T M A X
Units - pu flux
Resolution - 0.01
Range - 0.0 to 0.5

Definition - Maximum clamp on the output of the crosstie regulator. It defines the maximum reduction that can be applied to the flux command by the crosstie regulator. Normal setting is 0.4 pu.

CTMAX = max pu cross-tie flux reduction

The related software variable definition is

CRSST MAX LMT E = 8192 * CTMAX

---

C T T C
Units - seconds
Resolution - 0.001
Range - 0.001 to 0.5

Definition - Time constant used to stablilize the cross-tie regulator. Normal setting is 0.03 seconds

CTTC = time constant (seconds)

The related software variable definition is

CRSST PRP GAIN E = 8 * CTGAIN * CTTC

---

---

D R H P P U

| | |
|---|---|
| Units | - horsepower |
| Resolution | - 1.0 |
| Range | - not clamped |

Definition - Defines the drive's 1 pu horsepower rating.

DRHPPU = rated drive horsepower

There is no related software variable.

---

D R I P U

| | |
|---|---|
| Units | - amps |
| Resolution | - 1.0 |
| Range | - not clamped |

Definition - Defines the drive's 1 pu rms current rating.

DRIPU = rated drive rms current

There is no related software variable.

---

D R M C

| | |
|---|---|
| Units | - pu capacitance |
| Resolution | - 0.01 |
| Range | - .5 to 3.0 |

Definition - Defines the size of the motor filter capacitors.

DRMC = pu motor filter capacitance

(per phase on motor base)

The related software variable definition is

MOT CAP E = 35.96 / (DRZRAT * DRMC)

---

---

D R M F R Q

| | |
|---|---|
| Units | - hertz |
| Resolution | - 1.0 |
| Range | - 45 to 60 |

Definition - Defines the motor base frequency. Normally set to 60 hz.

DRMFRQ = motor base frequency (hz)

There is no related software variable.

---

D R M R P M

| | |
|---|---|
| Units | - rpm |
| Resolution | - 1.0 |
| Range | - not clamped |

Definition - Defines synchronous speed at 60 hz.

DRMRPM = motor 60 hz synchronous speed (rpm)

There is no related software variable.

---

D R M V P U

| | |
|---|---|
| Units | - volts |
| Resolution | - 1.0 |
| Range | - not clamped |

Definition - Defines rated 1 pu rms line-to-line motor voltage.

DRMVPU = 1pu rms motor voltage

There is no related software variable.

---

D R S F R Q

| | |
|---|---|
| Units | - hertz |
| Resolution | - 1.0 |
| Range | - 50 to 60 |

Definition - Defines the source line frequency.

DRSFRQ = line frequency (hz)

There is no related software variable.

---

D R S V P U
Units - volts
Resolution - 1.0
Range - not clamped

Definition - Defines rated 1 pu rms source line to line voltage.

DRSVPU = 1 pu rms line voltage

There is no related software variable.

---

D R V C F G
Units - bit coded
Resolution - n/a
Range - not clamped

Definition - Control configurations and optional features are selected with this bit coded variable. The related software variables are assigned based on the bit patterns in DRVCFG:

| Bit | | Related Software Variable |
|---|---|---|
| 0 | ___ | SRC CONFIG E |
| 1 | | |
| 2 | | |
| 3 | ___ | DCLC SERIES CONFIG P |
| 4 | | |
| 5 | ___ | DCLC SHUNT CONFIG P |
| 6 | | |
| 7 | | |
| 8 | ___ | SERIES SHUNT E |
| 9 | ___ | ICAP FIL ENABLE P |
| A | ___ | FLX ERROR EN P |
| B | ___ | TORQ VS SPD EN P |
| C | ___ | START OPTION E |
| D | ___ | unused |
| E | ___ | unused |
| F | ___ | unused |

The related software variable bit pattern definitions are:

SRC CONFIG E: selects the source bridge configuration

00 - 6 pulse channel A
01 - 6 pulse channel B (factory test only)
10 - 12 pulse channel B lags
11 - 12 pulse channel B leads

DCLC SERIES CONFIG P: selects the number of series GTOs present in the series

000 - none
001 - 1 cell
010 - 2 series cells
011 - 3 series cells
100 - 4 series cells
101 - invalid
110 - invalid
111 - invalid

SERIES SHUNT E: CONFIGURES THE CONTROL FOR EITHER SHUNT OR SERIES/SHUNT POWER CONFIGURATION.
0 - shunt only
1 - series/shunt combination

ICAP FIL ENABLE P: low pass filter on calculated capacitor current (normally ENABLED)
0 - disable
1 - enable

FLX ERR FIL EN P: low pass filter on the flux error signal (normally DISABLED)
0 - disable
1 - enable

TORQ VS SPD EN P: function to limit torque command as a function of speed
0 - disable
1 - enable

START OPTION E: selects startup options
00 - disable BRAKE function
01 - enable BRAKE function

---

D R V T S T

| | |
|---|---|
| Units | - n/a |
| Resolution | - n/a |
| Range | - not clamped |

Definition - Control test modes are selected with this variable. The related software variables are assigned based on the value of DRVTST.

| Value | Related test variable set true |
|---|---|
| 0 | none |
| 1 | IO TEST MODE E - background is disabled, i/o can be exercised from monitor |
| 2 | CROWBAR TEST MODE E - diametrics are gated on the inverter while current is regulated at value specified by SUMTCH |

| | |
|---|---|
| 3 | DCLC TEST E - load side diametrics are gated using the dc link commutatorl Link commutator on and off times are controlled by TEST$ON$P and TEST$OFF$P. Current is regulated at value specified by CRMIN. |
| 4 | MANUAL FIX FIRE E - inverter is gated at frequency specified by SUMFRQ while current is regulated at value specified by SUMTCH |

---

D R Z R A T

| | |
|---|---|
| Units | - ratio |
| Resolution | - 0.01 |
| Range | - 0.5 to 1.5 |

Definition - Defines the ratio between the motor base impedance and the inverter base impedance.

DRZRAT = Zbmot / Zbinv

Zbmot = motor base impedance

Zbinv = inverter base impedance

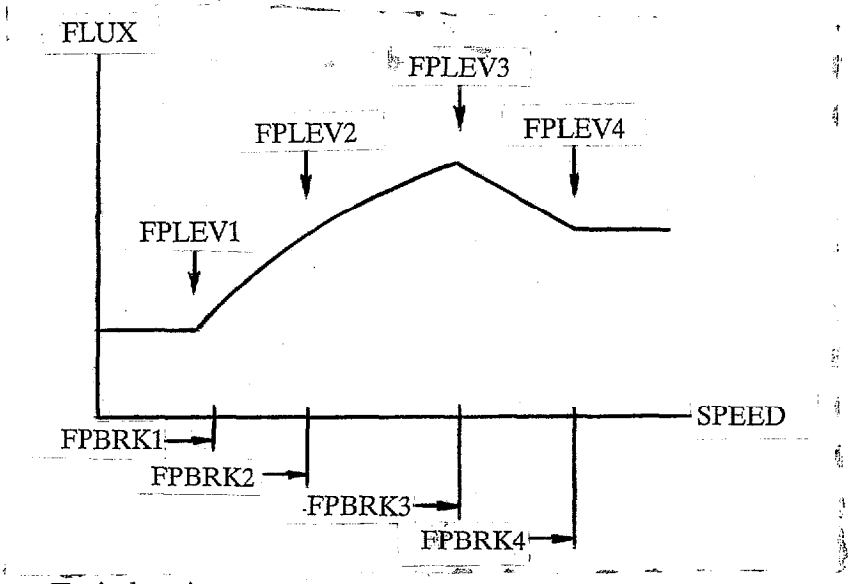
There is no related software variable.

---

F P B R K 1, 2,
3, 4
Units - pu motor speed
Resolution - 0.01
Range - 0.0 to 1.5

Definition - Group of variables used in the "Flux Setpoint vs Speed" function that define the speed breakpoints.

FPBRK? = desired pu speed




Typical settings are:

FLX BRK 1 = 0.20
FLX BRK 2 = 0.40
FLX BRK 3 = 0.85
FLX BRK 4 = 1.10

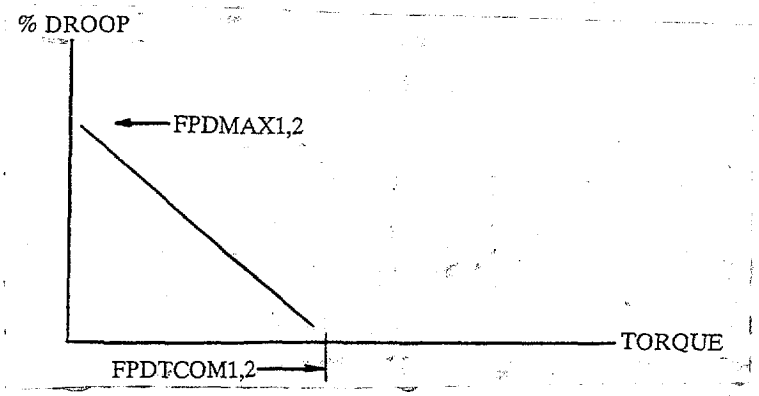
The related software variable definition is
FLX BRK? E = 683 * FPBRK? * (DRFRQM/60)

F P D M A X 1
F P D M A X 2
Units - percent
Resolution - 0.01
Range - 0.0 to 0.85

Definition - Used to program the "Flux Droop" profiles. They define the percentage that flux command will be reduced (drooped) when the drive is at no-load. Two different flux droop profiles can be programmed, typically one for low speed and one for high speed. Typical maximum low speed droop is 0.35 pu and maximum high speed droop is 0.45 pu.

FPDMAX1 = lo speed max flux droop (%)

FPDMAX2 = hi speed max flux droop (%)




The related software variable definition is

FLX DROOP MAX E (1) = 512 * FPDMAX1
FLX DROOP MAX E (2) = 512 * FPDMAX2

F P D S P D 1
F P D S P D 2
Units - pu speed
Resolution - 0.01
Range - 0.0 to 1.5

Definition - Used to program the "Flux Droop" profiles (refer to FPDMAX). Defines the motor speed thresholds for sctivating the different flux droop profiles. Below FPDSPD1, flux droop is not active. Between FPDSPD1 and FPDSPD2, flux droop profile #1 is active. Above FPDSPD2, flux droop profile #2 is active. Typical settings are 0.3 and 0.72 pu.

FPDSPD1 = pu speed to activate flux droop profile #1
FPDSPD2 = pu speed to activate flux droop profile #2

The related software variable definition is

FLX DROOP SPD1 T = 683 FPDSPD1 * (DRMFRQ/60)
FLX DROOP SPD2 T = 683 FPDSPD2 * (DRMFRQ/60)

F P D T C O M 1
F P D T C O M 2

| | |
|---|---|
| Units | - pu torque |
| Resolution | - 0.01 |
| Range | - 0.0 to 0.6 |

Definition - Used to program the "Flux Droop" profiles (refer to FPDMAX). They define the light load torque thresholds below which the flux droop functions become active. Two different flux droop profiles can be programmed, typically one for low speed and one for high speed. Typical low speed light load torque threshold is 0.2 pu and high speed light load torque threshold is 0.45 pu.

FPDTCOM1 = lo speed light load pu torque threshold
FPDTCOM2 = hi speed light load pu torque threshold

The related software variable definition is

FLX DROOP TCOM E (1) = 1024 *FPDTCOM1
FLX DROOP TCOM E (2) = 1024 *FPDTCOM2

---

F P L E V 1, 2,
3, 4

| | |
|---|---|
| Units | - pu flux |
| Resolution | - 0.01 |
| Range | - 0.5 to 1.45 |

Definition - Group of variables used in the "Flux Setpoint vs Speed" function that define the flux setpoint levels. See FPBRK?.

FPLEV? = desired pu flux

Typical settings are:

FLX LEV 1 = 0.50
FLX LEV 2 = 0.80
FLX LEV 3 = 1.15
FLX LEV 3 = 0.90

The related software variable definition is

FLX LEV? E = 1024 * (FPLEV-FPX1/2)

---

F P X 1
Units - pu impedance
Resolution - 0.01
Range - 0.0 to 0.30

Definition - Defines motor parameter X1 per unit. It is used to compensate for the voltage drop across X1 in the motor volts calculation and in the flux profile levels.

FPX1 = motor parameter X1 pu

The related software variable definition is

XM1 E = 128 * FPX1

---

F R F E T C
Units - seconds
Resolution - 0.001
Range - 0.018 to 0.26

Definition - Time constant of the low pass filter applied to flux error. The filter can be disabled through DRVCFG. Typically set for 0.018 seconds and disable filter using DRVCFG.

FRFETC = filter time constant (sec)

The related software variable definition is

FLX ERR FIL GAIN E = 16 / (FRFETC*DRSFRQ)

---

F R G A I N
Units - gain
Resolution - 0.1
Range - 2.0 to 100.0

Definition - Gain of the flux regulator. The relation to system parameters is:

FRGAIN = Wc

Wc = desired crossover (10 radians)

The related software variable definition is

FLX INT GAIN E = 256 * FRGAIN / (2*DRSFRQ)

---

F R I C T C

| | |
|---|---|
| Units | - seconds |
| Resolution | - 0.001 |
| Range | - 0.018 to 0.26 |

Definition - Time constant of the low pass filter applied to calculated capacitor current. The filter can be disabled through DRVCFG. Typical setting is for 0.083 seconds and enable filter using DRVCFG.

FRICTC = filter time constant (sec)

The related software variable definition is

ICAP FLTR GAIN E = 16 / (FRICTC*DRSFRQ)

---

F R L L I

| | |
|---|---|
| Units | - pu dc link current |
| Resolution | - 0.01 |
| Range | - 0.0 to 0.50 |

Definition - Defines the threshold level for the test to determine light load operating condition based on level of commanded dc link current. Below this level light load operation is flagged. Typical setting is 0.17 pu.

FRLLIH = desired light load threshold
(pu dc link current)

The related software variable definition is

FLX LL IDC E = 1024 * FRLLIH

---

F R L L I H

| | |
|---|---|
| Units | - pu dc link current |
| Resolution | - 0.01 |
| Range | - 0.0 to 0.2 |

Definition - Defines the hysteresis for the test to determine light load operating condition based on level of commanded dc link current. The hysteresis is centered about the threshold level defined by FRLLI. Typical setting is 0.05 pu.

FRLLIH = desired hysteresis
(pu dc link current)

The related software variable definition is

FLX LL IDC HYST E = 1024 * FRLLIH

---

F R L L S
Units - hertz
Resolution - 1.0
Range - 15 to 90

Definition - Defines the inverter fundamental frequency above which light load operation is exited regardless of level of commanded dc link current. This is to prevent overvoltage conditions on the motor. Typical settings are for approximately 12Hz above the self-excitation frequency:

FRLLS = 12 + 1/(2*pi*sqr(Lm*Cm))

Lm = per phase motor magnetizing inductance (henries)
Cm = per phase motor filter capacitance (farads)

The related software variable definition is

FLX LL SPD E = 683 * FRLLS / 60

---

F R T C
Units - seconds
Resolution - 0.001
Range - 0.2 to 2.0

Definition - Time constant used to stablilize the flux regulator. It is calculated to cancel one of the poles in the motor transfer function.

FRTC = (Lm + L2) / R2

Lm = magnetizing inductance
L2 = referred rotor leakage inductance
R2 = referred effective rotor resistance

The related software variable definition is

FLX PRP GAIN E = 4 * FRTC * FRGAIN

---

F R X M
Units - pu magnetizing inductance
Resolution - 0.01
Range - 0.5 to 6.0

Definition - Defines the pu motor magnetizing inductance.

FRXM = Xmpu

Xmpu = pu motor magnetizing inductance

The related software variable definition is

XM GAP E = 4 * FRXM * DRZRAT

---

---

I F A D V L

Units
- degrees
Resolution
- 1.0
Range
- 0 to 90

Definition - Defines the advance limit clamp for the inverter firing angle (measured from alpha = 0). Typical setting is 15 degrees.

IFADVL = desired maximum advance angle

The related software variable definition is:

LD ANG MAX ADV E = (IFADVL-180) * 1024/360

---

I F I N I T

| | |
|---|---|
| Units | - degrees |
| Resolution | - 1.0 |
| Range | -18 0 to 270 |

Definition - Variable sets the angle of the first inverter firing in runfire. Typical setting is 240 degrees.

IFINIT = desired initial firing angle

The related software variable definition is:

LD INIT ANG E = (IFINIT-180) * 1024/360

---

---

I F L C A N G
Units - degrees
Resolution - 1.0
Range - 90 to 180

Definition - Sets the inverter firing angle (measured from alpha=0) threshold for using the dc link commutator to accomplish inverter commutation. The link commutator will be used at firing angles greater than this threshold. Typical setting is 160 degrees.

IFLCANG = desired angle (degrees)

The related software variable definition is

LD DIVERT ANG E = (IFLCANG-180) * 1024/360

---

I F L C A N G H
Units - degrees
Resolution - 1.0
Range - 0 to 30

Definition - Sets the amount of hysteresis centered around the threshold defined by IFLCANG. Typical setting is 10 degrees.

IFLCANGH = desired hysteresis (degrees)

The related software variable definition is

LD DIVERT ANG HYST E = IFLCANGH * 1024/360

---

I F L C S P D
Units - hertz
Resolution - 1.0
Range - 0 to 50

Definition - Defines the inverter frequency below which the drive is always using the dclc to commutate the inverter regardless of commanded inverter firing angle.

IFLCSPD = desored frequency (hertz)

The related software variable definition is

LD DIVERT SPD E = 683 * IFLCSPD / 60

---

---

I F R E T L

| | |
|---|---|
| Units | - degrees |
| Resolution | - 1.0 |
| Range | - 180 to 270 |

Definition - Defines the retard limit clamp for the inverter firing angle (measured from alpha = 0). Typical setting in shunt only configuration is 216 degrees. Typical setting is series/shunt configuration is 270 degrees.

IFRETL = desired maximum retard angle (degrees)

The related software variable definition is

LD ANG MAX RET E = (IFRETL-180) * 1024/360

---

I F R E T L A

| | |
|---|---|
| Units | - degrees |
| Resolution | - 1.0 |
| Range | - 180 to 270 |

Definition - Defines the retard limit clamp for the inverter firing angle (measured from alpha = 0) when the control is notching to eliminate the 5th and 7th harmonics. Typical setting in shunt only configuration is 216 degrees. Typical setting is series/shunt configuration is 270 degrees.

IFRETLA = desired maximum retard angle (degrees)

The related software variable definition is

LD ANG MAX RETA E = (IFRETL-180) * 1024/360

---

I F S P I L G

| | |
|---|---|
| Units | - gain |
| Resolution | - 0.1 |
| Range | - 0 to 2 |

Definition - The gain used to control the spillover of the inverter firing angle into link commutator "on" time when the inverter firing runs into the retard limit clamp. A gain of 1 would provide one degree of additional link commutator "on" time for every degree of spillover. The gain is typically set to 1.

IFSPILG = desired gain

The related software variable definition is

ANG SPILL GAIN E = 16 * IFSPILG

---

---

I F S R L
Units - degrees per firing
Resolution - 1.0
Range - 1 to 30

Definition - Defines the rate limit applied to the inverter firing angle command when the firing angle quadrant changes. Typical setting is 5.

IFSRL = desired rate limit (deg/fire)

The related software variable definition is

LD SLOW RATE LIM E = 1024 * IFSRL / 360

---

L C A T H E T A
Units - degrees
Resolution - 0.1
Range - 1 to 20

Definition - Controls the notching widths in the 5th-7th harmonic elimination mode. recommended setting is 9.0 degrees.

LCATHETA = desired notch width (degrees)

The related software variable definition is

DCLC CHOPA THETA E = LCATHETA * 1024/360

---

L C B 2 T H E T A
Units - degrees
Resolution - 0.1
Range - 0 to 20

Definition - Controls the notching widths in the 5th harmonic elimination mode. recommended setting is 12.0 degrees.

LCB2THETA = desired notch width (degrees)

The related software variable definition is

DCLC CHOPB 2THETA E = LCB2THETA * 1024/360

---

L C C A
L C A B
L C B C
Units - hertz
Resolution - 1.0
Range - 0 to 50

Definition - These variables define the transition frequency threshold between the various harmonic elimination notching modes. The transition trequencies are based on the system resonant frequency defined by

Fres = 1 / [2 * pi * sqr(Cm*Lst)]

Cm = per phase motor filter capacitance
Lst = per phase motor starting inductance

The transition frequencies are given by

LCCA = Between square wave and 5th-7th notching
(set to 0 hertz for shunt only config)
(set to Fres/7 - 5 hertz for series/shunt)

LCAB = between 5th-7th notching and 5th notching
(set to Fres/5 hertz)

LCBC = between 5th notching and square wave
(set to Fres/5 + 9 hertz)

The related software variable definition is

SPD BRK CHOP CA E = LCCA * 603/60
SPD BRK CHOP AB E = LCAB * 603/60
SPD BRK CHOP BC E = LCBC * 603/60

---

L C I C O M P A
L C I C O M P B
L C I C O M P C
Units - gain
Resolution - 0.01
Range - 1.0 to 2.5

Definition - Defines the gains applied to the dc link current command in the various harmonic elimination notching modes. The gains are intended to compensate for dc link current that is lost through the shunt arm of the link commutator.

LCICOMPA = desired gain in 5th-7th harm elim
(typical setting is 1.4)
LCICOMPB = desired gain in 5th harm elim
(typical setting is 1.8)
LCICOMPC = desired gain in commutation only
(typical setting is 1.0)

The related software variable definition is

DLC IDC COMP E (1) = 32 * LCICOMPA
DLC IDC COMP E (2) = 32 * LCICOMPB
DLC IDC COMP E (3) = 32 * LCICOMPC

---

L C M A X D N A
L C M A X D N B
L C M A X D N C
Units - degrees
Resolution - 1.0
Range - 0 to 60

Definition - Defines the maximum conduction angle of the shunt arm of the link commutator in the various harmonic elimination notching modes.

LCMAXONA = max angle in 5th-7th mode
(typical setting is 14 degrees)
LCMAXONB = max angle in 5th mode
(typical setting is 40 degrees)
LCMAXONC = max angle in com only mode
(typical setting is 50 degrees)

The related software variable definition is

MAX GTO ON ANG A E = LCMAXONA * 1024/360
MAX GTO ON ANG B E = LCMAXONB * 1024/360
MAX GTO ON ANG C E = LCMAXONC * 1024/360

---

---

L C M I N O N
Units - mSec
Resolution - 0.01
Range - 0.25 to 3.0

Definition - Minimum clamp placed on the conduction time od the shunt arm of the link commutator. Typical setting is 0.7 mSec.

LCMINON = minimum on time (mSec)
The related software variable definition is

GTO MIN ON CLMP E = 3993 * LCMINON

---

L C O N A N G A
L C O N A N A G A 1
Units - degrees
Resolution - 0.01
Range - 0 to 20

Definition - Defines the minimum conduction angles for the shunt arm of the link commutator when notching to eliminate the 5th and 7th harmonics.

LCONANGA = min conduction ang 1
(typical setting is 9.9 degrees)
LCONANGA1 = min conduction ang 2
(typical setting is 9.9 degrees)

The related software variable definition is

GTO COM ANGA E = LCONANGA * 1024/360
GTO TUF ANGA E = LCONANGA1 * 1024/360

---

L C O N T I M B C
Units - mSec
Resolution - 0.01
Range - 0.20 to 3.0

Definition - Defines the minimum conduction time for the shunt arm of the link commutator when notching to eliminate the 5th and notching for inverter commutation only. Typical setting is 0.7 mSec.

LCONTIMBC = min conduction time (mSec)
The related software variable definition is

GTO COM TIME BC E = 3993 * LCONTIMBC

---

L C S R C
Units - hertz
Resolution - 1.0
Range - 0 to 5

Definition - Defines the speed at which the transition from source forced commutation to link commutator commutation is made during a dead start sequence when in the shunt only configuration. Typical setting is 8 Hz.

LCSRC = desired transition freq (hertz)

The related software variable definition is

SPD BRK SRC COM E = LCSRC * 683/60

---

P O 1, 2, 3, 4, 5, 6
Units - n/a
Resolution - n/a
Range - 0 to 19

Definition - This collection of variables selects the six items to be displayed by the data logger on the door module printer. The number of the information to be printed is entered as data into each of the variables. If no data for a particular entry is required, then it should be set to zero. The selections are

| Setting | Data |
|---|---|
| 00 | no print out |
| 01 | Motor Speed |
| 02 | Speed Command |
| 03 | Motor Current |
| 04 | Motor Voltage |
| 05 | Drive Horsepower |
| 06 | Drive Kilowatts |
| 07 | Link Voltage |
| 08 | Link Current |
| 09 | Source Current |
| 10 | Source Voltage |
| 11 | Source Firing Angle |
| 12 | Inverter Firing Angle |
| 13 | Inverter Power Factor |
| 14 | Source Power Factor |
| 15 | Source KVA |
| 16 | Source KVAR |
| 17 | Torque Limit |
| 18 | Current Limit |
| 19 | Torq Lim, Crosstie, Turning Gear Status |

There is no related software variable.

---

---

P O E
Units - logical
Resolution - n/a
Range - T or F

Definition - Variable used to enable the "auto log" feature of the data logger. In this mode the printer prints the selected information at time intervals selected by the printer time interval setting.

POE = "T"rue enables auto log
= "F"alse disables auto log

There is no related software variable.

---

P T L C S T A T
Units - μec
Resolution - 1.0
Range - 50 to 400

Definition - Defines the time delay associated with looking at the status of the gyo gate drivers after every transition of the gate driver command. Note, the counter that impliments the delay only has resolution to incriments of 65 mSec. Typical set 195 μsec.

PTLCSTAT = desired delay (μsec)

The related software variable definition is

DCLC LK DLY E = PTLCSTAT * 0.0156

---

P T L O C
Units - pu dc link current
Resolution - 0.01
Range - 0.5 to 2.0

Definition - Defines the trip threshold for the load software detected overcurrent function. Typical setting is 1.5

PTLOC = desired trip level
(pu dc link current)

The related software variable definition is

LOD OVER CUR E = 1024 * PTLOC

---

---

P T L O C C
Units - n/a
Resolution - n/a
Range - 1 to 60

Definition - Controls number of consecutive load overcurrents allowed before a fault is declared. A penalty function is implemented such that 6 good firings will cancel one fault firing. This allows recovery from non-repetitive faults. Typical setting is 3.

PTLOCC = number of consecutive overcurrents

The related software variable definition is

LOD OVER CUR CNT E = 6 * (PTLOCC-1)

---

P T L P L L C
Units - n/a
Resolution - n/a
Range - 1 to 150

Definition - Controls number of consecutive out-of-tolerance load phase-lock-loop calculations allowed before a fault is declared. A penalty function is implemented such that 6 good calculations will cancel one bad calculation. This allows recovery from non-repetitive faults. Typical setting is 60.

PTLPLLC = # of consecutive out-of tol calcs

The related software variable definition is

LOD PLL CNT E = 6 * (PTLPLLC-1)

---

P T L P L L T
Units - degrees
Resolution - 1.0
Range - 0 to 60

Definition - Defines the out-of tolerance threshold for the load-side phase-lock-loop calculation. Typical setting is 22 degrees.

PTLPLLT = out-of-tol threshold (degrees)

The related software variable definition is

LOD PLL ERROR TOL E = 1024 * PTLPLLT / 360

---

P T S O C
Units - pu dc link current
Resolution - 0.01
Range - 0.5 to 2.0

Definition - Defines the trip threshold for the source software detected overcurrent function. Typical setting is 1.5

PTSOC = desired trip level
(pu dc link current)

The related software variable definition is

SRC OVER CUR E = 1024 * PTSOC

---

P T S O C C
Units - n/a
Resolution - n/a
Range - 1 to 60

Definition - Controls number of consecutive source overcurrents allowed before a fault is declared. A penalty function is implemented such that 6 good firings will cancel one fault firing. This allows recovery from non-repetitive faults. Typical setting is 3.

PTSOCC = number of consecutive overcurrents

The related software variable definition is

SRC OVER CUR CNT E = 6 * (PTSOCC-1)

---

P T S O C L C
Units - pu dc link current
Resolution - 0.01
Range - 0.25 to 2.0

Definition - Defines the trip threshold for the source software detected overcurrent function when the control is in a link commutated mode. Typical setting is 0.75

PTSOCLC = desired trip level
(pu dc link current)

The related software variable definition is

OVER CUR SRC1 E = 1024 * PTSOCLC

---

P T S P E T
Units - pu speed (60 hz base)
Resolution - 0.001
Range - 0.0 to 0.5

Definition - Defines the maximum speed error allowed before a "loss of speed control" alarm is declared. Typical setting is 0.1 pu

PTSPET = max allowed pu speed error (on 60 hz base)

The related software variable definition is

SPD REG ERR TOL E = 1024 * PTSPET

---

PTSPLLC
Units - n/a
Resolution - n/a
Range - 1 to 150

Definition - Controls number of consecutive out-or-tolerance source phase-lock-loop calculations allowed before a fault is declared. A penalty function is implemented such that 6 good calculations will cancel one bad calculation. This allows recovery from non-repetitive faults. Typical setting is 60.

PTSPLLC = # of consecutive out-of tol calcs

The related software variable definition is

SRC PLL CNT E = 6 * (PTSPLLC-1)

---

PTSPLLT
Units - degrees
Resolution - 1.0
Range - 0 to 60

Definition - Defines the out-of tolerance threshold for the source-side phase-lock-loop calculation. Typical setting is 22 degrees.

PTSPLLT = out-of-tol threshold (degrees)

The related software variable definition is

SRC PLL ERROR TOL E = 1024 * PTSPLLT / 360

---

PTSUVT
Units - seconds
Resolution - 0.1
Range - 0 to 20

Definition - Defines the maximum time a source undervoltage may exist before a source uv fault is declared. If the undervoltage exists for less than this time, then it is considered a bus transfer and the control will restart when the source voltage returns.

PTSUVT = max bus transfer time (sec)

This variable is application dependent and the customer should be consulted to determine its setting. In the absence of other guidlines, set at 7.2 seconds

The related software variable definition is

UV TIME OUT LIM E = 862 * PTSUVT

---

RRACC0, 1, 2,
3, 4, 5 3, 4, 5
Units - pu speed per second
Resolution - 0.001
Range - 0.001 to 0.097

Definition - Family of variables controlling the rate-of-change of the drive speed. Selection of a particular speed acceleration rate is controlled in the background. Consult ladder diagram to determine mode.

RRACC? = pu speed change per second

The ralated software variable definition is

SPD RATE ACCL E (?) = 186499 * RRACC? * DRMFRQ/DRSFRQ

---

RRBP0, 1, 2, 3
Units - pu speed
Resolution - 0.01
Range - 0.0 to 1.87

Definition - Selects regions of drive speed where different acceleration/deceleration rates may be selected. 1 pu is defined as motor rated speed.

RRBP? = desired pu speed

The related software variable definition is

SPD REF BREAK POINT E (?) = 683 * RRBP? * DRMFRQ / 60

---

RRDEC0, 1, 2, 3, 4, 5
Units - pu speed per second
Resolution - 0.001
Range - 0.001 to 0.097

Definition - Family of variables controlling the rate-of-change of the drive speed. Selection of a particular speed deceleration rate is controlled in the background. Consult ladder diagram to determine mode.

RRDEC? = pu speed change per second

The related software variable definition is

SPD RATE DECEL E (?) = 186499 * RRDEC? * DRMFRQ/DRSFRQ

---

RRFMAX
RRFMIN
Units - pu speed
Resolution - 0.001
Range - 0.0 to 1.33

Definition - Variables controlling the maximum and minimum speed reference scaling in speed mode 3. Note : 1 pu speed is motor rated speed.

RRFMAX = maximum desired pu speed
RRFMIN = minimum desired pu speed

The related software variable definition is

SPD REF MAX/MIN LMT E = 683 * RRFMAX/MIN * DRMFRQ / 60

---

RRGMAX
RRGMIN
Units - pu speed
Resolution - 0.001
Range - 0.1 to 1.33

Definition - Define the maximum and minimum speed reference clamps. Note: 1 pu speed is rated motor speed.

RRGMAX = max pu speed ref clamp
RRGMIN = min pu speed ref clamp

The related software variable definition is

SPD REG REF MAX/MIN LMT E = 683 * RRGMAX/MIN * DRMFRQ / 60

---

RRMOD
Units -n/a
Resolution -n/a
Range -0 to 7

Definition - Variable selecting the speed reference for normal operation (not turning gear or E-stop).

| RRMOD | Reference Selected |
|---|---|
| 0 | Remote 0-10 volt reference |
| 1 | Remote 4-20mA reference |
| 2 | Remote -10 to 10 volt reference |
| 3 | Local 0-8.2 volt reference |
| 4 | Remote contact reference |
| 5 | Turning gear reference |
| 6 | Emergency speed request |
| 7 | Special product line reference |

The related software variable definition is

SPD REF MODE E = RRMOD

---

RRPMAX
RRPMIN
Units -pu speed
Resolution -0.001
Range -0.0 to 1.33

Definition - Variables controlling the maximum and minimum speed reference scaling in speed modes 0, 1, 2, and 4.
Note: 1 pu speed is motor rated speed.

RRPMAX = maximum desired pu speed
RRPMIN = minimum desired pu speed

The related software variable definition is

SPD REF PROCESS MAX/MIN E =
683 * RRPMAXMIN * DRMFRQ / 60

---

RRTG
Units - pu speed
Resolution - 0.001
Range - 0.1 to 1.33

Definition - Variable setting motor fixed speed when speed set point mode 05 is selected. This mode is normally used for turning-gear function.

RRTG = desired pu speed

Note: 1 pu sped is rated motor speed

The related software variable definition is

SPD REF TURNING GEAR E = 683 * RRTG * DRMFRQ / 60

---

SQBLWR
Units - bit coded
Resolution - n/a
Range - not clamped

Definition - Controls the desired blower alarm/fault configuration. The low order byte sets the alarm configuration, the high order byte sets the fault configuration. If the sum of the running blowers is less than or equal to the number represented in the alarm byte, then a blower alarm will be declared. The blower fault works in the same manner.

The related software variable definition is

BLOWER CNT E = SQBLWR

SQOPT1

| | |
|---|---|
| Units | - bit coded |
| Resolution | - n/a |
| Range | - not clamped |

Definition - Individual bits are used to enable/disable various functions or inputs in the application program (background). See the ladder diagram for specific usage. Typically a "1" in the bit position indicated, will force the indicated state.

| Bit | Function |
|---|---|
| 0 | no motor overtemperature |
| 1 | no bridge differential pressure alarm |
| 2 | mask blower 1 starter energized |
| 3 | mask blower 2 starter energized |
| 4 | mask blower 3 starter energized |
| 5 | mask blower 4 starter energized |
| 6 | no user defined alarm |
| 7 | no transformer alarm |
| 8 | no dc link reactor alarm |
| 9 | no pcr alarm |
| A | no blown bridge filter fuse |
| B | mask run permissive energized |
| C | mask start permissive energized |
| D | blower fault is not system fault |
| E | unused |
| F | reserved |

The related software variable definition is

SQ OPT1 E = SQ OPT1

SQOPT2
Units - bit coded
Resolution - n/a
Range - not clamped

Definition - Individual bits are used to enable/disable various functions or inputs in the application program (background). See the ladder diagram for specific usage. Typically a "1" in the bit position indicated will force the indicated state.

| | |
|---|---|
| 0 | unused |
| 1 | gto overcurrent (0 = alarm, 1 = fault) |
| 2 | no master/slave current imbalance fault |
| 3 | unused |
| 4 | unused |
| 5 | unused |
| 6 | disable gto gate driver fault (1 = disable) |
| 7 | unused |
| 8 | unused |
| 9 | mask fan door closed |
| A | unused |
| B | unused |
| C | unused |
| D | unused |
| E | unused |
| F | reserved |

The related software variable definition is

SQ OPT2 E = SQ OPT2

---

SQSP 1, 2, 3, 4
Units - pu speed
Resolution - 0.001
Range - 0.0 to 1.875

Definition - Defines uncommited speed breakpoint thresholds for general use in the application program (background). See the ladder diagram for specific usage. 1 pu is defined as motor rated speed.

SQSP? = desired pu speed

The related software variable definition is

BKGND SPD POINT? E = 683 * SQSP? * DRMFRQ / 60

---

S Q T 1, 2, 3, 4, 5, 6
Units - seconds
Resolution - 0.01
Range - 0.0 to 200.0

Definition - Defines time delay durations in uncommitted timers used by the application program (background). See the ladder diagram for specific usage.

SQT? = desired time (seconds)

The related software variable definition is

BKGND TIMER? LIMIT E = SQT? * DRSFRQ

---

S R E R R F
Units - bit coded
Resolution - n/a
Range - 0,1,2

Definition - Selects the break of the low pass filter applied to the speed error signal. Typically set at 2.

SRRERRF = 0 — filter is bypassed
= 1 — double at 17 rad (60Hz src)
double at 14 rad (50Hz src)
= 2 — double at 40 rad (60Hz src)
double at 28 rad (50Hz src)

The related software variable definition is

SPD ERR FILTER E = SRRERRF

---

S R G A I N
Units - gain
Resolution - 0.1
Range - 0.1 to 20.0

Definition - Speed regulator gain. The relation to system parameters is:

$$SRGAIN = \frac{N^2 * WK^2}{1076731.2 * HP * Tr^2} * \frac{60^2}{DRMFRQ^2}$$

N = rated motor rpm
Tr = regulator response time (1 sec)
$WK^2$ = total drive $WK^2$ at motor shaft ($lb\text{-}ft^2$)
HP = drive horse power

The related software variable definition is

SPD G2 E = 1024 * SRGAIN / (2*DRSFRQ)

---

S R L L A D J
Units - bit coded
Resolution - n/a
Range - 0,1,2,3,4,5,6,7,8

Definition - Variable for selecting the speed feedback time constant. This time constant is normally coordinated with the speed regulator crossover so as to give critical damping. Typically set to 7.

| SRLLADJ | lag T.C. 60Hz | lag T.C. 50Hz | speed reg crossover |
|---|---|---|---|
| 0 | filter is bypassed | filter is bypassed | |
| 1 | 0.0125 | 0.0104 | 80 |
| 2 | 0.0292 | 0.0243 | 34 |
| 3 | 0.0625 | 0.0521 | 16 |
| 4 | 0.1292 | 0.1077 | 8 |
| 5 | 0.2629 | 0.2191 | 4 |
| 6 | 0.5300 | 0.4417 | 2 |
| 7 | 1.0640 | 0.8867 | 1 |
| 8 | 2.1300 | 1.7750 | 0.5 |

The related software variable definition is

LEAD LAG ADJUST E = SRLLADJ

---

S R S L
Units - pu slip
Resolution - 0.001
Range - 0.0 to 0.1

Definition - Defines full load motor slip.

SRSL = f.1 slip pu

The related software variable definition is

SLIP GAIN E = 683 * SRSL

---

S R T C
Units - seconds
Resolution - 0.01
Range - 0.05 to 3.0

Definition - Time constant used to stabilize the speed regulator.

SRTC = desired time constant (seconds)

The related software variable definition is

SPD$G1$E = 8 * SRGAIN * SRTC

---

---

S R T C D
Units - pu torque
Resolution - 0.01
Range - 0.0 to 1.0

Definition - Sets the default torque command level when the speed regulator is opened. Typically set to 0.2 pu.

SRTCD = desired pu torque

The related software variable definition is

TCOM DEFAULT E = 1024 * SRTCD

---

S R T C M A X
Units - pu torque
Resolution - 0.01
Range - 0.5 to 1.5

Definition - Defines the maximum motoring torque command clamp applied to the output of the speed regulator.

SRTCMAX = desired pu torque

The related software variable definition is

TCOM HI CLMP E = 1024 * SRTCMAX

---

S R T C M I N
Units - pu torque
Resolution - 0.01
Range - 0.0 to 1.0

Definition - Defines the minimum motoring torque command clamp that is calculated in the "Torque Clamp vs Speed" function (see SRTVSG).

SRTCMIN = desired pu torque

The related software variable definition is

TCOM LO CLMP E = 1024 * SRTCMIN

---

S R T C R E G

| | |
|---|---|
| Units | - pu torque |
| Resolution | - 0.01 |
| Range | - 0.0 to 1.0 |

Definition - Defines the maximum regenerative torque command clamp applied to the output of the speed regulator.

SRTCREG = desired pu torque

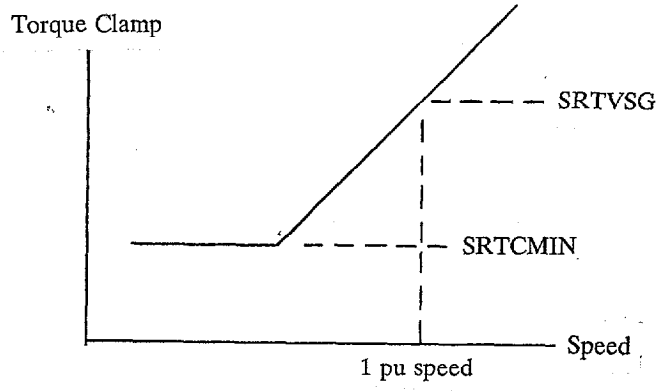
The related software variable definition is

TCOM REGEN CLMP E = -1024 * SRTCREG

S R T V S G

| | |
|---|---|
| Units | - pu torque |
| Resolution | - 0.01 |
| Range | - 0.1 to 1.87 |

Definition - Defines the "Torque Clamp vs Speed" function that limits torque as a function of speed. This variable specifies the desired torque limit a 1 pu speed. The torque limit at other speeds will be directly proportional.




SRTVSG = desired pu torque limit

The related software variable definition is

TORQ VS SPD GAIN E = 24 * SRTVSG

S U B T C M
Units - pu dc link current
Resolution - 1.0
Range - 0.1 to 1.25

Definition - Defines the current command when the control is in braking mode. . Typically set to 0.5 pu.

SUDSTL = desired time (seconds)

The related software variable definition is

BRAKE TCOM E = 1024 * SUBTCM

---

S U B T L
Units - seconds
Resolution - 1.0
Range - 1 to 30

Definition - Defines the maximum length of time that the control will attempt to brake the motor to a standstill. After this time has expired, if the run command is still present, the control will attempt a dead start. Braking is only entered if attempting to decelerate below min speed. Typically set to 10 seconds.

SUBTL = desired max brake time (seconds)

The related software variable definition is

BRAKE TMR LMT E = SUBTL

---

S U D S T L
Units - seconds
Resolution - 1.0
Range - 1 to 20

Definition - Defines the maximum length of time that the control will attempt to start the motor from a standstill. Typically set to 7 seconds.

SUDSTL = desired time (seconds)

The related software variable definition is

DEAD START TMR LMT E = SUDSTL

---

S U F T M R
Units - seconds
Resolution - 0.1
Range - 0.0 to 20.0

Definition - Defines the time allowed for the load-side flux integrators to settle before an attempt is made to run the load-side phase-lock-loop. Typically set to 0.7 seconds.

SUFTMR = desired time (seconds)

The related software variable definition is

FLX DELAY TIME E = 360 * SUFTMR

---

S U M F R Q
Units - hertz
Resolution - 0.1
Range - 0.2 to 75.0

Definition - Defines the inverter frequency when the control is in test mode Manual Fix Fire selected in DRVTST. Typically set to 5Hz.

SUMFRQ = desired frequency (hertz)

The related software variable definition is

MANUAL FREQ E = 6 * SUMFRQ

---

S U M I N F L X
Units - pu flux
Resolution - 0.01
Range - 0.1 to 0.75

Definition - Defines the flux magnitude required before the control will start to ramp the inverter frequency up during a dead start attempt. Typically set to 0.5 pu.

SUMINFLX = desired pu flux

The related software variable definition is

MIN START FLUX E = 1024 * SUMINFLX

---

S U M T C M

| | |
|---|---|
| Units | - pu dc link current |
| Resolution | - 0.01 |
| Range | - 0.1 to 1.0 |

Definition - Defines the current command when the control is in test mode Manual Fix Fire selected in DRVTST. Typically set to 0.2 pu.

SUMTCH = desired pu dc link current

The related software variable definition is

MANUAL TCOM E = 1024 * SUMTCM

---

S U R E G S P D

| | |
|---|---|
| Units | - hertz |
| Resolution | - 0.1 |
| Range | - 1.0 to 30.0 |

Definition - Defines the inverter frequency above which the speed and flux regulator loops are closed. This value should always be greater than SUTRNFRQ. Typically set to 10Hz for series/shunt configuration. Set to 7Hz for shunt only.
SUREGSPD = desired frequency (hz)

The related software variable definition is

SPD BRK CLOSE REG E = 683 * SUREGSPD / 60

---

S U R U P R

| | |
|---|---|
| Units | - pu speed per second |
| Resolution | - 0.001 |
| Range | - 0.001 to 0.1 |

Definition - Defines the rate at which the inverter fundamental frequency ramps up from the initial frequency of SUSFRQ when the control is attempting a dead start. Typical setting is 0.0078 pu/sec.

SCRDNR = desired ramp rate (pu/sec)

The related software variable definition is

RAMP UP E = 1 / SCRUPR

---

S U S F R Q
Units - hertz
Resolution - 0.1
Range - 0.5 to 3.0

Definition - Defines the inverter frequency at which the control initially starts firing in order to establish flux when attempting a dead start. Typically set to 2 hz.

SUSFRQ = desired frequency (hz)

The related software variable definition is

START FREQ E = 6 * SUSFRQ

---

S U S T C M
Units - pu dc link current
Resolution - 0.01
Range - 0.1 to 1.0

Definition - Defines the level of dc link current command when the control is attempting a dead start. Typically set to 0.85 pu.

SUSTCM = desired pu current

The related software variable definition is

START TCOM E = 1024 * SUSTCM

---

S U T R N F R Q
Units - hertz
Resolution - 0.1
Range - 1.0 to 15.0

Definition - Defines the minimum frequency at which the control can turn on the phase-lock-loop and transition to closed loop inverter angle control when attempting a dead start. Typically set to 8 hz for series/shunt config., 5 hz for shunt only config.

SUTRNFRQ = desired frequency (hz)

The related software variable definition is

MIN TRANS FREQ E = 6 * SUTRNFRQ

---

T S M A X
T S M I N
Units - amps
Resolution - 1.0
Range - no clamps

Definition - Used during the QC FINAL TEST GTO OFF TEST. These values define maximum and minimum values of 1 pu current that the test man is allowed to enter. Typically use 1250 amps max and 100 amps min.

TSMAX = max allowed current (amps)
TSMIN = min allowed current (amps)

The related software variable definition is

QC CUR UPPER LIMIT E = TSMAX
QC CUR LOWER LIMIT E = TSMIN

---

T S B T O L
T S V T O L
Units - %
Resolution - 1
Range - not clamped

Definition - Used during the QC FINAL TEST GTO OFF TEST. This value defines the maximum allowable difference between actual and commanded current in percent. Typically use 5% (5).

TSBTOL = desired tolerance (%)
TSVTOL = desired tolerance (%)

The related software variable definition is

QC CUR BALANCE TOL E = TSBTOL
QC CUR VALUE TOL E = TSVTOL

---

## 4-2. PUBLIC VARIABLES

Public variables are those software variables which may be accessed by name, using the DCLC -IMD diagnostic monitor. These variables may be byte, integer or bit coded quantities. The byte variables are used to show logic states in the sequencing and their value is shown as "T"rue or "F"alse when displayed by the DCLC - IMD diagnostic monitor. Integer variables are used to show the value of proportional quantities (such as voltage or current) used in the regulator software. The value of integer variables is shown as a positive or negative decimal number when displayed by the DCLC-IMD diagnostic monitor. Bit coded variables are displayed as positive decimal numbers which must be converted to binary to see which bits are true. Each bit represents some logic state or operating condition.

Acomplete list of public variables is given in the system elenentary along with their definitions.

# CHAPTER 5

# SECTION A
# DIAGNOSTIC MONITOR SPECIFICATIONS

## 5-1. INTRODUCTION

The diagnostic monitor for the DCLC-IMD product provides the user with a variety of functions. Via an RS-232 terminal, the user may display regulator and logic variables and system fault and status information. The monitor also provides the ability to modify system variables and regulator constants, which facilitates troubleshooting and tuning procedures.

Communication with the monitor is handled through the RS232 port on the drive control. Hence, the communication device may be any RS232 compatible terminal, including handheld, hard copy and CRT types. Provisions are made to specially format the monitor output when a handheld terminal is being used.

All communication via the RS232 port are 7-bit data, even parity, with one stop bit. Acceptable baud rates are 300, 1200, 2400, 4800, and 9600 bits per second (baud). Upon power up, the monitor searches for carriage return characters to lock onto the baud rate used by the terminal device.

Monitor functions are accessed by entering a valid command (from the monitor's instruction set) at the diagnostic terminal. Figure 1-1 illustrates the basic command structure for the drive. Many of the commands are called directly from the MONITOR level, '*>' prompt, and control is returned to this level after execution of the requested process. The ENGINEERING and TUNEUP functions, however, are actually extended modes of operation that have their own instruction sets. After execution of one of the ENGINEERING mode commands, for example, control is returned to the ENGINEERING mode, 'E>' prompt. To return to the MONITOR level again, the QUIT command must be invoked -- this is also true for the TUNEUP mode.

In sections 1-3 and 1-4, the purpose and syntax of each monitor command is described in detail. Table 5-1 defines the conventions used in specifying the command formats.

**TABLE 5-1 Definitions for Command Syntax Notation**

| Symbol | Definition |
|---|---|
| ::= | "Is defined to be" |
| [ ] | Items between brackets are optional |
| \| | Separates items in a list - choose one |
| ::= | "Is defined to be" |
| {} | Braces delimit items for the purpose of minimizing ambiguity |
| ( ) | Optional characters in a command orparameter are in parentheses |
| **HIGHLITED** | User entries (Highlited) |

<decimal no> ::= <1-5 decimal digits> [T]

<hex no> :: = <1-4 hex digits> [H]

<new value> :: = [+>-]<1-5 decimal digits w/ or w/o dec. pt.>

<pointer> ::= [{CS | DS | <hex no>} : ] <hex no>

<register> ::= [CS | DS]

CS ::= Code Segment

DS ::= Data Segment

## 5-2. OPERATING INSTRUCTIONS

**5-2.1. General Information.** To initiate communication with the Monitor, the user should enter several "carriage return" characters at the terminal keyboard. The terminal will then respond with a banner message and the monitor prompt character (*>) to indicate that it is ready to accept command inputs. At this point, after passing the PASSCODE. The operator may enter any of the MONITOR level commands or select one of the extended modes of operation (for example TUNEUP) as described in detail in sections 3 and 4.

When finished using the diagnostic monitor, the terminal session may be terminated by entering QUIT at the terminal keyboard (see section 5-3.9). The logoff message is then printed at the terminal, indicating that the session is over, and the monitor will idle, waiting for the carriage return characters that indicate the start of another terminal session.

Terminal Format:

The Diagnostic Monitor is designed to work with either a handheld terminal or a standard keyboard type. The default (at the start of the terminal session) is the handheld format. In this mode of operation, the output messages are sent one line at a time to accommodate the limited display length of the handheld terminal (16 characters max per line). Entering a carriage return causes the next line of output to be sent to the terminal. The TERMINAL command (section 5-3.8) may be used to switch to an 80 column

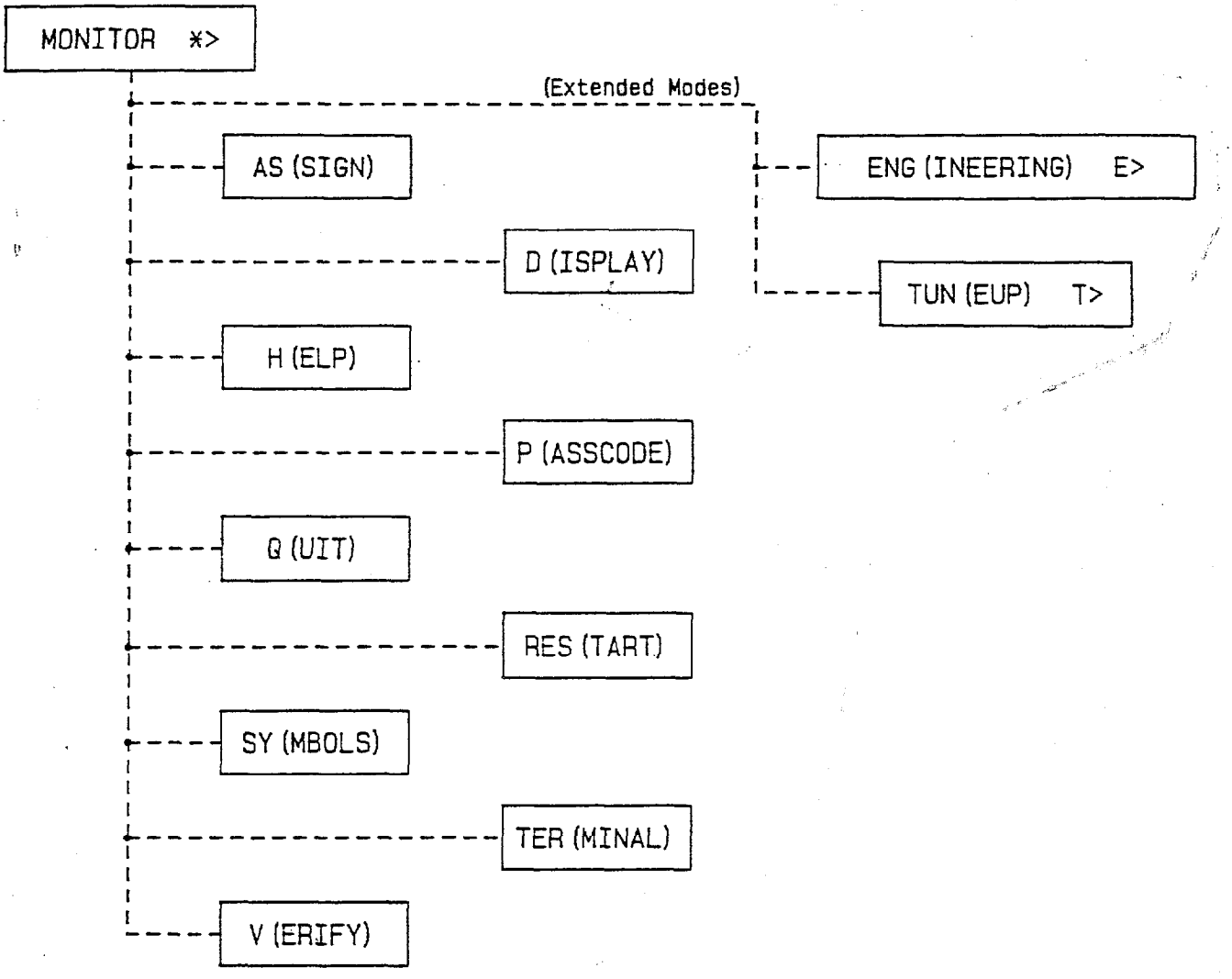



*Figure 5A-1. Monitor Command Structure*

listing format with no pauses for use with a standard CRT or hardcopy terminal.

Power-up Status Messages:

When power is first applied to the drive control, a sequence of self-tests and initialization procedures is performed. Any failure status of these tests is automatically sent to the panel printer. Should the printer fail, diagnostic messages will be rerouted to the diagnostic terminal.

**5-2.2. Editing Functions.** The diagnostic monitor provides text editing functions for correcting errors entered within a command string. Each function is listed below with a brief description.

NOTE: In the examples below, the symbol *> is the diagnostic monitor prompt (not typed by the user), and the symbol [CUR] represents the cursor position at the end of the sequence shown.

DEL/RUBOUT -

Removes the previous character from the message and echoes the deleted character to the terminal.

Example: Typed - *> HEKP<DEL><DEL>LP

Echoed - *> HEKP /K P LP [CUR]

CONTROL-R

Causes all characters currently in the message to be transmitted to the terminal on a new line. This function is convenient after using DEL/RUBOUT because it shows the state of the message after the editing.

Example:

Typed - *> HEK<DEL>LP<CONTROL-R>

Echoed - *> HEK /K LP
HELP [CUR]

NOTE: Typing subsequent consecutive CONTROL-R keys will invoke the monitor recall function. This feature calls up previous commands that were typed at the terminal, starting with the most recent and progressing to the oldest command remembered. A total of 20 command lines are remembered. Using the recall feature helps the user avoid retyping a command that he wishes to execute repetitively.

CONTROL-X

Deletes the entire message that has been entered so far.

NOTE: This has the same effect as using the DEL/RUBOUT function to remove all the characters from the message.

Example: Typed - *> HEKL<CONTROL-X>HELP
Echoed - *> HEKL
HELP [CUR]

CONTROL-Y

Aborts the current command sequence and returns the monitor to the current mode of operation (e.g., ENGINEERING, TUNEUP or the main level).

NOTE: CONTROL-Y or CONTROL-C may be entered at any time to abort the current process--even during the listing of a message at the terminal.

Example:

Typed - T> HE<CONTROL-Y>

Echoed - T> HE^ Y

T> [CUR]

CONTROL-C

Aborts the current command sequence and returns the monitor to the outermost (main) level of operation. When a CONTROL-C is received, the monitor also displays a message identifying the product as a DCLC-IMD Drive as in the login sequence.

Example:

Typed - T> HE<CONTROL-C>
Echoed - T> HE^ C
-- DCLC MONITOR--
*> [CUR]

CONTROL-S

Causes the current output to be halted until it is restarted using the CONTROL-Q switch.

CONTROL-Q

Causes the current output to be restarted after having been stopped using the CONTROL-S switch.

**5-2.3. Command Syntax.** The general syntax for a monitor command is shown below:

<command> ::= <instruction>
[ <delimiter> <parameter> ...] <terminator>

where <instruction> is one of the valid monitor instructions (defined in section 5-3) and <terminator> is a carriage return (<cr>). The <delimiter> and <parameter> entries, if necessary, depend on the particular instruction being called. Each instruction has its own set of valid delimiters and parameters. Also, some commands may require several <delimiter> <parameter> pairs, while some require no parameters at all.

As an aid to the operator, and to facilitate handheld terminal use, all the <delimiter> <parameter> pairs need not be entered on the same line as the instruction -- each needed parameter will be requested by the monitor. The syntax for this type of sequence is:

<command> ::= <instruction> <terminator>
<parameter> <terminator>.

After each <terminator> entry, the monitor displays an appropriate prompt that asks the user for the necessary parameter.

The use of abbreviated commands is another feature designed to aid the handheld user. Rather than typing the complete instruction word, only the first one or two letters need to be entered (see the syntax definition tables in sections 3 and 4 for the minimum acceptable abbreviations).

**5-2.4. Protected Commands.** Some monitor commands allow the user to modify system variables and change tuning parameters. These functions, because of their potentially destructive nature, are available only after executing a privilege enabling sequence. This is so the user will not inadvertantly make modifications that could inhibit the proper operation of the drive control.

The privileges are enabled by entering a password at the monitor terminal. Privileges are automatically disabled when: (1) the monitor is logged off (QUIT).

**5-2.5. Help.** The user may print out a list of available commands by asking for HELP. From either the MONITOR mode, or one of the extended modes, a HELP request will print the list of commands available at that mode only.For example, if in the ENGINEERING mode, only the ENGINEERING mode commands, and not the MONITOR mode commands, will be listed.

## 5-3. MONITOR COMMANDS

The MONITOR level commands are intended to be the primary tools used in interrogating the drive control and monitoring its status. The instruction set provides display and modify functions, the ability to reset the control, and the power to direct variables to analog outputs for strip chart monitoring, among other things. Table 5-2 shows the complete list of instructions and their correct syntax. The commands are defined in their single line entry format, although the "prompt-through" type sequence (as discussed in section 2.3) is also valid.(See Table 5-2)

Assign

Function: To designate that a particular variable be assigned to a circular list and/or D/A converter.

Using the ASSIGN function, the user may modify the configuration of the internal circular lists by changing the number of entries (maximum of 8) to be stored at each sample interval and/or selecting which entries are being captured. A total of 2K bytes of memory are available for storage of the diagnostic list information. This memory is evenly divided among the parameters within the list. As an example, if 4 word parameters are monitored in the circular a total of 256 events will be recorded.

The list may be triggered either by a level trigger or mask trigger. In either case the trigger is compared against the variable selected in list position no. 1.. In a level (>, <, =). When the bit trigger mode is selected, any bit of variable 1 becoming true causes the list to be taken. Bits of variable 1 may be masked using the mask variable option of the list. The list can only be triggered by the level trigger or the bit trigger. When more than one trigger has been activated only the last will be accepted by the list function.

*Table 5-2. Syntax Definitions for Monitor Level Commands*

| | |
|---|---|
| ASSIGN: | A(SSIGN) {LIST\| DA} |
| DISPLAY: | {D(ISPLAY) { <varname> [C]<br>\| FAULTS \| LIST \| RELAYS } \|<br>DF \|<br>DL \|<br>DR } <cr> |
| HELP: | H(ELP) <cr> |
| PASSCODE: | P(ASSCODE) <cr> |
| QUIT: | {Q(UIT) \| EX(IT) <cr> |
| RESTART: | RES(TART) <cr> |
| SYMBOLS: | SY(MBOLS) [ <symbolname> ] <cr> |
| TERMINAL: | TER(MINAL) <cr><br><current format> CHANGE ? [ Y\| N ] <cr> |
| VERIFY: | V(ERIFY) [ <symbol group name> ] <cr> |

**5-3.1.** . The user may also direct drive variable contents to a D/A converter for observation on an oscilloscope or chart recorder. The analog outputs are available from test points on the front edge of the printed circuit board HLCB. When assigning a variable to a D/A output, the operator selects the two parameters, SCALE and OFFSET. Since the possible range of a variable often exceeds the range of the D/A converter, a function is provided to scale the variable contents before sending data to the D/A converter.

The following table gives the possible SCALE parameters; beside each one is the value by which the variable contents are multiplied
before being converted if that SCALE factor is selected.

An OFFSET may also be selected during the ASSIGN process -- this value is subtracted from the variable contents before the scaling process. This function may be used when the slight variations around a fixed value need to be examined.

SCALE PARAMETERS

| SCALE | MULTIPLIER | SCALE | MULTIPLIER |
|---|---|---|---|
| 15 | 1/32768 | 0 | 1 |
| 14 | 1/16384 | -1 | 2 |
| 13 | 1/8192 | -2 | 4 |
| 12 | 1/4096 | -3 | 8 |
| 11 | 1/2048 | -4 | 16 |
| 10 | 1/1024 | -5 | 32 |
| 9 | 1/512 | -6 | 64 |
| 8 | 1/256 | -7 | 128 |
| 7 | 1/128 | -8 | 256 |
| 6 | 1/64 | -9 | 512 |
| 5 | 1/32 | -10 | 1024 |
| 4 | 1/16 | -11 | 2048 |
| 3 | 1/8 | -12 | 4096 |
| 2 | 1/4 | -13 | 8192 |
| 1 | 1/2 | -14 | 16384 |
| 0 | 1 | -15 | 32768 |

**Examples:**

*>**ASSIGN LIST**

List Setup:
0 – quit
1 – samples ; selects variables to be sampled
2 – trig level ; sets up level triggered list
3 – trig mask ; sets up bit triggered list
4 – site ; selects site where data is taken

Choice? 1
NO OF ENTRIES=1 ; previously was set to capture 1 parameter
NEW VALUE ? **3<cr>** ; set to capture 3 parameters
ENTRY 1 = FLUXFB ; previously set to capture FLUXFB
NEW ENTRY? **SPDREF<cr>** ; speed reference (no. 1)
ENTRY 2 = 0000H ; null value
NEW ENTRY? **SPDFB<cr>** ; speed feedback (no. 2)
ENTRY 3 = 0000H ; null value
NEW ENTRY? **IAFB<cr>** ; armature current feedback (no. 3)

```
*>ASSIGN LIST

List Setup:
  0 – quit
  1 – samples
  2 – trig level
  3 – trig mask
  4 – site
Choice? 2
TRIGGER LEVEL = 0
NEW VALUE? 200                    ; level of trigger in counts
TRIGGER CONDITION? OFF
NEW VALUE? <                      ; choices are <, >, =

*>ASSIGN LIST
List Setup:
0 – quit
1 – samples
2 – trig level
3 – trig mask
4 – site

Choice? 4
TRIGGER SOURCE = OFF              ; present data site
CHOICES:                          ; available data sites
0 - OFF
1 - SPLL
2 - SDRV
3 - LPLL
4 - LDRV
5 - FLXR
6 - SPDR
7 - TCMR
8 - BKGND

NEW VALUE? 1 <cr>
*>ASSIGN LIST <cr>
List Setup:
0-quit
1-samples
2-trig level
3-trig mask
4-site

Choice? 3
PRESENT MASK FF                   ;you must turn off level trigger
NEW VALUE? OF                     ;value must be in hexadecimal
*>
*>ASSIGN LIST                     ; go to choice 4 to select site
```

```
*>ASSIGN LIST
*>                                  ;because choice 3 turned off list

*> ASSIGN DA1 <cr>                  ; choose DA1 (TP2) on HLCB (slot 2E)
ENTRY = SPDREF                      ; the current setup is shown
SCALE = 8
OFFSET = 0
ENTRY = IAFB<cr>                    ; set new parameters -- I feedback
SCALE = 6<cr>                       ; divide by 64 to get in range
OFFSET = <cr>                       ; leave the offset the same (at 0)
*>

*> AS DA2<cr>                       ; choose the DA2 (TP3) on HLCB (slot 2E)
ENTRY = SPDFB                       ; the current setup is shown
SCALE = 8
OFFSET = 0
ENTRY = <cr>                        ; still look at speed feedback
SCALE = -2 <cr>                     ; multiply 4 to accent variations
OFFSET = 1000<cr>                   ; look at fluctuations around 1000
*>
```

The following notes on using the diagnostic list apply.

1. When using the bit trigger option the mask must always be entered in hexadecimal. The bit (s) to trigger the list must be set to a "1" in the appropriate bit location.
2. In either the bit trigger or magnitude trigger mode, entry "1" in the variable selections is compared against either the magnitude or bit trigger word.
3. When using the bit trigger option, the word must be re-entered, wherever selection 3 of "ASSIGN LIST" is accessed, otherwise the mask is reset to "0".
4. The list has completed taking data when the trigger source has returned to "OFF" after entering another selection.
5. List sites must be selected with care. The list may not necessarily fill up if a site is selected which doesn't operate during the event to be captured. For example, LPLL is disabled when the drive "RUN" command is removed.
6. If variable is by address rather than name, use offset only (converter if necessary to data segment DS: [F000] base).

**5-3.2. Display.** Function: to display the contents of select drive control variables and circular lists.

Several options are allowed in the DISPLAY command:

1. DISPLAY LIST, to display the circular list information.

2. DISPLAY FAULTS, to display the status of the fault and alarm logic.

3. DISPLAY RELAYS, to display a concise list of the drive's background relay status.

The DISPLAY FAULTS command will produce a list of the most recent STATUS messages (stored in a circular "fault log"). This list includes current status indications (shown also with the DISPLAY STATUS command).

Drive variables are displayed in their respective engineering units. Generally they are in per unit (of flux, current, speed, etc...). The general format in which variable contents are displayed is:
<varname> = <value>
A particular variable may be displayed repetitively by simply typing a "C" (for continuous) following the variable name on the command line.

Some DISPLAY command options may be abbreviated to two letters. These abbreviations and their equivalents are listed as follows:

DF ::= DISPLAY FAULTS
DL ::= DISPLAY LIST
DR ::= DISPLAY RELAYS

**Examples:**
*> **DISP SPDSP <cr>**
SPDSP = 472
*> **D <cr>**
NAME? **SPDSP <cr>**
SPDSP = 472
*>

*> **D RELAY <cr>**
<drive relay status list>
*>

*> **DISPLAY LIST <cr>**
<listing of circular list>
*>

*> **DISPLAY FAULTS <cr>**
<listing of circular fault log>
*>

**5-3.3. Help.** Function: to display at the terminal a list of the commands available in the current mode of monitor operation.

Examples:

*> **H <cr>**
<list of available commands>
*>

*> **HELP <cr>**
<list of available commands>
*>

**5-3.4. Passcode.** Function: to enable the monitor's privileged functions.

*** NOTE ***

**For security, the characters are not echoed when entering the password at the terminal. The present passcode is: [GESALEM 9][20:222, 224] "PASS"**

Examples:

*> **PASSCODE <cr>**
PASSWORD : **<password>** <cr> ; password not echoed
PRIVILEGES ENABLED
*>
To change the passcode, see SET password under TUNEUP

**5-3.5. Quit.** Function: to terminiate the current terminal session. Upon execution of the QUIT command, the monitor will print a logout message and then begin idling, waiting for the carriage return inputs that indicate the start of a new terminal session.

Examples:
*> **Q <cr>**
MONITOR OFF

*> **EX <cr>**
MONITOR OFF

**5-3.6. Restart.** Function: to execute a software reset of the microprocessor.

Executing the RESTART command is essentially a soft reset. The diagnostic checks are run the same as during power-up, and the drive control is re-initialized to restore the hardware (counters,etc.) to its normal running state.

*** NOTE ***

**To execute the RESTART command, the drive must be stopped. If the drive is running, the RESTART routines will not run, and a warning will be printed at the terminal.**

Examples:

*> **RES <cr>**
<selftest results messages>

*> **RESTART <cr>** <selftest results messages>

**5-3.7. Symbols.** Function: to display at the diagnostic terminal the symbolic names that may be used to access drive control variables. If no qualifier is given, the complete list will be printed; if a partial name is specified, however, all symbols that match that partial name are listed. See system elementary sheets 1DE-1DK for lists of available symbol names.

Examples:

*> **SY <cr>**
<complete symbol list>
*>

*> **SYMBOLS TS <cr>**
TSMAX
TSMIN
TSVTOL*>

*> **SYM SFDBK <cr>**
SFDBK
*>

**5-3.8. Terminal.** Function: to examine and possibly modify the terminal format. The monitor will display the current format (either "handheld" or "standard") and then ask the operator whether a change is required. Entering Y(ES) will cause the

monitor to toggle to the other terminal format. Any other entry will cause no change.

Examples:

*> **TERMINAL <cr>**
HNDHLD-CHANGE? **Y <cr>**
STANDARD FORMAT
*>

*> **TER <cr>**
STD - CHANGE? **<cr>**
STANDARD FORMAT
*>

**5-3.9. Verify**. Function: displays the current state of all drive EEPROM settings and adjustments to allow the user to verify factory settings. Individual groups of variables may be displayed for verification by appending one or more group names to the VERIFY command (see example). The groups will be listed in the order specified. Following are the possible group names:

DR - Drive configuration
CR - Current Regulator
FP - Flux Profile
FR - Flux Regulator
IF - Inverter Firing
LC - Link Commutator
PO - Printer Output
PT - Protective Parameters
RR - Speed Ramp Rate Conditioning
SQ - General Sequencing
SR - Speed Regulator
SU - Startup Parameters
TS - Test

Examples:

*> **VERIFY <cr>**
<grouped list of all software jumpers and adjustments>
*>

*> **V SR, FR, SQ <cr>**
(list of speed reg. software jumpers and adjustments>
(list of flux reg. software jumpers and adjustments>
(list of software sequencing jumpers>
*>

## 5-4. EXTENDED MODES

**5-4.1. Engineering.** The ENGINEERING mode provides primitive functions like examining and modifying memory, and input and output ports. Absolute addresses, rather than symbolic names, are used to access memory locations and I/O ports. Table 5-3 summarizes the command syntax for this mode of monitor operation.

**Table 5-3 Syntax Definitions for ENGINEERING Level Commands**

| | |
|---|---|
| DISPLAY: | D[W] <start addr>[,<end addr>] <cr> |
| HELP: | H <cr> |
| INPUT: | I[W] <port addr>,[, ...] <cr> |
| PUT: | P[W] <port addr>,[, ...] <cr> |
| MOVE: | M <start addr>, <end addr>, <dest addr> <cr> |
| OUTPUT: | O[W] <port addr>, <data>[,<data> ...] <cr> |
| QUIT: | {Q \|EX} <cr> |
| SUBSTITUTE: | S[W] <addr>,[<new value>, ...] <cr> |
| <start addr>:: | = <pointer> |
| <port addr >:: | = <hex no> |
| <end addr > :: | = <pointer> |
| <new value>:: | = <hex no> |
| <dest addr> :: | = <pointer> |
| <data> :: | = <hex no> |
| <addr> :: | = <pointer> |

**5-4.2. Display**. Function: to display the contents of a memory location or a block of memory. The current value of each byte (or word) is listed in hexadecimal form. Additionally, the ASCII equivalent of the memory shown on a line is displayed in a field to the right of the standard display. Only printable ASCII characters are shown --others are displayed as a period (.). The ASCII display function is suppressed when operating in the handheld terminal format.

Format:

D[W] <start addr>[,<end addr>] <cr>

Examples:

E> **D DS:800 <cr>**
F800:0800 41 A
E>

E> **DW 200, 210 <cr>**
0000:0200 55AA 00FF 1233 5256 4147 4E49
2020 0020 .U..3.VRGAIN .
0000:0210 4243 CB
E>

**5-4.3. Help -.** Function: to list the available command set for the ENGINEERING mode.

Examples:

E> H <cr>
<list of available commands>
E>

**5-4.4. Input .** Function: to display a byte or word at an input port.

Format:

I[W] <port addr> <cr>

Examples:

E> **I 9008 <cr>**
7F ,
7F ,
7F **<cr>**
E>

E> **IW 9008 <cr>**
007F **<cr>**
E>

**5-4.5. Put .** Function: to modify the contents of selected memory locations. Typing a comma causes the monitor to remain in the PUT mode and increment to the next sequential memory address. A carriage return at the end of a line executes the current command (if any) and then returns control to the ENGINEERING level. This command is very useful in writing to EEPROM because no verification of the modified information is made.

Format:

P[W] <addr> <cr>

Examples:

E> **P DS:A40 <cr>**
FC00:0A40 A5- ,
FC00:0A41 5A- **AA <cr>**
E>

E> **PW DS:50 <cr>**
FC00:0050 0FFF- **7FF,**
FC00:0051 5A5A- **<cr>**
E>

**5-4.6. Move .** Function: to move a block of data from one area of memory to another. Memory contents are transferred sequentially from low memory to high.

Format:

M <start addr>, <end addr>, <dest addr> <cr>
CB

Examples:

E> **M DS:50,59,CS:00 <cr>**
E>

E> **M 800,900,DS:42 <cr>**
E>

**5-4.7. Output .** Function: to output a byte or word to an output port.

Format:

O[W] <port addr>, <data><cr>[<data>, ... <cr>]

Examples:

E> **O 8010, FF <cr>**
- **F8,**
- **F0 <cr>**
E>

**5-4.8. Quit .** Function: to return from the ENGINEERING mode to the MONITOR mode.

Examples:

E> **Q <cr>**
*>
E> **EX <cr>**
*>

**5-4.9. Substitute.** Function: to exam and modify the contents of selected memory locations. Typing a comma causes the monitor to remain in the SUBSTITUTE mode and increment to the next sequential memory address. A carriage return at the end of a line executes the current command (if any) and then returns control to the ENGINEERING level. This command is very useful in writing to EEPROM because no verification of the modified information is made.

Format:

**S[W] <addr> <cr>**

Examples:

E> **S DS:A40 <cr>**
FC00:0A40 A5- ,
FC00:0A41 5A- **AA <cr>**
E>

E> **SW DS:50 <cr>**
FC00:0050 0FFF- **7FF,**
FC00:0051 5A5A- **<cr>**
E>

## 5-5. TUNEUP

In the TUNEUP mode of monitor operation, the user is able to adjust the performance of the drive by modifying regulator parameters and configurations. A step-by-step tuneup procedure is contained in Chapter 6 . Following are descriptions of the commands that are used as part of this procedure. Table 5-4 summarizes the command syntax for the TUNEUP mode.

For on-line tuneup of any regulator, the background sequencing RUN logic must be satisfied. Monitor commands alone will not allow the drive to run. This mode is intended to be used primarily when the drive is initially installed.

**Table 5-4 Syntax Definitions for TUNEUP Level Commands**

| | |
|---|---|
| APPLY: | AP(PLY) <cr> |
| ASSIGN: * | AS(SIGN) {LIST \| DA} |
| DISPLAY: * | {D(ISPLAY) { <varname> [C]\| |
| | STATUS \| |
| | LIST <listname> \| } \| |
| | DS \| |
| | DL <listname>} <cr> |
| HELP: | H <cr> |
| MODIFY: | M(ODIFY) <varname> <cr> |
| | <old value message> |
| | <new value prompt> [ I \| D \| <new value> ] <cr> |
| MODIFY REFERENCE: | M(ODIFY) REF(ERENCE) <cr> or MREF |
| QUIT: | {Q \| EX } <cr> |
| REMOVE: | REM(OVE) <cr> |
| RESTART | RES(TART) <cr> |
| SAVE: | SAV(E) <cr> |
| | <verify query> [ Y \| N ] <cr> |
| SAVE D/A | SDA <cr> <verify query>[ Y \| N ]<cr> |
| SET PASSCODE | Set (PASSCODE) <cr> |
| SYMBOLS: * | SY(MBOLS) [ <symbol name qualifier> ] <cr> |
| TERMINAL: * | TER(MINAL) <cr> |
| | <current format> CHANGE ? [ Y \| N ] <cr> |
| TUNEUP SYMBOLS: | TSY(MBOLS) <cr> |
| TUNEUP: | TUN(EUP) [symbol group name] <cr> |
| VERIFY: * | V(ERIFY) [ <symbol group name> ] <cr> |

* - Also available in MONITOR mode -- see section 5-3 for descriptions.

**5-5.1.**

**Apply -.** Function: to apply the test generator reference to the regulator being tuned up.

Examples:

T> **AP <cr>**
T>

**5-5.2. Help-.** Function: to list the available command set for the TUNEUP mode.

Examples:

T> **H <cr>**
<list of available commands>
T>

**5-5.3. Modify -.** Function: to allow the operator to modify the contents of select drive control variables. A new value may be entered, or the current value may be incremented or decremented repeatedly. Also,the increment (or decrement) interval may be selected by placing a new interval after the I or D entry (see

example below). If none is specified, the increment interval will be one unit of the least significant digit. Values are displayed and entered in appropriate engineering units.

*** NOTE ***

**Because modifying certain memory locations could have adverse effects on the drive control operation, the MODIFY command may only be executed after completing the privilege enabling sequence.**

Examples:

```
T> MOD VRGAIN <cr>
VRGAIN = 1.1
NEW VALUE ? 15 <cr>
VRGAIN = 15.0
NEW VALUE ? <cr>
T>
T> M VRGAIN <cr>
VRGAIN = 1.1
NEW VALUE ? I<cr>
VRGAIN = 1.2
NEW VALUE ? I .3<cr>
VRGAIN = 1.5
NEW VALUE ? I<cr>
VRGAIN = 1.8
NEW VALUE ? D<cr>
VRGAIN = 1.5
NEW VALUE ? <cr>
T>
```

*** NOTE ***

**When a tuneup case is selected (section 5-4.20),a numbered list of symbols associated with that case is displayed at the terminal. While running that particular test, a symbol's NUMBER (from the list) rather than its name may be used as the argument in the DISPLAY and MODIFY commands. This eliminates the need to laboriously type out long symbol names at the terminal during a tuneup session.**

Examples:

```
T> MOD 3 <cr>
FRKGAIN = 1.1
NEW VALUE ? 1.3 <cr> ;change gain from 1.1 to
                       1.3
FRKGAIN = 1.3
NEW VALUE ? <cr>
T>
```

**5-5.4. Modify Reference -.** Function: to modify (customize) an internal reference generator. When a tuneup case is selected, its corresponding reference generator is automatically selected (see Section 5-4.20). The user then has the option to modify the characteristics of the reference generator as shown in the following examples. A reference generator is a series of two steps, each with selectable amplitude and duration. Step durations are entered in seconds (i.e., .05 for 50 milliseconds) and reference levels are specified in per unit levels (i.e., for a current test, the level is in per unit current and for a speed test in per unit speed, etc.).

After the steps are fully defined and the test generator is enabled by the APPLY command, the regulator specified by the TUNEUP command will be excited with a periodic signal corresponding to the test reference levels defined. A fixed constant level may be obtained by setting the amplitudes of both steps to the same value.The REMOVE command disables the test generator and removes all excitation to the specified regulator.

Examples:

(Fixed Reference)

```
T> MOD REF <cr>

STEP      LEVEL(PU)      TIME(SEC)
  1       0.100            1.00
  2       0.000            1.00

STEP 1 LEVEL = <cr>
STEP 1 DURATION = <cr>
STEP 2 LEVEL = 0.10 <cr>  ;make levels agree
STEP 2 DURATION = <cr>
T>
```

(Stepped Reference)

```
T> MOD REF <cr>

STEP      LEVEL(PU)      TIME(SEC)
  1       0.100            1.00
  2       0.100            1.00

STEP 1 LEVEL = <cr>
STEP 1 DURATION = <cr>
STEP 2 LEVEL = 0.00 <cr>  ;set for 10% step
STEP 2 DURATION = <cr>
T>
```

**5-5.5. Quit -.** Function: to return from the TUNEUP mode to the MONITOR mode. On leaving the TUNEUP mode, the monitor asks the user whether hewould like to have any new constant values transferred into the non-volatile memory (i.e., execute a SAVE command). This is so the operator will not inadvertantly forget to save the new drive status after finishing the TUNEUP sequence.Examples:

```
T> Q <cr>
DO YOU WANT TO SAVE NEW CONSTANTS?
```

? Y <cr>
CONSTANTS SAVED
*>

T> **EX <cr>**
DO YOU WANT TO SAVE NEW CONSTANTS ?
? N <cr>
CONSTANTS NOT SAVED
*>

**5-5.6. Remove -.** Function: to remove the test generator signal from the regulator being tuned up.

Examples:

T> **REM <cr>**
T>

**5-5.7. Set Passcode -.** Function: to to allow the user to put in his own password — up to 6 characters long. For this to be a permanent change it must be saved.

*** NOTE ***

**For security, passcodes are not echoed**
**:Password address**

Example:

OLD PASSCODE =: **<OLD PASSCODE>** **<cr>**
; password not echoed
NEW PASSCODE =: **<NEW PASSCODE>** **<cr>**
; password not echoed
NEW PASSCODE =: **<NEW PASSCODE>** **<cr>**
; verify to enable
T > ; can now save if want permanent change.

**5-5.8. Save -.** *Function: to store the present values* of the regulator variables in the system's nonvolatile memory. After modifying system parameters (for example, gains and time constants during a tuneup session), the new values may be transferred into permanent storageusing the SAVE command. The previously stored values, of course, are lost during this process; hence, the monitor will prompt the user for confirmation before actually executing the command.

*** NOTE ***

**Privileges must be enabled to execute the SAVE command. Also, the hardware write enable switches on the DMPB cards must be in the "store enable" position before values may be stored in the nonvolatile memory.Examples:**

*> **SAVE <cr>**
ARE YOU SURE? **Y<cr>**
CONSTANTS SAVED ; saved if DMPB switch on
*>
*> **SAVE <cr>**
ARE YOU SURE? **<cr>**
CONSTANTS NOT SAVED
*>

**5-5.9. Save D to A (SDA).** Function: To store the present values of the D to A assignment variables in the system's nonvolotile memory. During tuneup mode, the D to A converters (1 and 2) are continually reassigned according to the default values required for the regulator being turned. If the D to A values were included in the global "SAVE" command, the intended setup values would be lost. A separate "SAVE" command is therefore included.

When it is required to modify, the D to A assignments permanently, the "ASSIGN DA" command used first be used to setup the required parameters.

The hardware write enable switches on the DMPB card must be in the "write enable" position before values may be stored in the nonvolatile memory.

Examples:

*> **SDA <cr>**
ARE YOU SURE? Y **<cr>**
CONSTANTS SAVED
* >

*>**SDA <cr>**
ARE YOU SURE? **<cr>**
CONSTANTS NOT SAVED
*>

**5-5.10. Tuneup Symbols -.** Function: to list all tuneup variables associated with the current tuneup case.

Examples:

T> **TSY <cr>**
TUNEUP VARIABLES ARE:
<numbered symbol list> or NONE
T>

**5-5.11. Tuneup -.** *Function: to initiate a particular* regulator tuneup sequence. The user must specify which of several regulator is to be tuned. He is then allowed to select the desired tuneup case. Examples of valid commands are:

CR - Current Regulator
FR - Flux Regulator
SR - Speed Regulator

Example:

(Current Regulator Tuneup)
T>**TUNE CR<cr>**

```
1 - CRGAIN      = 338
2 - CRTC        = 0.020
3 - CRVDC       = 0.96
4 - CRMIN       = 0.20
5 - CRMAX       = 1.00
6 - CRLIMP      = 0.12
7 - CRMARG      = 17
T>
```

**5-5.12. Monitor Commands -.** The following main-level monitor commands are also available in the TUNEUP mode:

ASSIGN
DISPLAY
SYMBOLS
TERMINAL
VERIFY

The syntax definitions and functional descriptions for these commands are contained in section 5-3.

## 5-6. LOADING DISPLAY MESSAGES HEADERS

Customized display messages can be loaded into EEPROM to identify the user, drive, personnel to contact, actions to be taken, etc. Five (5) different messages can be loaded - starting at the locations below. Each message can be up to 64 characters long (minus 1 for end of message = 63).

| Starting Address | Message type |
|---|---|
| 20:IFO | Header - message printed at power-up |
| 20:22F | Repetition - message printed at beginning of "data logging" |
| 20:270 | Requisition - message printed when "Status" P.B. used |
| 20:2BO | Fault - message printed at beginning of "Fault" messages |
| 20:2FO | Alarm - message printed at beginning of "Alarm" messages |

Procedure:

1. Write the message out on the form in Figure 5B-I (remember to leave 2 spaces) for <cr> <LF> (by word) if you don't need to force a new line.
2. Convert the characters to their ASCII equivalent using Table 5B-1. Add after the last conversion 00 to signal end-of-message.
3. Using the monitor Put command, load the ASCII equivalents.
4. Go to "TUNE" mode, then do "SAVE".

*** NOTE ***

**You can use a "," instead of a <cr> after each ASCII code to increment to the next location - rather than reloading the address (faster).**

Suggestions
Always include user's name and the drive/channel #. For the fault, alarm, and repetition messages, a reminder such as "Log items in maint book' may be useful. The default messages should be okay . It should only be necessary to add the user's name and drive I.D. to the beginning (if sufficient space) or to the end (will need to move "00" of the default message.

```
MONITOR OFF <cr>    ; Form 5B-1 or 5B-2 should
                      be filled out.

IGTO MONITOR
*>PASSWORD  <cr>
PASSWORD ? PASS <cr>

PRIV. ENABLED
*> ENG<cr>

E>PUT 3000:200<cr>  ; Enable EEPROM write
                      switch
20:200 ?? 53,
20;201 ?? 43,       ; Continue till all characters
done
    :
    :
20:2?? ?? 00<cr>
```

Figure 5B-1 : MESSAGE HEADER FORM

Message Type_____

Starting Address - 3000:_____

| Message Character | ASCII (HEX) | Message Character | ASCII (HEX) |
|---|---|---|---|
| 01____ | ____ | 33____ | ____ |
| 02____ | ____ | 34____ | ____ |
| 03____ | ____ | 35____ | ____ |
| 04____ | ____ | 36____ | ____ |
| 05____ | ____ | 37____ | ____ |
| 06____ | ____ | 38____ | ____ |
| 07____ | ____ | 39____ | ____ |
| 08____ | ____ | 40____ | ____ |
| 09____ | ____ | 41____ | ____ |
| 10____ | ____ | 42____ | ____ |
| 11____ | ____ | 43____ | ____ |
| 12____ | ____ | 44____ | ____ |
| 13____ | ____ | 45____ | ____ |
| 14____ | ____ | 46____ | ____ |
| 15____ | ____ | 47____ | ____ |
| 16____ | ____ | 48____ | ____ |
| 17____ | ____ | 49____ | ____ |
| 18____ | ____ | 50____ | ____ |
| 19____ | ____ | 51____ | ____ |
| 20____ | ____ | 52____ | ____ |
| 21____ | ____ | 53____ | ____ |
| 22____ | ____ | 54____ | ____ |
| 23____ | ____ | 55____ | ____ |
| 24____ | ____ | 56____ | ____ |
| 25____ | ____ | 57____ | ____ |
| 26____ | ____ | 58____ | ____ |
| 27____ | ____ | 59____ | ____ |
| 28____ | ____ | 60____ | ____ |
| 29____ | ____ | 61____ | ____ |
| 30____ | ____ | 62____ | ____ |
| 31____ | ____ | 63____ | ____ |
| 32____ | ____ | 64<EOM> | 00 |

| Character | ASCII | Character | ASCII |
|---|---|---|---|
| A | 41 | a | 61 |
| B | 42 | b | 62 |
| C | 43 | c | 63 |
| D | 44 | d | 64 |
| E | 45 | e | 65 |
| F | 46 | f | 66 |
| G | 47 | g | 67 |
| H | 48 | h | 68 |
| I | 49 | i | 69 |
| J | 4A | j | 6A |
| K | 4B | k | 6B |
| L | 4C | l | 6C |
| M | 4D | m | 6D |
| N | 4E | n | 6E |
| O | 4F | o | 6F |
| P | 50 | p | 70 |
| Q | 51 | q | 71 |
| R | 52 | r | 72 |
| S | 53 | s | 73 |
| T | 54 | t | 74 |
| U | 55 | u | 75 |
| V | 56 | v | 76 |
| W | 57 | w | 77 |
| X | 58 | x | 78 |
| Y | 59 | y | 79 |
| Z | 5A | z | 7A |
| | | | |
| 0 | 30 | <space> | 20 |
| 1 | 31 | <cr> | 0D |
| 3 | 32 | <1f> | 0A |
| 3 | 33 | # | 23 |
| 4 | 34 | & | 26 |
| 5 | 35 | ' | 2C |
| 6 | 36 | - | 2D |
| 7 | 37 | . | 2E |
| 8 | 38 | : | 3A |
| 9 | 39 | @ | 40 |

TABLE 5B-1: ASCII Character Set

# CHAPTER 6

# INSTALLATION AND STARTUP GUIDE (OUTLINE)

I. Keep detailed installation log for each drive.

II. Install Equipment
(Align motor, setup coupling to run uncoupled from driven device. Setup motor protectives - lubrication, thermocouples, RTDs, vibration detectors. Check-out transformers and reactors - fault devices, alarm devices; source and load breakers - close and trip circuits, overcurrent devices.).

III. Interconnect equipment

IV. Wire check

V. Check-out source and load breakers

VI. Megger

VII. Run motor in "Bypass".

VIII. Check Berg jumpers, daughter board components.

IX. Apply control power
Check starters and blowers
Check power supplies
Check relays, lights

X. Operate in gate-test-mode
Check source and load firing
Check process inputs and outputs
Set current feedback

XI. Apply source voltage
Check attenuator signals
Set scaling and trip levels

XII. Simulate load voltage
Check attenuator signals
Set scaling and trip levels

XIII. Perform crowbar test

XIV. Perform "locked" rotor test

XV. Run motor uncoupled
Check bearings, vibration
Check acceleration, deceleration, regeneration

XVI.Check transfer schemes

XVII. Run motor coupled
Tune control
Check loaded transfer schemes
Evaluate control

XVIII. Clean-Up
Send back to factory corrected prints
Make 3 lists of tune up settings
(Customer, Service Engineering, Factory)

## 6-1. DETAILED INSTALLATION INSTRUCTIONS

It is assumed that all interconnecting wire and cables are in place and are connected. The erection and alignment of the motors is not described here, nor is the routine checkout of the circuit interrupters. It is advised that a detailed installation log be maintained for each drive.

## 6-2. MEGGER TEST

a. Short all power buses together in the IMD panel. L1[A/B], L2[A/B], L3[A/B], P2, N1, N2, M1, M2, M3)
b. Pull the ribbon cables that connect to the front edge of the HPTK cell firing cards.
c. Disconnect ribbon cable to each HPTK card in the Leg module of the bridge compartment. Also remove the ground (GND) stab on this card and on the XPTN.
d. Also remove the XPTN ground (GND) stab located in the motor capacitor bank compartment.
e. Confirm that both the source side isolator switch and the load side motor isolator are open.
f. Perform the 1000V megger test and record the test values. Indicated resistance should exceed 100 megohms.
g. Remove jumpers and reconnect the ribbons and stab connections.
h. Jumper the load side motor isolator load side terminals (T1, T2, T3) together and megger to ground. Indicated resistance should exceed 100 megohms.
i. Remove jumpers. Megger test is now completed.

## 6-3. RUN MOTOR UNCOUPLED ON BY-PASS CONTACTOR

CONSIDERATIONS:
Bearing monitoring protection should be functioning at this point to protect the motor. The motor bearing temperature rise, vibration, and oil rings (slingers) should be checked.

a. Stop the drive and rack-out and lock-out the drives load circuit interrupter. Rack-in the bypass interrupter. Source interrupter should be closed. (Drive's isolation switch should be open).
b. Place INVERTER/BYPASS switch in BY-PASS. Start motor with start switch. Check motor direction of rotation - matches drive. Check motor vibration, magnetic center, bearing and stator temperature rises, etc. Stop motor.

## 6-4. PRELIMINARY CONSIDERATIONS

a. Verify the variable component values in the SEM per sheet DA of the drive elementary.
b. Verify Berg jumper locations as shown on drive elementary sheet DA.
c. Inspect all compartments for loose hardware/debris removing any such material prior to applying power.
d. Inspect all stab connections in both SCR and GTO for tightness.
e. Open all circuit breakers; refer to drive elementary.
f. Open or turn off all switches and power supplies.
g. Partially extract all PWB's from the SEM.
h. Check each SRC, diodes, and GTOs for shorts.

## 6-5. CHECK SCR AND GTO CONTINUITY

a. Using a multimeter, check across each SCR in the source bridge. Measurements should indicate approximately 10K ohms. If greater, check the equalizing resistors (R1) in the snubber boxes. If less than 100 ohms check for defective snubber capacitor (C1) Reference Figure 6.1.

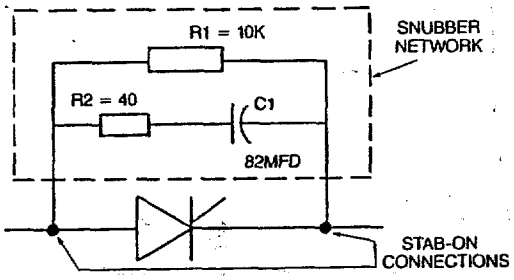



Figure 6.1

b. Using a multimeter, check across each diode and GT0 in load side bridge. See Figure 6.2. Parts 1-2 should read approximately the same reading for each pair of diodes and GTO. Parts 2-3 should read approximately the same reading for each pair of diodes and GTO.

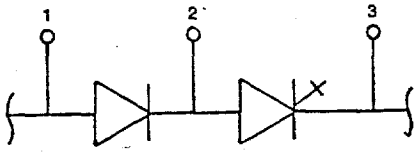



Figure 6.2

## 6-6. AUXILIARY POWER

a. Open the power supplies circuit breakers.
b. Apply the 480Vac and/or 120Vdc control power.
c. Check bridge cooling fans for proper direction of rotation, air should enter the front of each of the bridge compartments and exit the top of the blowers when the doors are closed.

## 6-7. POWER SUPPLIES

1. PSFF

a. Turn on control power supply, PSFF.
b. Verify presence of P5, P15, N15 and P28 at the electronics module backplane.
c. Insert the PWBs and verify that the power supply remains constant.
d. With all PWBs in, adjust the e power supply to obtain the following voltages at the MEM.

- Adjust R1 on NPPB until P5 = + 5.0 +/- 0.1 volts.
- Adjust R6 on NPSK until P15 = +15 +/- 0.2 volts.
- Adjust R7 on NPSK until N15 = - 15.0 +/- 0.2 volts.

2. CPS

   a. Turn on chopper power supply CPS.
   b. Monitor the voltage across the dc filter cap.
   c. Adjust R20 and see that voltage across the cap changes.
   d. Set voltage at 162Vdc.
   e. At XDCE, verify that hot side of R154 (12KΩ) is at ≈ 115 Vac to chassis. This is the side nearest the molex JY connector. If it is at ground potential, swap 400 Hz connection @ the chopper power supply.

3. PS7

   f. Turn on source (70V) gating supply PS7
   g. Verify output is 85Vdc (+5V -10V).

## 6-8. ELECTRONICS MODULE INITIALIZATION

Null all meters.

**6-8.1. Set-up Panel Printer.** Consult the panel printer's manual to enable the battery backup (clock). Then set the time and date. Also, set the interval time greater than 60 minutes.

**6-8.2. Run IMD Selftest:.** Turn-on (up) NLCB test switch. Reset HMPG by jumpering TP1(RST) to TP2(DCOM). Follow the indicated instruction and investigate any fault messages. Repeat till unit passes all tests.

**6-8.3. Operating Set-up Adjustments and EEPROM Memory Initialization:.** Measure and set the voltages for all the OV, SUPP, RTD, UVR, OSP test rings as tabulated in the system elementary sheet 1DC.

Jumper testpoint TINO on HLCB-2E to DCOM, turn-on(up) WREN switch on DMPB(HMPG), WREN LED on the DMPB card should come on. Reset the HMPG. Remove the jumper and turn off WREN switch. Reset the HMPG again. Dump EEPROM memory, using VERIFY, on a printing terminal and compare to values shown on system elementary (sheet 1DE - 1DH).

**6-8.4. Auxiliary Outputs.** If one or more of the auxiliary 4-20 mA outputs are via DAHI1 or DAHI2 (see elementary sheet CL), (From the D to A port), then use the "ASSIGN DA"command to assign the correct variable. Then use "SDA" command to save the variable in the nonvoltile memory. (See elementary sheet BF).

**6-8.5. Message Headers.** Load Display message headers per 5-6 load headers to identify special user and drive for motor (especially if multi-motor).

## 6-9. BRIDGE CURRENT FEEDBACK CALIBRATION (MVB) :

Verify resistors and Berg jumpers on the XTFH, against system elementary (sheet DA). Remove CT lead plugs JD and JT from the XTFH. Measure the resistance across each of the burden resistors on the XTFH; it should equal the "R BURDEN SETTING" value per the elementary. Adjust the corresponding pot to fine tune. Re-connect plugs JR and JT.

## 6-10. GATE TEST MODE OPERATION

Refer to the paragraph entitled GATING TESTS in Chapter 7 for a description of procedure.

Verify that the source-side circuit interrupt and load-side circuit interrupter are open.

Follow the procedure described in GATING TESTS.

## 6-11. CHECK PROCESS INPUTS

Verify that the source-side circuit interrupter and load side circuit interrupter are open and locked out.

The purpose of the tests in this section is to operate each external input shown on system elementary sheets BD, BE into the SEM. Operate each of these inputs by the most convenient means and confirm that the correct LED lights with a closed circuit in each case.

## 6-12. CHECK PROCESS OUTPUTS

Verify that the source contactor and load isolator are open and locked out.

Connect the hand terminal (or equivalent) to the HMPG and MODIFY EEPROM variable (DRVTST ) to "1" to temporarily disable the interrupts. (See Chapter 5, section MONITOR - section EXTENDED MODES for procedure). Save the modification and reset the HMPG.

Using the OUTPUT command [ENG. MODE], operate the output relays shown on system elementary sheets BG, BJ in accordance with the table 6-1. Confirm that the relays operate the proper device(s) per the system elementary.

Relays may be operated in any desired combination by adding the appropriate DATA values from table 6-1.

Re-MODIFY DRVTST to original value.

TABLE 6-1
SLOT 1A

| ADDRESS | DATA | LED ON |
|---|---|---|
| 8040 | 01 | 0 |
| 8040 | 02 | 1 |
| 8040 | 04 | 2 |
| 8040 | 08 | 3 |
| 8040 | 10 | 4 |
| 8040 | 20 | 5 |
| 8040 | 40 | 6 |
| 8040 | 80 | 7 |

| ADDRESS | DATA | LED ON |
|---|---|---|
| 8044 | 01 | 8 |
| 8044 | 02 | 9 |
| 8044 | 04 | A |
| 8044 | 08 | B |
| 8044 | 10 | C |
| 8044 | 20 | D |
| 8044 | 40 | E |
| 8044 | 80 | F |

## 6-13. NO-LOAD FULL-VOLTAGE TEST

The tests in this section require that the source power be connected and available. Open and lock-out the load side circuit interrupter. (If no interrupter is available disconnect leads to motor.)

Apply the control power, reset the SEM and check that all the IMOK LEDs are on. Check for any fault indications. Investigate and clear any indicated faults. Null all meters.

**6-13.1. DC Offset Check.** This procedure should be performed whenever the NLIB card is replaced by a new one.

a. With source power off and the control not in GATE-TEST-MODE, measure TP4 (FACR), TP5 (FCBR), and TP6 (FACE) on NLIB's in slots 1C and 1D with a DVM in the millivolts dc range. Adjust dc offset pots R1, R2 and R3, respectively on the NLIB MOTHER BOARD so that TP4, TP5, and TP6 read 0.0 millivolts dc (±5).

**6-13.2. VOLTAGE PHASING AND BALANCING**

a. Close source side circuit interrupter. Observe that all source side bridge neon lamps are illuminated.

b. Connect an oscilloscope to simultaneously observe the waveforms on testpoints TP7 (VBA), TP8 (VCB), and TP9 (VAC) on the NSFC-2C. Confirm that the phase sequence is 1-2-3 and that all these are appromimately equal. ($V_{BA}$ leads $V_{CB}$ leads $V_{AC}$)

The actual system voltage =
(ATTN RATIO) X VAC

where ATTN RATIO is given on the system elementary sheet EG.

Using a DVM, measure the dc voltage offset at TP7, TP8 and TP9. This value should be less than 5 millivolts.

NOTE: Must measure between TP and ACOM testpoint on the card.

c. Connect an RMS-measuring voltmeter to TP1 (FAB) on the NLIB-1C. Adjust R1(FLXAB) to obtain voltage as follows:

$$\text{FAB} = \frac{\text{ACTUAL SYSTEM VOLTAGE}}{\text{NOMINAL SYSTEM VOLTAGE}} \times 4.75 \text{ VAC RMS}$$

Nominal system voltage is given on the system elementary Sh. EA.
Now connect the voltmeter to TP2 (FBC) and adjust R2(FLXBC) for the same rms voltage as above. Repeat for TP3 (FCA) and R3(FLXCA). Balance signals as accurately as possible (+5 millivolts). Check for dc offset at TP4, TP5 and TP6 of NLIB (<10 millivolts). TP4 TP5 and TP6 should read the same voltage as TP1, TP2, and TP3 respectively.

d. Connect the voltmeter to measure TP13 (VABS) on the NLIB-1C and check that the (-)dc voltage level is approximately 2.229 X VAC at TP9 of NSFC-2C.

**6-13.3. Set Overvoltage:.** Connect the voltmeter to TP18 (OVR) on the NLIB-1C. Turn R8 (OV) slowly counterclockwise until the "OV" LED comes on. The source side circuit interrupter should trip.

Observe the dc voltage on TP-18 and re-adjust R8 so that the voltage is [per unit source overvoltage] times the trip value. (If incoming line is high or low, compensate by the ratio of RATED LINE VOLTS/ACTUAL LINE VOLTS.) See elementary sheet DC for per unit source overvoltage.

**6-13.4. Set Undervoltage..** Clear any faults and close the line contactor. Connect voltmeter to TP19 (UVR) of NLIB-1C.

Connect an oscilloscope to backplane pin 55-1C; this should be a logic zero. Turn R9 (UV) slowly cw until pin 55 just transitions high (may alternate high and low).

Observe TP19-1C and adjust UV(R9)-1C so that the measured value is per unit source undervoltage X trip level. (If incoming line is high or low, compensate by the ratio of RATED LINE VOLTS/ACTUAL LINE VOLTS.) See system elementary sheet 1EC for per unit source undervoltage.

**6-13.5. Slave Bridge:.** Using an oscilloscope, observe the waveforms on NSFC (2A) VBA (TP7), VCB (TP8), and VAC (TP9). Confirm that the phase sequence is TP7, TP8, TP9.

Use a DVM to measure the voltage - record each voltage.

TP7 = _____ TP8 = _____ TP9 =_____

The actual L-L system voltage = (ATTN RATIO) X V1-n avg.

where ATTN RATIO is given in system elementary (sheet EA)

**6-13.6. Check Phasing Between master and Slave:**

Using an oscilloscope, observe TP7 (VBA) - NSFC(2C) and TP7 (VBA) - NSFC(2A). Verify master(2C) leads slave(2A) by 30 degrees. If not, MODIFY DRVCFG.

The actual system voltage =
(ATTN RATIO) X VAC

where ATTN RATIO is given on the system elementary sheet EP.

Using a DVM, measure the dc voltage offset at TP7, TP8 and TP9. This value should be less than 5 millivolts.

*** NOTE ***

**Must measure between TP and ACOM testpoint on the card.**
**Connect an RMS-measuring voltmeter to TP1 (FAB) on the NLIB-1C. Adjust R1(FLXAB) to obtain voltage as follows:**

$$FAB = \frac{\text{ACTUAL SYSTEM VOLTAGE}}{\text{NOMINAL SYSTEM VOLTAGE}} \times 4.75 \text{ VAC RMS}$$

Nominal system voltage is given on the system elementary Sh. EA.
Now connect the voltmeter to TP2 (FBC) and adjust R2(FLXBC) for the same rms voltage as above. Repeat for TP3 (FCA) and R3(FLXCA). Balance signals as accurately as possible (+5 millivolts). Check for DC offset at TP4, TP5 and TP6 of NLIB (< 10 millivolts). TP4, TP5 and TP6 should read the same voltage as TP1, TP2, and TP3 respectively.

Connect the voltmeter to measure TP13 (VABS) on the NLIB-1C and check that the DC voltage level is approximately 2.229 X VAC at TP9 of NSFC-2C.

## 6-14. LOAD SIDE BRIDGE VOLTAGE TESTS WITH SIGNALS FROM SOURCE SIDE BRIDGE

Trip source circuit interrupter. Disconnect ribbon cable JA from NSFC-2B and move the ribbon cable JA from NSFC-2C to NSFC-2B. Connect a test jumper from 1C-55 to 1C-1 (disables source undervoltage fault).

**6-14.1. Voltage Signal Phasing and Balancing.** Close source circuit interrupter.

a. Connect oscilloscope to simultaneously observe the waveforms on testpoints TP7 (VBA), TP8 (VCB), and TP9 (VAC) on the NSFC-2B. Confirm that the phase sequence is 1-2-3 and all balanced. The testpoints should measure the same as for the source-side. Using a DVM, measure the dc voltage offset at TP7, TP8 and TP9. This value should be less than 5 millivolts.
b. Connect an RMS-measuring voltmeter to TP1 (FAB) on the NLIB-1D. Adjust R1 (FLXAB) to obtain voltage as follows:

$$FAB = \frac{\text{ACTUAL VOLTAGE}}{\text{NOMINAL MOTOR VOLTAGE}} \times 4.75\text{VAC RMS} \times \frac{\text{LOAD ATTN RATIO}}{\text{SRC ATTN RATIO}}$$

Rated frequency and nominal motor volts from elementary sheet EA.
Now connect the voltmeter to TP2 (FBC) and adjust R2 (FLXBC) for the same rms voltage as above. Repeat for TP3 (FCA) and R3 (FLXCA). Balance signals as accurately as possible (+5 millivolts). Check for DC offset at TP4, TP5 and TP6 of NLIB (< 10 millivolts).

*** NOTE ***

**Temporarily unplug NCCC card at 1F slot, TP4, TP5 and TP6 at NLIB -1D. It should read the same voltage as TP1, TP2 and TP3 respectively.**

c. Check for the presence of zero-crossing pulses at TP9 (SFT2) on the HLCB-2E. These are narrow 1.25 Vdc pulses evenly spaced at 2.78 millisecond intervals.
d. Connect the voltmeter to measure TP13 (VABS) on the NLIB-1D and check that the dc voltage level is the same as for the source-side 2.229 X Vac at TP9 of NSFC-2B

**6-14.2. Set Overvoltage .** Connect voltmeter to TP18 (OVR) on the NLIB-1D. Turn R8 (OV) slowly ccw until the "OV" LED comes on. The motor contactor should trip. Observe the dc voltage on TP18 and re-adjust R8 so that the voltage is as follows:

$$\text{"OVA" volts} = 1.2 \times \text{trip value} \times \frac{\text{RATED MOTOR VOLTS}}{\text{ACTUAL LINE VOLTS}}$$

**6-14.3. Set Speed and Voltage Indicators.** Since speed is derived from the motor voltage frequency, the source 60 Hz can be used to set the overspeed.

Close-in line contactor. Connect a voltmeter to TP1-1E (SPD). Adjust R1 (SPD) until TP1 = 10.0 Vdc @ 60 Hz .

Motor Voltage Indication:

Adjust VM (R7) -1D on DLIB so that the motor voltmeter reads actual line voltage.

**6-14.4. Set Overspeed:.** Set OSP -1E to +11Vdc. Connect voltmeter to testpoint OSP (TP6)-1E and slowly turn OSP (R2)-1E ccw until the OSP LED comes on. The voltage on TP6 should be +10 Vdc at the trip point. See elementary sheet DC for per unit overspeed.

Adjust OSP until TP6 = [per unit overspeed] X trip level.

**6-14.5. Capacitor Current Calculation.**

1. Measure NSFC (2B) -Vac(TP9) and drive frequency(freq).
2. Calculate capacitor current Icap

$$\text{Icap} = \frac{\text{Vln}}{\text{Xc}} = \frac{\text{Vac*ATTNRATIO*2*PI*FREQ*C}}{1.732}$$

where ATTNRATIO and C are given on system elementary sh. 1EA.

3. Check that MIFB(TP11)-NCCD(1F) =

$$\frac{\text{Icap} * 3.887}{\text{Rated Inverter Current}}$$

where Rated Inverter Current is listed on system elementary sh. 1EA.
Reduce (increase) RO8 & RO9 on DCCC-1F if MIFB is too high (low).

*** NOTE ***

**Individual phase (motor) current is scaled 3.322 Vacrms on inverter base.**

4. Adjust GAIN-NLIB(1D) so that ammeter reads Icap.
5. Trip the source circuit interrupter.
6. Restore NSFC ribbon cables to correct configuration.
7. Remove jumpers.

## 6-15. CROWBAR TEST

The crowbar test permits controllable dc link current to be circulated without involving the motor windings. On the load-side, the dc current circulates though one or more diametric cell pairs bypassing the motor. The signals to be observed on the oscilloscope will indicate whether the source-side CTs are phased correctly.

*** NOTE ***

**Bypass must be open or STATUS disconnected to fine cells.**

a. Reset SEM and apply primary power. Clear all existing faults.
b. Connect hand terminal to source-side HMPG.
c. MODIFY the following EEPROM variables (Example 6-2):

```
CRMIN     to     .10
DRVTSTx   to     2
SUMTCM    to     .15

MONITOR OFF <cr>
*> PASSWORD
PASSWORD? PASS<cr>

PRIV. ENABLED
*>TUN<cr>

T>MOD DRVTST<cr>

DRVTST = 0
VALUE?2<cr>

T>MOD CRMIN<cr>
CRMIN = 0.13
VALUE? .10<cr>
CRMIN = 0.10
VALUE?<cr>
```

d. Connect channel one of the scope to NSFC-2C SIG1 (TP1).
Connect channel two of the oscilloscope to NLIB (1C) "OIA" (TP11). Trigger on channel one.
e. Initiate a run sequence; i.e., sequence the drive to pick-up the software logic ladder's "RUN". The current regulator should now

be active but regulating min current. Check that word MFLAG = 50H.(See elementary sheet 1FF for software logic "RUN").

f. Using the MODIFY command, increase (I) SUMTCM to .25.

```
T>MOD SUMTCM<cr>
SUMTCM = 0.75
VALUE?.15<cr>

SUMTCM = -.15      Start drive and check IFB
VALUE?I.05<cr>     ;Increase by 0.05
SUMTCM = 0.20      ;check IFB level & if continuous
VALUE?I<cr>        ;Increase till IFB = .25 PU
SUMTCM = 0.25      check CTs, Vdc(master & slave)
VALUE?I<cr>        ;adjust SUPP to min and
                   ;increase IFB till trip.
                   ;  approximately 0.9 pu
```

EXAMPLE 6-2

g. Verify that test point "IFB" on NLIB (1C) = 1.25 Vdc (+/- 25%) (current should be continuous - not going to zero).

h. Connect channel one to NLIB (1C), 0IA (TP11). Channel 2 to NSFC (2C), SIG1 (TP1). Compare the display to Figure 6-8.

i. Connect channel one to NLIB (1C), 0IB (TP10). Channel 2 to NSFC (2C), SIG3 (TP3). Compare the display to Figure 6-3.

j. Connect channel one to NLIB (1C), 0IC (TP9). Channel 2 to NSFC (2C), SIG5 (TP5). Compare the display to Figure 6-8.

k. Check NSFC (2C) - VDC - (TP11) for saw-tooth (360 Hz) wave form. Adjust pot VDC for minimum volts at TP(-VDC).

l. If there is no slave source bridge skip to 6-15.1., step f. to test for hardware suppression.

### 6-15.1. Check Slave Bridge.

a. Verify that testpoint IFB2(TP4) - NCCC(1F) = 1.25 Vdc (+/- 30%).

b. Connect channel one to NCCC (1F), 0IA2

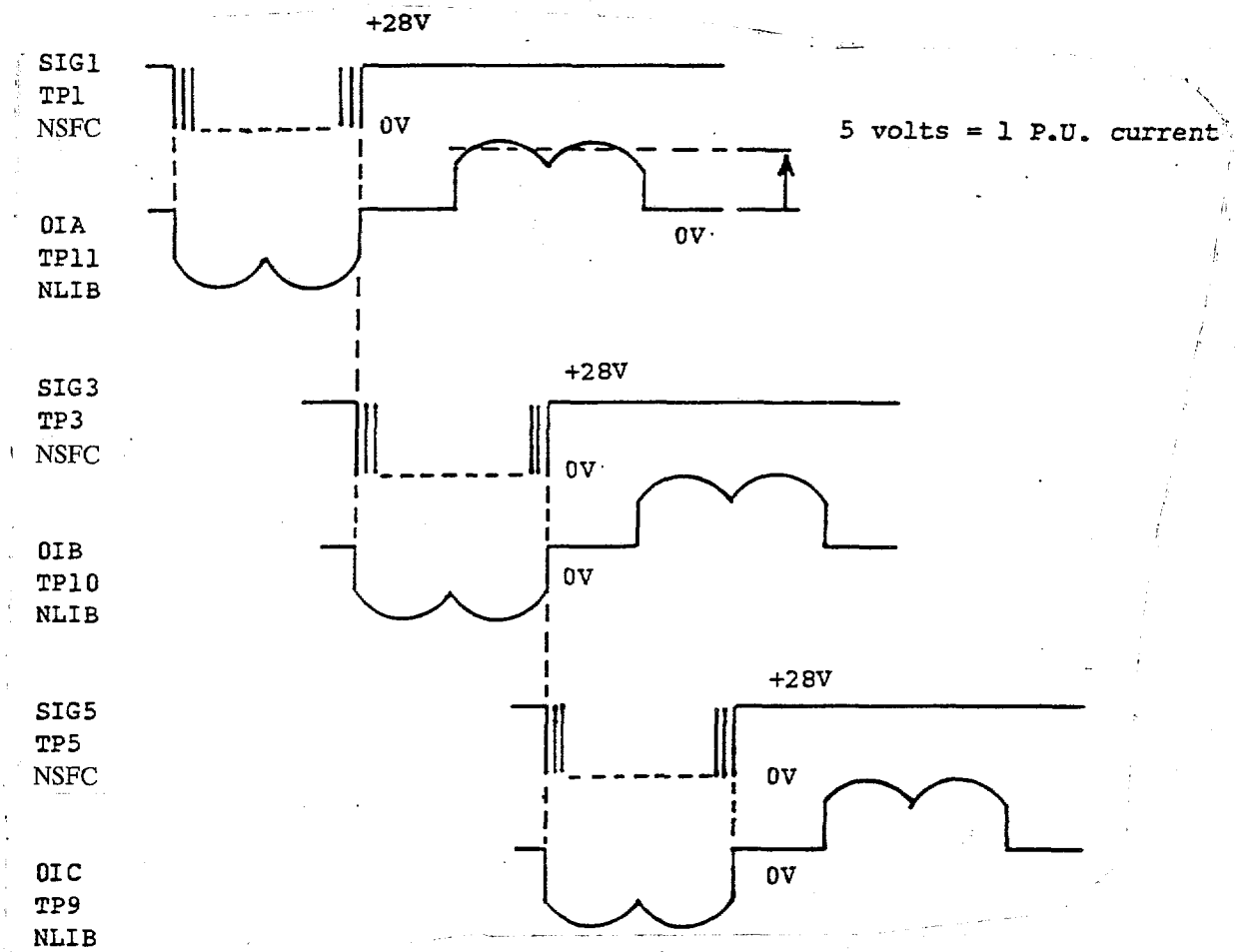



*Figure 6-3. Firing Signals and Current Signals*

(TP1). Channel 2 to NSFD (2C), SIG1 (TP1). Compare the display to Figure 6-3.

c. Connect channel one to NCCC (1F), 0IB2 (TP2). Channel 2 to NSFC (2B), SIG3 (TP3). Compare the display to Figure 6-3.

d. Connect channel one to NCCC (1F), 0IC2 (TP3). Channel 2 to NSFC (2B), SIG5 (TP5). Compare the display to Figure 6-3.

e. Check NSFC(2A) - Vdc - (TP11) for saw-tooth (360 Hz) wave form. Adjust the pot VDC for minimum volts at TP(-VDC).

f. Adjust suppression (R6) 1C on NLIB to minimum (fully clockwise). Testpoint SUPPR should read approximately 3.0 volts.

g. Increase SUMTCM to 0.75 so that suppression LED on NLIB-1C comes on. The source breaker should trip.

h. Readjust suppression (R6) 1C for values as shown on system elementary sheet 1EC.

i. Re-MODIFY CRMIN, DRVTST, and SUMTCM to original values.

## 6-16. "LOCK ROTOR"

The following test is designed to check the phasing of the load-side CTs. The motor may start and turn slowly, therefore, the lubrication system should be operating at this time.

a. Reset the SEM and apply primary power.

b. MODIFY the following EEPROM variables:

| | | |
|---|---|---|
| DRVTST | to | 3 |
| SUMFRQ | to | 3 |
| SUMTCM | to | .09 |

c. Connect a brush recorder to the following signals; channel A to 0IA-1D (TP11) and channel B to SIG1-2B (TP1) Start brush recorder and then drive. Move brush to 0IB-1D (TP10) and SIG3-2B (TP3). Move brush to 0IC-1D (TP9) and SIG5-2B (TP5). Verify traces against Figure 6-3.

d. After a few seconds move channel B of the brush to OIB-1D (TP10). After a few seconds, move channel B to OIA-1D (TP11). Compare current brush traces against Figure 6-4.

e. Verify IFB-1C = 0.65 Vdc (+/- 12.5%). Stop drive.

f. Return DRVTST, SUMFRQ, and SUMTCM to original values.

## 6-17. RUN MOTOR UNCOUPLED CONSIDERATIONS:

Bearing monitoring protection should be functioning at this point to protect the motor. The motor should be run to 50% speed once started and bearing temperature rise, vibration, and oil rings(slingers) checked. Note that drive will not operate at low speed when uncoupled in the shunt only DCLC configuration.

### 6-17.1. Test Connections

1. Make the following connections to a six channel chart recorder:

| | |
|---|---|
| CH1 - DA1(TP1) | - 2E |
| CH2 - OIA(TP11) | - 1D |
| CH3 - FCBR(TP5) | - 1D |
| CH4 - OIC(TP9) | - 1D |
| CH5 - IFB(TP7) | - 1C |
| CH6 - SPD(TP10) | - 1E2 |

2. MODIFY the following EEPROM variables:

| | | |
|---|---|---|
| SRGAIN | to | 1.0 |
| SRTC | to | 1.0 |
| SUSTCM | to | .25 |
| RRGMAX | to | .5 |
| RRGMIN | to | .15 |
| CRMIN | to | .09 |

ASSIGN LSTAT(MODE) to D/A #1
(scale = 0)
(offset= 0)

### 6-17.2. Start and Run Motor

1. Check that auxiliary and control power is applied.
2. Clear all alarms and faults.
3. Close source circuit interrupter and rack-in load circuit interrupter (bypass remains racked-out).
4. Select local speed reference and set it to minimum.
5. Initiate run.
   Monitoring LSTAT(MODE) changes on DA1-2E: starts from IDLE mode 0 Vdc when run initiated
   switches to mode FIX FIRE (open loop) 1.25 Vdc
   at approx. 5Hz

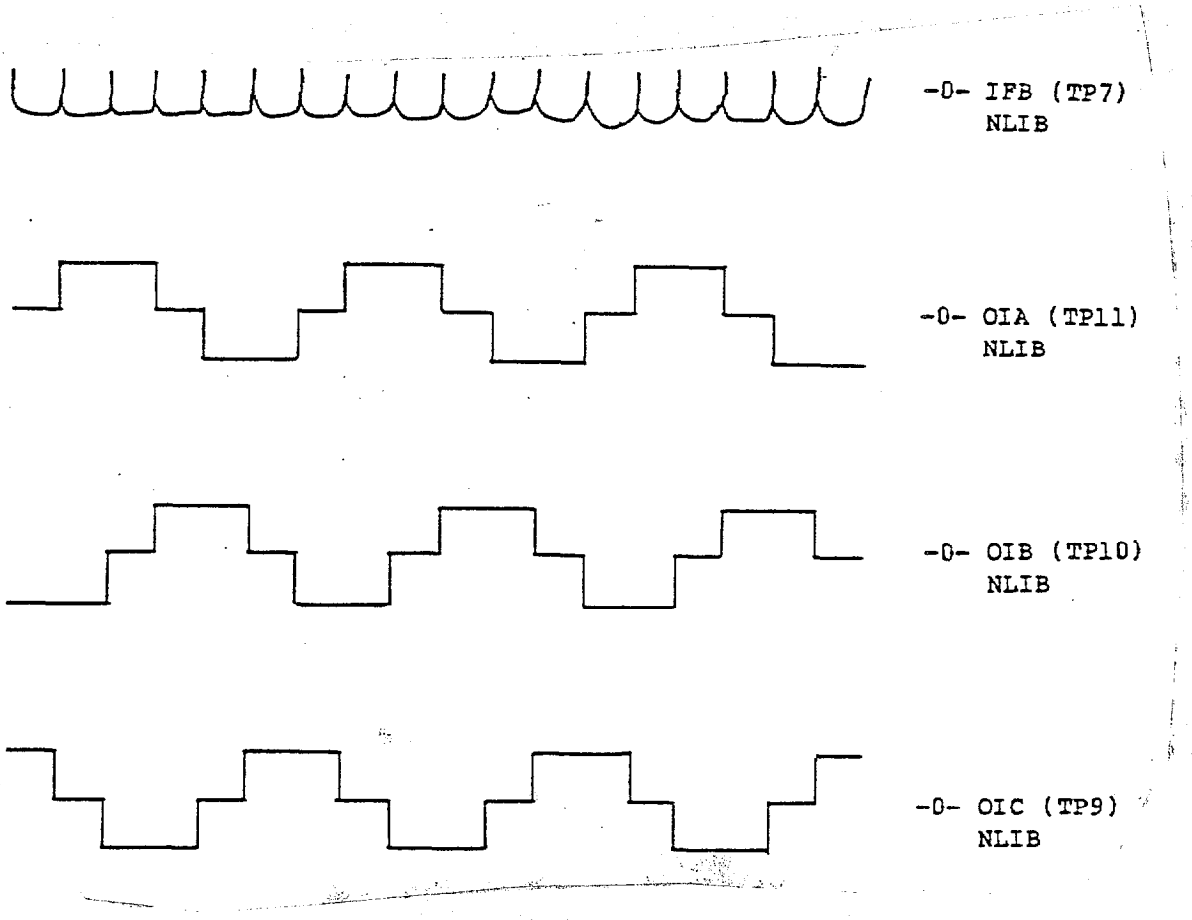



*Figure 6-4.*

switches to NO HARMONIC ELIMINATION W/DCLC mode 6.3 Vdc
at approx. 15Hz
switches to ELIMINATE 5TH and 7TH mode 3.75 Vdc
at approx. 25 Hz.
switches to ELIMINATE 5TH mode 5.0 Vdc
at approx. 30Hz
switches to NO HARMONIC ELIMINATION mode 6.3 Vdc
at approx. 33Hz
switches to SELF COMMUTATION mode 7.5 Vdc

6. If motor fails to start, then decrease SUSFRQ by steps of 0.5 (minimum of 1) until motor starts, and/or increase SUSTCM.
7. Check motor rotation. If motor rotates backward, check IMD to motor wiring; swap two motor leads if necessary.
8. Initiate stop and motor should coast stop.

**6-17.3. Speed Range Test.** Start drive and gradually increase the drive speed to 50% speed . Check magnetic center, vibration, and bearing temperature rise. Verify panel speed meter reads 1/2 rated rpm.

**6-17.4. Speed Ramp Test.** While monitoring speed with the brush recorder, decelerate the drive as fast as possible by changing the speed reference to minimum as quickly as possible.
Then check acceleration by increasing reference as fast as possible.

**6-17.5. Remote Operation**

. Place drive control in remote and verify drive follows remote reference and remote stop/start commands.

## 6-18. OPERATE WITH MOTOR COUPLED

**6-18.1. Test Connections.** Make the following connections to a six channel chart recorder.

| | |
|---|---|
| CH2 - DA1 | TP2-2E |
| CH2 - 0IA | TP11-1D |
| CH3 - FCBR | TP5-1D |
| CH4 - 0IC | TP9-1D |
| CH5 - IFB | TP7-1C |
| CH6 - SPD | TP10-1E |

Assign ALPHLD to D/A # 1
(Scale = 2)
(Offset = 0)

**6-18.2. Starting Tune-up.** The start-up of the IMD should be optimized by varying EEPROM variables SUSFREQ and SUSTCM.

The procedure is as follows:

1. Attempt to start drive and monitor chart recorder.
2. If you fail to get load flux, start decreasing SUSFREQ until system starts(starts getting flux) (to minimum .5).
3. If IMD still fails to start, return SUSFREQ to original value, increase SUSTCM, and repeat step 2.

**6-18.3. Motor Flux Check**

1. Start the drive and run up to some speed between 50% and 75%. Connect oscilloscope to testpoint "VAC" (TP7-2B). The waveform should appear as shown in Figure 6-5:
   Use DVM to measure $V_{AC}$ RMS

   $$f = \frac{1}{T}$$

   Calculate motor volts/Hz (flux).

   $$\overline{V/Hz} = \frac{(\text{attn ratio})(V_{AC}\text{RMS})/f}{(\text{rated motor volts})/60\text{Hz}}$$

   The flux integrators on 1D (TP1, TP2 and TP3) should then have RMS voltages of:

   $$Vf = (\overline{V/Hz}) * 4.75 = _____ V_{rms}$$

2. Check for accurate reconstruction. See Figure 6-4 for illustration.

**6-18.4. Capacitor Current**

1. Measure NSFC(2B)- $V_{AC}$(TP9) and drive frequency(freq)

2. Calculate capacitor current $I_{cap}$

   $$I_{cap} = \frac{V_{LN}}{X_C} = \frac{V_{AC} * \text{ATTNRATIO} * 2 * \text{PI} * \text{FREQ} * C}{1.732}$$

   where ATTNRATIO and C are given on system elementary sheet 1EA.

3. MODIFY EEPROM variable DRMC so that RAM variable ICAP

   $$\text{ICAP} = \frac{I_{CAP} * 1024}{\text{Rated Drive Current}} (\pm 5\%)$$

4. Save DRMC if changed.

**6-18.5. Current Regulator.** The current regulator for the IMD drive is calculated for a 150 (±25) radian response (Check system elementary sheet 1EP for desired responses.) The parameters involved in this calculation are fairly well known and measurement of the response is impractical.
The stability of the regulator can be checked by the following procedure:

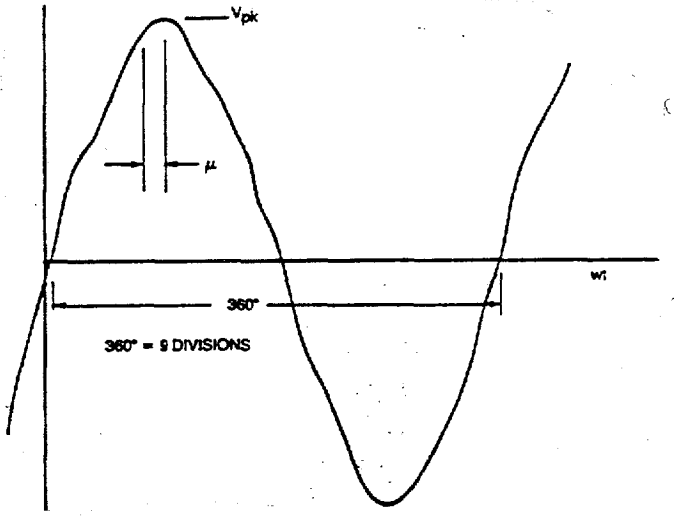



*Figure 6-5. Waveform for Motor Flux Check (Motor Volts)*

Run drive to the middle of the operating (between the speed limits) range. The current, IFB-1C, should be greater than 25% (1.25 Vdc).

*** NOTE ***

**In "TUN CR" DA1 has been temporarily assigned to current feedback (ISRC, divided by 8) so that the step reponse can be examined. DA2 has been temporarily assigned to CURCMP (divided by 64) so that feedforward can be adjusted.**

**6-18.6. Check Feedforward (VDCGAIN).** Using the monitor commands - MODIFY (see example 6-3), MODIFY CRVDC till the output of DA2-HLCB (TP2-2E) is approximately zero (but positive).

SAVE CRVDC if changed.

Apply a 2-5% step for 0.1-0.5 seconds to the current gulator using the monitor commands - MODIFY REFERENCE, APPLY, and REMOVE (TUNEUP MODE - CURRENT REGULATOR). See Example 6-3.

The brush recorder should be running at its fastest speed.

Verify IFB-1C does not overshoot or oscillate on both the up-step to a higher current and speed, and the return step to the lower current and speed.

*>**TUN** <cr>
T >**TUN CR** <cr>

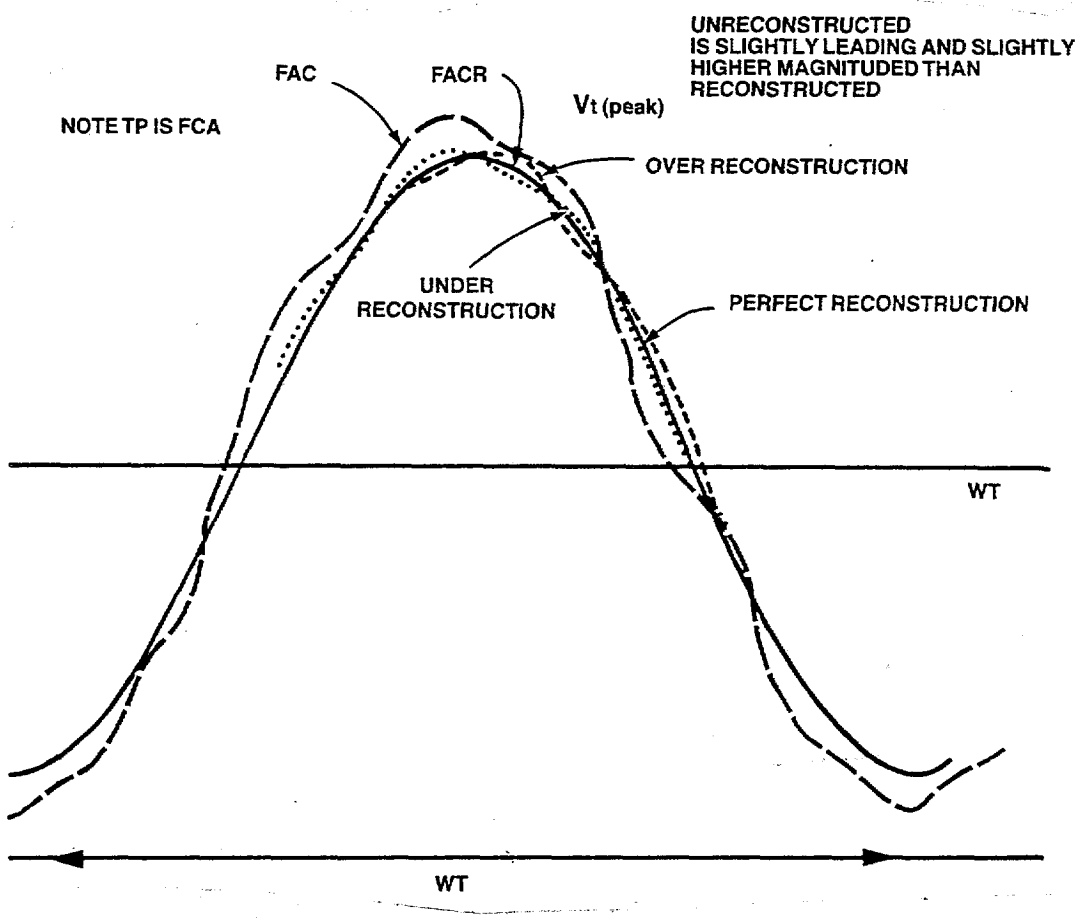



*Figure 65A. The Voltage at Testpoints FBAR, FCBR, and FACR should appear as follows: It is important that these be balanced as closely as possible.*

Available Variables are:
(List of Variables)

T >**MOD REF <cr>**

| Step | Level(PU) | Time(sec) |
|---|---|---|
| 1 | | |
| 2 | | |
| Step 1. | Level. = | **0.02 <cr>** |
| Step 1 | Time = | **.1 <cr>** |
| Step 2 | Level = | **0 <cr>** |
| Step 2 | Time = | **.1 <cr>** |

T > **APPLY <cr>** ; take brush recording of up +down response

T > **REMOVE <cr>** ;remove step function

EXAMPLE 6-3

**6-18.7. Flux Regulator.** The response of the flux regulator should be critically damped at 20 (+/-2) radian. (Check system elementary sheet EP for desired response).

Connect TP1(DA1) -2E and TP7(IFB) -1C to chart recorder. Run drive to the middle of the operating (between the speed limits) range. The current, IFB-1C, should be greater than 25% (1.25 Vdc).

In "TUN FR" DA1 has been temporarily assigned to flux error (FLXERR, divided by 32) so that the step response can be examined. DA2 has been temporarily assigned to FLXPI (divided by 32) so that the flux regulator feedforward can be adjusted.

**6-18.8. Checking Flux Feedforward (XMGAP).** Using the monitor, in the TUNEUP MODE - FLUX REGULATOR case and MODIFY (see example 6-4), MODIFY FRXM till the output of DA2-HLCB (TP2-2E) is approximately zero (but positive).

SAVE FRXM if changed.

a. Apply a 5-10% step for 3-5 seconds to the flux regulator using the monitor commands - MODIFY REFERENCE, APPLY, and REMOVE (TUNEUP MODE - FLUX REGULATOR) . See Example 6-4.
b. Observe the response on the chart recorder. Response should be critically damped and non-oscillating. Measure and record response.
c. If the response is too fast (or to slow), decrease (increase) EEPROM variable FRTC until the nearest desired response is obtained.
d. If the response is overdamped or underdamped, increase or decrease FRGAIN to obtain critical damping. If damping is now correct but response is not, return to step c.
e. SAVE EEPROM variables if changed.
f. Verify the NSFC(2B)-VDC(-) TP11) for sawtooth (6Xload frequency) waveform.

*TUN FR <cr>
T> TUN <cr>

Available Variables Are:
(List of Variables)

T> **MOD REF <cr>** ;MODIFY FRXM as needed

| Step | Level(PU) | Time(sec) |
|---|---|---|
| 1 | | |
| 2 | | |
| Step 1. | Level(pu) = | **0.02 <cr>** |
| Step 1 | Time = | **5 <cr>** |
| Step 2 | Level = | **0 <cr>** |
| Step 2 | Time = | **5 <cr>** |

t> **APPLY <cr>** :Take brush of up + down response

T> **REMOVE <cr>** ;Remove step function

EXAMPLE 6-4

**6-18.9. Speed Regulator.** Run the drive speed up to middle of operating speed range. IFB-1C should be greater than 1.25 Vdc. MODIFY DRVCFG to disable torque vs speed limit.

Verify that CRSTIE = 0. Apply a step size (2-5% forr 5 seconds) to obtain as large a step as possible without hitting current limit (observe IFB on chart recorder). (Using TUNEUP mode-SPDREG: commands MODIFY REFERENCE, APPLY, REMOVE) See Example 6-5.

*** NOTE ***

**It may be necessary to buck-out the speed signal with a battery voltage to obtain enough amplification. If step down response is different from step, use step down response.**

Observe the step response and adjust SRGAIN and SRTC for a 1 radian critically damped response as shown in Figure 6-7. (Check system elementary sheet EP for desired response). It should only be necessary to modify SRGAIN at all times.

Save the values for SRGAIN, SRTC, into EEPROM. Record

*** NOTE ***

**DA1 has been temporarily assigned to speed error, so that the step response could be examined. DA2 has been temporarily assigned to load firing angle (ALPHLD), divided by 8.**

List of Variables are:
*> **TUN <cr>**
T> **TUN SR <cr>**

T> **MOD REF <cr>**

| Step | Level(pu) | Time(sec) |
|---|---|---|
| 1 | | |
| 2 | | |
| Step 1 | Level(pu)= | **.05 <cr>** |
| Step 1 | Time = | **2 <cr>** |
| Step 2 | Level = | **0 <cr>** |
| Step 2 | Time = | **2 <cr>** |

t> **APPLY <cr>** :Take brush of up + down response
T> **REMOVE <cr>** ;Remove step function

EXAMPLE 6-5

**6-18.10. Source-side Line Impedance.** Normally the source-side operates in the rectifying quadrant. However, when the drive is forced to slow down, the source converter may have to regenerate (invert). To avoid commutation failures, the angle of retard must be programmed as a function of current. Since a regenerating condition is difficult to hold steady-state, a measurement of system impedance will be made and this value will be programmed into EEPROM.

Run the drive to some convenient speed (the higher the better). Connect channel 1 of the oscilloscope to testpoint "VAC"-2C. Connect channel 2 to testpoint "0IA"-1C. The signal should be similar to those shown in Figure 6-10:

$\overline{VI\text{-}1}$ = (attn ratio)*(Vpk/1.414)/Vnom where Vnom = transformer nominal voltage

$\overline{Id}$ = I/5

$\overline{Xc}$ = ($\overline{VI\text{-}1}$ $\overline{Id}$) [cos $\gamma$ - cos ($\gamma$ + $\mu$)]

CRLIMP = ($\overline{Xc}$)

MODIFY and SAVE into EEPROM.
Take drive to top spd, then decelerate as fast as possible. If drive trips on overcurrent then increase CRLIMP.

**6-18.11. Horsepower Meter**

1. Run drive to middle of operaing range and measure the drive horsepower (using chapter 6 - Section 6 - APPENDIX -A.2.II and B.3)
2. Adjust NLCB(1E) -HPM so that horsepower meter reads measured horsepower.

**6-18.12. Ammeter Check.**

1. Measure motor current (IMM) by means of clamp-on ammeter at motor lead if possible and check that MIFB(TP11) -NCCC(IF) =

$$\frac{\text{IMM} * 3.887}{\text{Rated Inverter Current}}$$

where rated inverter current is listed on system elementary sheet 1EP

*** NOTE ***

**Individual phase (motor) current is scaled 3.322 Vacrms on inverter base.**

2. Adjust GAIN-NLIB(1D) so that ammeter reads correct motor amps

**6-18.13. Speedmeter**

1. Run motor to full speed and measure motor rpm and speed reference input variable SPD_STR.

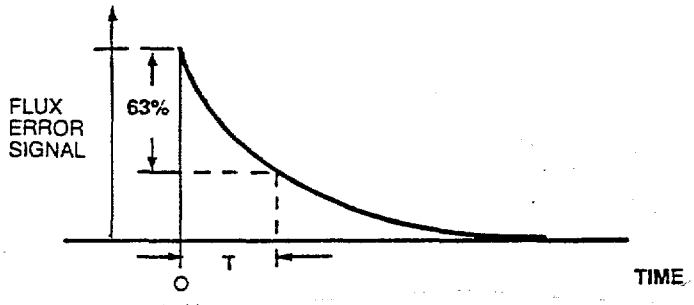



*Figure 6-6. Flux Error Signal*

2. MODIFY EEPROM variable SRSL so that motor rpm = requested rpm

$$RPM = \frac{SPD_STR}{1024}* \text{ (Rated RPM @ 60 Hz)}$$

3. Adjust gain-NLIB(IC) so that speedmeter reads correct motor rpm

**6-18.14. Re-start on fly .** It is possible to catch the motor while it is still spinning and to return to requested speed. Monitor the following on a brush recorder:

1. SPD (TP1)-1E
2. FCBR (TP5)-1D
3. IFB (TP7)-1C
4. Run drive to full speed, stop and immediately re-start (before flux dies) drive should 'catch' and control the motor
5. Run drive to full speed stop, wait 15 secs and restart (flux will be zero and speed signal will be zero). After 8-15 sec spd signal will go to ≈ 9-10 volts then ramp down. As speed ramps down flux will peak. After flux peaks drive should be "caught" if not, decrease SCFLUX by .1 (min of 0.10)

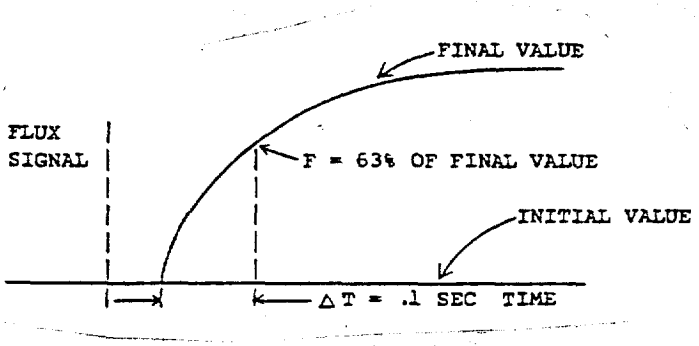



*Figure 6-7.*

## ADDITIONAL INFORMATION POWER READINGS

A. PCM input power factor and power readings (per channel) use Figure 6-10.

1. Power factor (PF)

   a. Measure μ
   b. Measure γ

$$\text{Power factor} = \frac{\cos(\gamma - \mu) + \cos\gamma}{2}$$

2.I. Kilowatts (KW)

   a. Measure power factor (PF).
   b. Measure OIA (peak average) (OIA).
   c. Measure peak voltage of VAC (Vacpk).

$$\text{Kilowatts} = \frac{\text{Vacpk} \bullet \text{attn ratio}}{1000 \bullet 2} * \frac{\text{OIA}}{5\text{V/unit}}$$

* rated bridge ac current • PF.

2.II Alternative to 2 (eliminates P.F reading)

   a. Measure -Vdc
   b. Measure IFB

$$\text{Kilowatts} = \frac{\text{IFB}}{5\text{V/volt}} \bullet \text{rated bridge ac current}$$

$$\bullet\ 1.2825 * \frac{\text{VDC} \bullet \text{ATTN RATIO}}{1000}$$

B. Motor power factor and power readings - use Figure 6-9.

1. Power factor (PF)
   With an oscilloscope, observe the potential transformer (PT) signal -VA and the current transformer (CT) signal - IA (with a current probe).
   Measure the power factor
   PF = cos (PF°)
   Kilowatts (KW)
   a. Measure power factor (PF)
   b. Measure voltage peak ($V_{pk}$)
   c. Measure current peak - ($I_{pk}$)

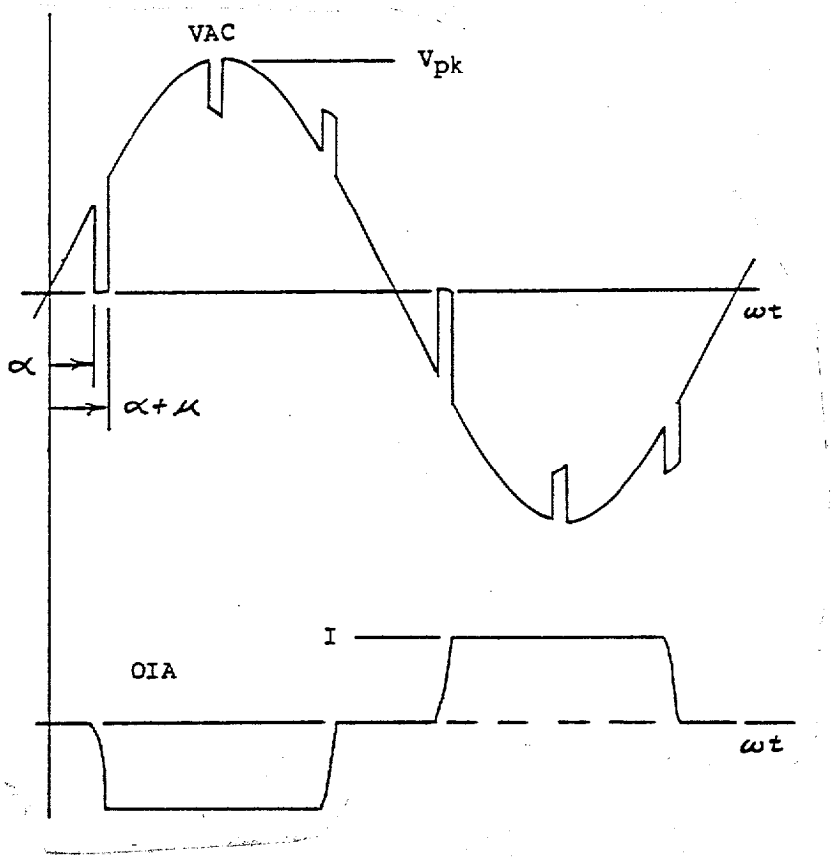



*Figure 6-8.*

$$\text{Kilowatts} = \left[\frac{I_{pk}}{\sqrt{2}} \bullet \text{CT Ratio}\right] \bullet \left[\frac{V_{pk}}{\sqrt{2}} \bullet \text{PT ratio}\right] \bullet \left[\frac{1}{1000}\right] \bullet \text{PF}$$

2. Horsepower (HP)

   a. Measure Kilowatts (KW)
   Horsepower = • KW • Eff./7457
   [Eff - efficiency can be estimated at 0.94 for large motors (see motor data sheets for accurate data)].

3. Torque ( τ )

   a. Measure horsepower (HP)
   b. Measure SPD (RPM)
   Put oscilloscope's time base on calibration for this entire measurement. Use as much as possible at the screen to display a half-cycle. Measure the time of this half cycle (T).
   SPD = [(rated frequency) . rated RPM at rated frequency)/(2-t)]

$$\text{Torque} = \frac{5256 \bullet \text{HP}}{\text{SPD}}$$

## ADDITIONAL INFORMATION FOR ALTERNATE AMMETER CHECKS

Alternate method to calculate motor current.

a. Calculate capacitor current
   ICAP = Vac * Att. ratio * 2 * PI * freq * C/1.732
b. Calculate drive current
   ID = (IFB * Drive rated current)/(5)
c. Calculate load firing angle
   use display command to display Variable FIREANGCL

Firing angle = FIREANGCL * .3516 degree
Where Vac is measured at TP 'Vac'-2B
Frequency is the motor stator frequency
IFB is measured at TP 'IFB' 1C
Attn. ratio is given on sheet EP

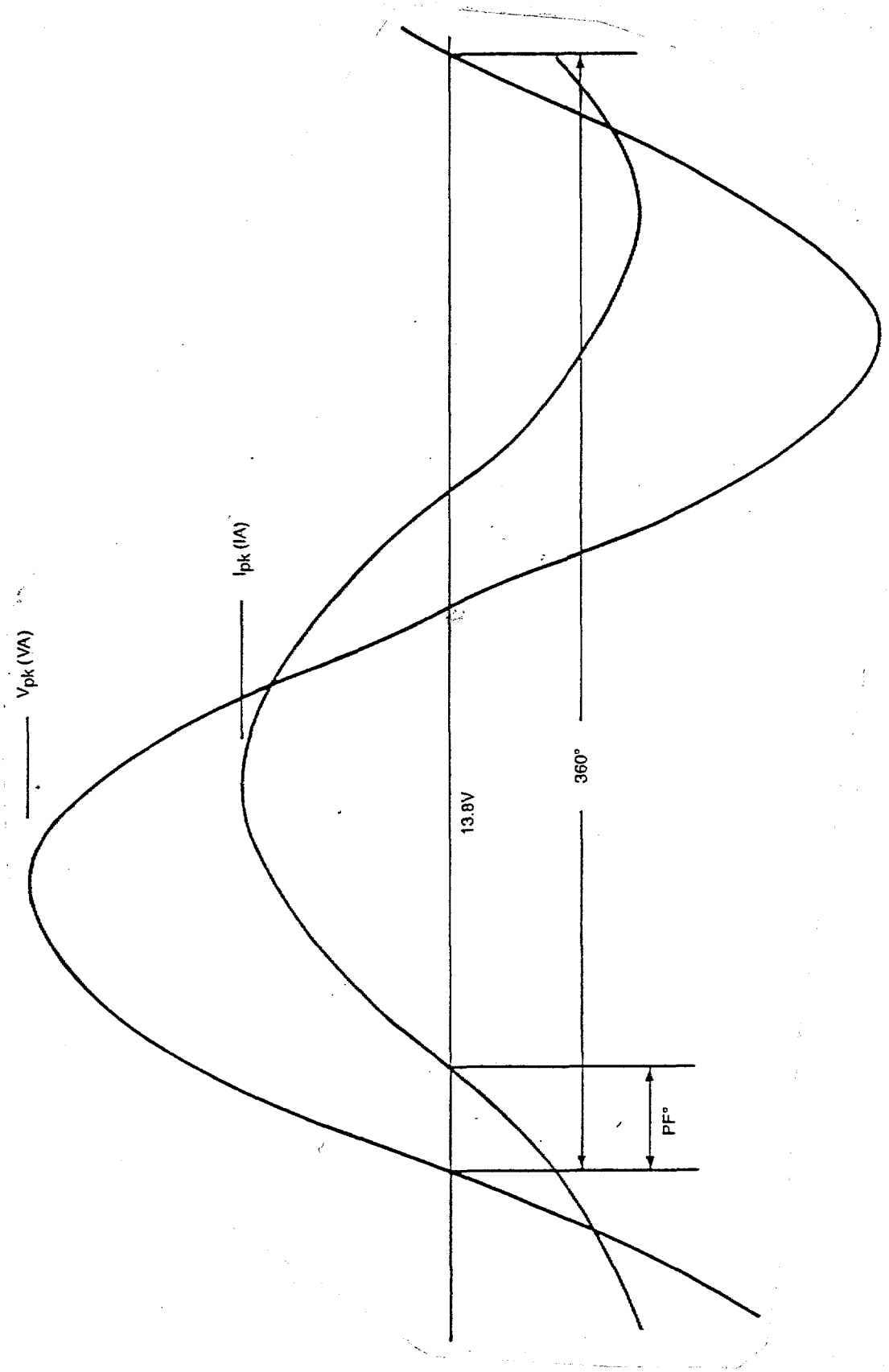



*Figure 6-9.*

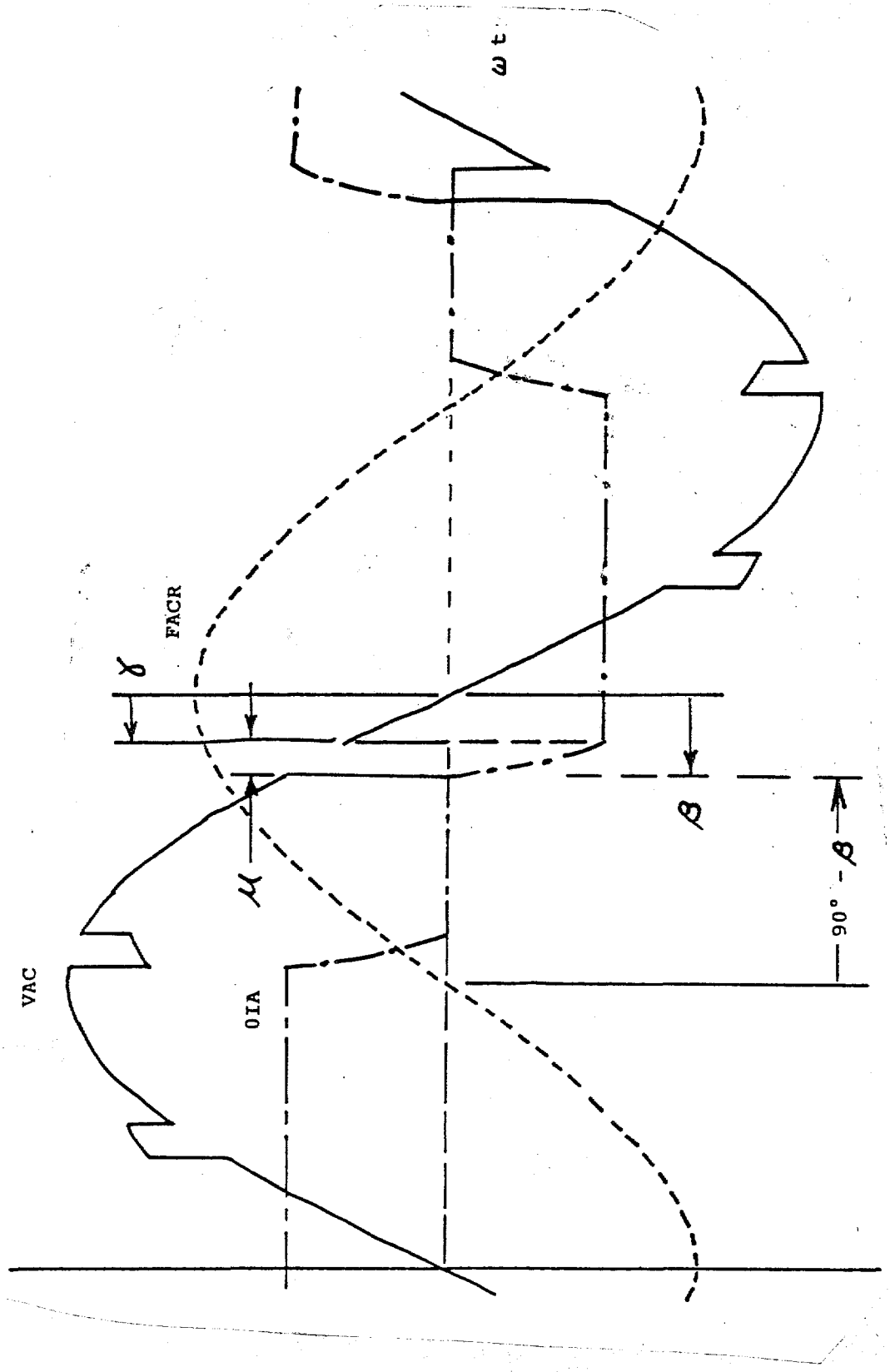



*Figure 6-10*

C is given on sheet EP
Drive rated current is given on sheet EP

Calculate motor current

$$IM = [ID * Cos \propto)^2 + (ID * SIN \propto -ICAP)^2]^{1/2}$$

Where $\propto$ is the firing angle in degree

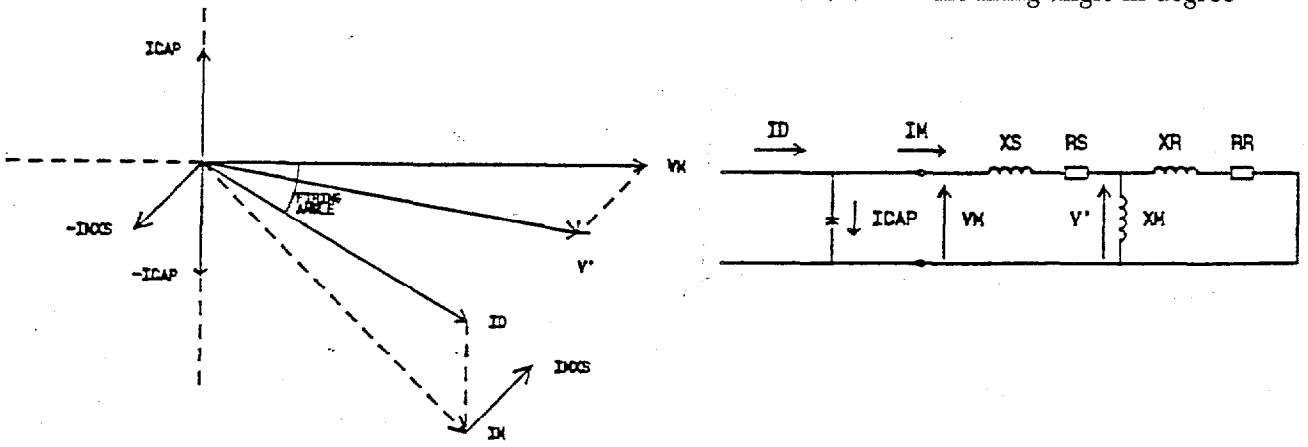



*Figure 6-11. Per Phase Equivalent Circuit*

# CHAPTER 7

# DIAGNOSTICS AND TROUBLESHOOTING

## 7-1. INTRODUCTION

Included in the drive power and electronic circuitry is a multi-layered diagnostic and troubleshooting capability. The main elements of this capability are the following:

1. Self testing of critical microcomputer system elements every time the control is reset or power applied.
2. A memory type fault monitor which displays the cause of shutdown in a printed message at the printer.
3. Lights and test points on front of cards provide the following information:
   a. Attenuated system bus voltages and currents on test points allow a check of magnitude and phasing.
   b. Gate pulse signals to converter bridge available at test points.
   c. Neon cell voltage indicating lights across each thyristor in power converters for indication of shorted cells.
   d. Overcurrent, overvoltage, and overspeed indicating lights in electronics module.
   e. Card front lights indicate status of all switching inputs and outputs to/from electronics module.
4. Gate-test-mode allows checkout of analog signal conditioning circuitry, microprocessor firing control, and cell gate signal integrity while off-line.
5. Serial monitor software which allows examination of system memory, inputs, and outputs-.(Chapter 5)
6. Capability to store diagnostic lists of system variables preceding and following fault conditions.(Chapter 4)

## 7-2. DIAGNOSTICS INITIALIZATION SELF TEST

SECTION A - SELFTEST

## 7-3. BASIC STRUCTURE

The Selftest routines are PLM86 procedures which may be executed by the HMPG resident in the Structured control Small Electronics Module (SEM) to test individual PWB's and functions. These routines are executed upon power-up if the TEST switch on the NLCB is in the up position. These routines may also be run when the drive is in a non-RUN condition by operator commands received via a terminal connected to the HMPG's RS-232 port or by depressing a test switch. The function of the Selftest is to test most of the control panel for the Structured DCLC-IMD, with great emphasis placed on the testing of the PWB's in the SEM.

Testing of the panel begins by performing tests of the hardware elements within a neighborhood of the HMPG's microprocessor and continues to tests of increasingly remote hardware. All of the tests incorporated within the selftest may be classed into 3 groups:
1. PWB-Specific tests
2. Paired PWB tests
3. Panel-level tests

## 7-4. PWB-SPECIFIC TESTS:

A great amount of effort has been placed on performing tests that will pinpoint problems on specific PWB's first. In this manner, if any easily diagnosed problem exists, it will be clearly spelled out to the operator, instead of vaguely defined by a list of possible solutions. Tests such as these comprise the PWB-Specific test group.

## 7-5. PAIRED PWB TESTS:

Tests which depend on the operation of several PWB's are classed within the Paired-PWB group. Diagnostic messages for failures of these tests will indicate a functional problem and also a list of possible solutions which may be general in nature.

| *CAUTION* |
|---|

***Selftest excercises the output relays; therefore, the output terminal boards should be disconnected from the I/O module, and/or source and load breakers should be racked-out and drive stopped (if 12-pulse).***

## 7-6. PANEL-LEVEL TESTS:

Panel-level tests also require correct operation of several PWB's or modules, however, these tests are segregated to allow the user to perform either a module-level test (involving only the control SEM) or a complete panel-level test which would include this last catagory of procedures. From a practical point of view, the sequence of testing within the control SEM progresses in the following manner:

1. HMPG
2. HLCB
3. HAIA
4. NSFC (source)
5. NSFC (load)
6. NSFC(DC LINK)
7. NLCB
8. NLIB (source)
9. NLIB (load)
10. HISA
11. HRDA

Further tests which include PWB's outside the SEM, progress in the following sequence:

1. Cables leading from the SEM.
2. the I/O Module.

## 7-7. EXECUTING THE TESTS

The self test routine may be initiated by resetting (toggle TP1 (RESET) to TP2 (DCOM) the HMPG with the TEST switch (on the NLCB) in the UP position. The entire test is completed in approx 20 seconds. At its completion, status information will be printed out at the door-mounted printer. In the present product, the entire self test routine is executed; however, provisions for executing small sections of the code are in place.

Upon completion of the self-test, the drive will operate in the gate-test-mode. To return to normal operation, the NLCB's test switch must be returned to the normal position and the HMPG must be reset (Toggle TP1 (RESET) to TP2 (DCOM)).

*CAUTION*

***Selftest excercises the output relays; therefore, the output terminal boards should be disconnected from the I/O module, and/or source and load breakers should be racked-out and drive stopped (if 12-pulse).***

## 7-8. DIAGNOSING FAILURES

As mentioned above, failure of any of the selftest procedures will produce a diagnostic message on the door-mounted printer. The format of the diagnostic message is:

| DATE TIME | |
|---|---|
| FAILED | |
| (aabb) DURING TEST OF: | aa=TEST NUMBER |
| description of test | bb=SUB-TEST NUMBER |
| PROBLEM: | |
| problem statement | |
| POSSIBLE SOLUTIONS | |
| solution 1 | |
| solution 2 | |
| solution 3 | |

Usually, a failure can easily be diagnosed by the problem statement. At least one solution will be suggested in the print out. It is suggested that troubleshooting be performed at the PWB level (Replace the PWB not its component). Although the diagnostic messages may indicate more specific failures, it is not recommended that PWB components be replaced or repaired (always write or attach the diagnosis to the card)

The TEST NUMBER and SUB-TEST NUMBER uniquely define a failure within the Selftest procedure. The TEST NUMBER is incremented from 1 to 31 (hex) as the tests progress. For different test numbers, the SUB-TEST parameter will vary from 1 up to the number of sub-tests within that major test. The Description of Tests section that follows details the complete Selftest procedure according to the TEST NUMBER and SUB-TEST NUMBER parameters.

**7-8.1. 6 DESCRIPTION OF TESTS.** The following lists all tests available with sub-test information:

| TEST | DESCRIPTION |
|---|---|
| 1 | HMPGROMTST: Reads all of ROM memory (F000:0 to F000:FFFF). Possible failures are: |
| .0 | BusHog error |
| .1 | - Read error |
| 2 | HMPGRAMTST: Reads all of RAM memory (0:0 to 0:3FFF).Possible failures are: |
| .0 | - BusHog error |
| .1 | - Read error |
| .2 | - Write error |
| 3 | HMPGINTRTST |
| .1 | - Verifies control and data lines to HMPG's 8259 |
| 4 | HMPG_TMR |
| .1 | - Verifies addr lines & data bits 6 & 7 |
| .2 | - Verifies addr lines & data bits 6 & 7 |
| .3 | - Verifies addr lines & data bits 6 & 7 |
| .4 | - Verifies data bits 5-0 with walking 1 pattern |
| .5 | - Verifies that all outputs are free to toggle. |
| .6 | - FOR TMR0: Checks for null count = 1 |
| .7 | - FOR TMR0: Loads TMR; checks for null count = 0 |
| .8 | - FOR TMR0: Verify TMR time out |
| .9 | - FOR TMR0: Verify TMR reloads |
| .A | - FOR TMR1: Checks for null count = 1 |
| .B | - FOR TMR1: Loads TMR; checks for null count = 0 |
| .C | - FOR TMR1: Verify TMR time out |
| .D | - FOR TMR1: Verify TMR reloads |
| .E | - FOR TMR2: Checks for null count = 1 |
| .F | - FOR TMR2: Loads TMR; checks for null count = 0 |
| .10 | - FOR TMR2: Verify TMR time out |
| .11 | - FOR TMR2: Verify TMR reloads |
| 4 | HMPG_SIO_PRES |
| .1 | - Check that trans and rec buffers are empty |
| 5 | HMPG_STALL |
| .1 | - Verifies software reset of Stall circuit. |
| .2 | - Initiate stall timer; checks for no STALL after 100 |
| .3 | - Verifies STALL asserted after 1.5 seconds. |
| 7 | HMPG_INTR |
| .1 | - Verify operation of BusHog interrupt. |
| .2 | - Verify that BusHog doesn't occur for legitimate read. |
| .3 | - Verify operation of Time Tic interrupt. |
| 8 | DMPB |
| .1 | - Checks that Write Enable Switch is not asserted. |
| .2 | - Checks for no BusHog during EEPROM reads. |
| .3 | - Checks for expected pattern in EEPROM. |
| 9 | HLCB_PRES |
| .1 | - Enables IMOK; checks for 0IMOK=0. |
| .2 | - Disables IMOK; checks for 0IMOK=1. |
| 9 | HLCB_DAK |
| .1 | - Checks for no BusHog during reads of HLCB. |
| A | HLCB_TMR_U22 |
| .1 | - Verifies addr lines & data bits 6 & 7 |
| .2 | - Verifies addr lines & data bits 6 & 7 |
| .3 | - Verifies addr lines & data bits 6 & 7 |
| .4 | - Verifies data bits 5-0 with walking 1 pattern |
| .5 | - Verifies that all outputs are free to toggle. |
| .6 | - FOR TMR0: Checks for null count = 1 |
| .7 | - FOR TMR0: Loads TMR; checks for null count = 0 |
| .8 | - FOR TMR0: Verify TMR time out |
| .9 | - FOR TMR0: Verify TMR reloads |
| .A | - FOR TMR1: Checks for null count = 1 |
| .B | - FOR TMR1: Loads TMR; checks for null count = 0 |
| .C | - FOR TMR1: Verify TMR time out |
| .D | - FOR TMR1: Verify TMR reloads |
| .E | - FOR TMR2: Checks for null count = 1 |
| .F | - FOR TMR2: Loads TMR; checks for null count = 0 |
| .10 | - FOR TMR2: Verify TMR time out |
| .11 | - FOR TMR2: Verify TMR reloads |

| TEST | DESCRIPTION |
|---|---|
| B | HLCB_TMRU23 |
| .1 | - Verifies addr lines & data bits 6 & 7 |
| .2 | - Verifies addr lines & data bits 6 & 7 |
| .3 | - Verifies addr lines & data bits 6 & 7 |
| .4 | - Verifies data bits 5-0 with walking 1 pattern |
| .5 | - Verifies that all outputs are free to toggle. |
| .6 | - FOR TMR0: Checks for null count = 1 |
| .7 | - FOR TMR0: Loads TMR; checks for null count = 0 |
| .8 | - FOR TMR0: Verify TMR time out |
| .9 | - FOR TMR0: Verify TMR reloads |
| .A | - FOR TMR1: Checks for null count = 1 |
| .B | - FOR TMR1: Loads TMR; checks for null count = 0 |
| .C | - FOR TMR1: Verify TMR time out |
| .D | - FOR TMR1: Verify TMR reloads |
| .E | - FOR TMR2: Checks for null count = 1 |
| .F | - FOR TMR2: Loads TMR; checks for null count = 0 |
| .10 | - FOR TMR2: Verify TMR time out |
| .11 | - FOR TMR2: Verify TMR reloads |
| .12 | - FOR TMR2: Initialize to Mode 0; verify status. |
| .13 | - FOR TMR2: Load with 10 ms delay; verify status. |
| .14 | - FOR TMR2: After 4 ms, verify status. |
| .15 | - FOR TMR2: After 15 ms, verify time out. |
| .16 | - FOR TMR2: Check that count = 0. |
| C | HLCB_INTR_PRE |
| .1 | - Verify addr and data lines to HLCB's 8259. |
| D | HLCB_SIO |
| .1 | - Initialize and check status of HLCB's 8251. |
| .2 | - Verify that single byte of data is send, by sensing status. |
| .3 | - Verify transfer of data pattern. |
| E | HAIA_PRES |
| .1 | - Checks that HAIA is present by reading single bit. |
| F | HAIA_DAK |
| .1 | - Checks for no BusHog during reads of HAIA. |
| .2 | - Checks for no BusHog during writes to HAIA. |
| 10 | HAIA_TMR |
| .1 | - Verifies addr lines & data bits 6 & 7 |
| .2 | - Verifies addr lines & data bits 6 & 7 |
| .3 | - Verifies addr lines & data bits 6 & 7 |
| .4 | - Verifies data bits 5-0 with walking 1 pattern |
| .5 | - Verifies that all outputs are free to toggle. |
| .6 | - FOR TMR0: Checks for null count = 1 |
| .7 | - FOR TMR0: Loads TMR; checks for null count = 0 |
| .8 | - FOR TMR0: Verify TMR time out |
| .9 | - FOR TMR0: Verify TMR reloads |
| .A | - FOR TMR1: Checks for null count = 1 |
| .B | - FOR TMR1: Loads TMR; checks for null count = 0 |
| .C | - FOR TMR1: Verify TMR time out |
| .D | - FOR TMR1: Verify TMR reloads |
| .E | - FOR TMR2: Checks for null count = 1 |
| .F | - FOR TMR2: Loads TMR; checks for null count = 0 |
| .10 | - FOR TMR2: Verify TMR time out |
| .11 | - FOR TMR2: Verify TMR reloads |
| 11 | HAIA_CONVERSION |
| .1 | - Start A/D conversion; check that BUSY=1. |
| .2 | - Check that 1DAVAL = 0. |
| .3 | - Wait for conversion to complete; check that BUSY=0. |
| .4 | - Check that 1DAVAL = 1. |
| 12 | HAIA_PSUPPLY |
| .1 | - Check that N15 supply is read correctly. |
| .2 | - Check that P15 supply is read correctly. |
| 13 | NSFC_S_PRES |
| .1 | - Set IMOK true; read back IMOK signal. |
| .2 | - Set IMOK false; read back IMOK signal. |
| 14 | NSFC_S_TMR_PRES |
| .1 | - Verifies addr lines & data bits 6 & 7 |
| .2 | - Verifies addr lines & data bits 6 & 7 |

| TEST | DESCRIPTION |
|---|---|
| .3 | - Verifies addr lines & data bits 6 & 7 |
| .4 | - Verifies data bits 5-0 with walking 1 pattern |
| .5 | - Verifies that all outputs are free to toggle. |
| 15 | NSFC_S_TMODE |
| .1 | - FOR TMR0: Checks for null count = 1 |
| .2 | - FOR TMR0: Loads TMR; checks for null count = 0 |
| .3 | - FOR TMR0: Verify TMR time out |
| .4 | - FOR TMR0: Verify TMR reloads |
| .5 | - FOR TMR1: Checks for null count = 1 |
| .6 | - FOR TMR1: Loads TMR; checks for null count = 0 |
| .7 | - FOR TMR1: Verify TMR time out |
| .8 | - FOR TMR1: Verify TMR reloads |
| .9 | - Perform operations to set pulse train to 0; verify. |
| .A | - In TEST Mode, write all on pattern, expect echo of same |
| .B | - In TEST Mode, walk 1 thru primary & firing latch. |
| .C | - In TEST Mode, see if pulse train is stuck low. |
| .D | - In TEST Mode, see if pulse train is stuck high. |
| .E | - In TEST Mode, check that pulse train is toggling. |
| 16 | NSFC_L_PRES |
| .1 | - Set IMOK true; read back IMOK signal. |
| .2 | - Set IMOK false; read back IMOK signal. |
| 17 | NSFC_L_TMR_PRES |
| .1 | - Verifies addr lines & data bits 6 & 7 |
| .2 | - Verifies addr lines & data bits 6 & 7 |
| .3 | - Verifies addr lines & data bits 6 & 7 |
| .4 | - Verifies data bits 5-0 with walking 1 pattern |
| .5 | - Verifies that all outputs are free to toggle. |
| 18 | NSFC_L_TMODE |
| .1 | - FOR TMR0: Checks for null count = 1 |
| .2 | - FOR TMR0: Loads TMR; checks for null count = 0 |
| .3 | - FOR TMR0: Verify TMR time out |
| .4 | - FOR TMR0: Verify TMR reloads |
| .5 | - FOR TMR1: Checks for null count = 1 |
| .6 | - FOR TMR1: Loads TMR; checks for null count = 0 |
| .7 | - FOR TMR1: Verify TMR time out |
| .8 | - FOR TMR1: Verify TMR reloads |
| .9 | - Perform operations to set pulse train to 0; verify. |
| .A | - In TEST Mode, write all on pattern, expect echo of same. |
| .B | - In TEST Mode, walk 1 thru primary & firing latch. |
| .C | - In TEST Mode, see if pulse train is stuck low. |
| .D | - In TEST Mode, see if pulse train is stuck high. |
| .E | - In TEST Mode, check that pulse train is toggling. |
| 19 | NSFE_PRES |
| .1 | - Set IMOK true; read back IMOK signal. |
| .2 | - Set IMOK false; read back IMOK signal. |
| 1A | NSFE_TMR_PRES |
| .1 | - Verifies addr lines & data bits 6 & 7 |
| .2 | - Verifies addr lines & data bits 6 & 7 |
| .3 | - Verifies addr lines & data bits 6 & 7 |
| .4 | - Verifies data bits 5-0 with walking 1 pattern |
| .5 | - Verifies that all outputs are free to toggle. |
| 1B | NSFE_TMODE |
| .1 | - FOR TMR0: Checks for null count = 1 |
| .2 | - FOR TMR0: Loads TMR; checks for null count = 0 |
| .3 | - FOR TMR0: Verify TMR time out |
| .4 | - FOR TMR0: Verify TMR reloads |
| .5 | - FOR TMR1: Checks for null count = 1 |
| .6 | - FOR TMR1: Loads TMR; checks for null count = 0 |
| .7 | - FOR TMR1: Verify TMR time out |
| .8 | - FOR TMR1: Verify TMR reloads |
| .9 | - Perform operations to set pulse train to 0; verify. |
| .A | - In TEST Mode, write all on pattern, expect echo of same. |
| .B | - In TEST Mode, walk 1 thru primary & firing latch. |

| TEST | DESCRIPTION |
|---|---|
| .C | - In TEST Mode, see if pulse train is stuck low. |
| .D | - In TEST Mode, see if pulse train is stuck high. |
| .E | - In TEST Mode, check that pulse train is toggling. |
| 1C | NSFE_DIAG |
| .1 | - Tests for dormant field status information. |
| 1D | NSFC_S_PLL |
| .1 | - FOR TMR2: Enables PLL; Checks for null count = 1 |
| .2 | - FOR TMR2: Loads TMR; checks for null count = 0 |
| .3 | - FOR TMR2: Verify TMR time out |
| .4 | - FOR TMR2: Verify TMR reloads |
| 1E | NSFC_L_PLL |
| .1 | - FOR TMR2: Enables PLL; Checks for null count = 1 |
| .2 | - FOR TMR2: Loads TMR; checks for null count = 0 |
| .3 | - FOR TMR2: Verify TMR time out |
| .4 | - FOR TMR2: Verify TMR reloads |
| 1F | NSFE_PLL |
| .1 | - FOR TMR2: Enables PLL; Checks for null count = 1 |
| .2 | - FOR TMR2: Loads TMR; checks for null count = 0 |
| .3 | - FOR TMR2: Verify TMR time out |
| .4 | - FOR TMR2: Verify TMR reloads |
| 20 | ROLL_CALL |
| .1 | - Test for ribbon cable connecting HAIA(JA) to SEM(JG). |
| .2 | - Enable NLCB IMOK; check for IMOK true. |
| .3 | - Disable NLCB IMOK; check for IMOK false. |
| .4 | - Enable NLIBS IMOK; check for IMOK true. |
| .5 | - Disable NLIBS IMOK; check for IMOK false. |
| .6 | - Enable NLIBL IMOK; check for IMOK true. |
| .7 | - Disable NLIBL IMOK; check for IMOK false. |

| TEST | DESCRIPTION |
|---|---|
| 21 | NLCB_ANALOG_DORMANT |
| .1 | - Test for 0 volts at REFO. |
| .2 | - Test for 0 volts at TREFO. |
| .3 | - Test for 0 volts at 12PREFO. |
| .4 | - Inject current in REFO circuit; test for change in output. |
| .5 | - Inject current in TREFO circuit; test for change in output. |
| .6 | - Inject current in 12PREFO circuit; test for change in output. |
| .7 | - Test for 0 volts at SP (speed). |
| .8 | - Reset exixting faults; test for 1OSP (overspeed) = 0. |
| 22 | NSFE_ANALOG |
| .1 | - Test for EI (exciter current) = 0 volts. |
| .2 | - Test for no current imbalance |
| .3 | - Inject signal in current circuit; test for output change. |
| .4 | - Again, test for no current imbalance. |
| .5 | - Excite current circuit with 55 Hz inputs; verify that currentimbalance is detected. |
| .6 | - Wait 100 ms; verify no current imbalance condition. |
| 23 | NLIB_S_ANALOG |
| .1 | - Reset existing faults; verify UV & not OV condition. |
| .2 | - Verify not SUPPression conditon. |
| .3 | - Verify 1UV70=0. |
| .4 | - Test C17 for time delay. (Overvoltage circuit) |
| .5 | - Test C17 for time delay. (Overvoltage circuit) |
| .6 | - Test for 0 volts at IFBS. |
| .7 | - Test for 0 volts at FLUX BAS. |
| .8 | - Inject current in source current condition circuit. Detect change in IFBS. |
| .9 | - Detect change in FLUX BAS. |
| .A | - For FLUX BAS: Check for 0 volts dc offset. |
| .B | - For FLUX BAS: Check level of positive peak. |
| .C | - For FLUX BAS: Check level of negative peak. |
| 24 | NLIB_L_ANALOG |
| .1 | - Reset existing faults; verify UV & not OV condition. |

| TEST | DESCRIPTION |
|---|---|
| .2 | - Verify not SUPPression conditon. |
| .3 | - Verify 1UV70=0. |
| .4 | - Test C17 for time delay. (Overvoltage circuit) |
| .5 | - Test C17 for time delay. (Overvoltage circuit) |
| .6 | - Test for 0 volts at IFBL. |
| .7 | - Test for 0 volts at FLUX BAL. |
| .8 | - Inject current in source current condition circuit. Detect change in IFBL. |
| .9 | - Detect change in FLUX BAL. |
| .A | - For FLUX BAS: Check for 0 volts dc offset. |
| .B | - For FLUX BAS: Check level of positive peak. |
| .C | - For FLUX BAS: Check level of negative peak. |
| .D | - For FLUX CBS: Check for 0 volts dc offset. |
| .E | - For FLUX CBS: Check level of positive peak. |
| .F | - For FLUX CBS: Check level of negative peak. |
| .10 | - For FLUX ACS: Check for 0 volts dc offset. |
| .11 | For FLUX ACS: Check level of positive peak. |
| .12 | For FLUX ACS: Check level of negative peak. |
| 25 | HLCB_INTR |
| .1 | - Enable test oscillator; verify occurance of source zero crossing interrupt (slave intr 0). |
| .2 | - Verify frequency of source zero crossing intr. |
| .3 | - Enable test oscillator; verify occurance of load zero crossing interrupt (slave intr 1). |
| .4 | - Verify frequency of load zero crossing intr. |
| .5 | - Verify occurance of source firing interrupt. |
| .6 | - Verify occurance of load firing interrupt. |
| .7 | - Verify occurance of field firing interrupt. |
| 26 | NLCB_ANALOG_ACTIVE |
| .1 | - Enable test oscillator; verify speed circuits reads 55 Hz. |
| .2 | - Verify no overspeed condition. |
| 27 | HISA_PRES |
| .1 | - Disable IMOK; verify by reading back bit. |
| .2 | - Enable IMOK; verify by reading back bit. |
| 28 | CABLE_CONT |
| .1 | - Tests for continuity in ribbon cable from SEM,JA to I/O Module. |
| .2 | - Tests for continuity in ribbon cable from SEM, JKto I/O Module. |
| .3 | - Tests for continuity in ribbon cable from SEM, JHto I/O Module. |
| .4 | - Tests for continuity in ribbon cable from SEM, JB to Door Module. |
| 29 | HISA_IO |
| .1 | - Set all inputs from I/O Module high; verify by reading back inputs to HISA. |
| .2 | - Set all inputs from I/O Module low; verify by reading back inputs to HISA. |
| 2A | P105_SUPPLY |
| .1 | - Check that 105 volt dc power supply is OK. |
| 2B | HROA_1A_IO: In sequence, pick-up single relays on the HROA in slot 1A of the I/O module. Expect feedback signal to indicate change. |
| .1 | -Force all HROA(1A) relays to drop-out; verify by feedback. |
| .2 | -Force 105 Vdc relays to pick-up; verify by feedback. |
| .3 | -Drop out 105 Vdc relays, wait 40 ms, verify by feedback. |
| .4 | -Pick-up single 28 V relay; verify by feedback. |
| .5 | -Drop-out all relays; verify by feedback. |
| .6 | -Pick-up single 28 V relay; verify by feedback. |
| .7 | -Drop-out all relays; verify by feedback. |
| .8 | -Pick-up single 28 V relay; verify by feedback. |
| .9 | --Drop-out all relays; verify by feedback. |
| .A | --Pick-up single 28 V relay; verify by feedback. |

| TEST | DESCRIPTION |
|---|---|
| .B | --Drop-out all relays; verify by feedback. |
| 2C | HROA_1B_IO: In sequence, pick-up single relays on the HROA in slot 1B of the I/O module. Expect feedback signal to indicate change. |
| .1 | -Force all HROA(1A) relays to drop-out; verify by feedback. |
| .2 | -Force 105 Vdc relays to pick-up; verify by feedback. |
| .3 | -Drop out 105 Vdc relays, wait 40 ms, verify by feedback. |
| .4 | -Pick-up single 28 V relay; verify by feedback. |
| .5 | -Drop-out all relays; verify by feedback. |
| .6 | -Pick-up single 28 V relay; verify by feedback. |
| .7 | -Drop-out all relays; verify by feedback. |
| .8 | -Pick-up single 28 V relay; verify by feedback. |
| .9 | -Drop-out all relays; verify by feedback. |
| .A | -Pick-up single 28 V relay; verify by feedback. |
| .B | -Drop-out all relays; verify by feedback. |
| .C | -Pick-up single 28 V relay; verify by feedback. |
| .D | -Drop-out all relays; verify by feedback. |
| .E | -Pick-up single 28 V relay; verify by feedback. |
| .F | -Drop-out all relays; verify by feedback. |
| 2D | NLCB_R_DVR |
| .1 | - Verify presence of HROA in slot 1B of I/O Module. |
| .2 | - Verify presence of HROA in slot 1A of I/O Module. |
| .3 | - Force relay drivers on NLCB on; verify response from I/O Module via feedback. |
| .3 | - Force relay drivers on NLCB on; verify response from I/O Module via feedback. |
| 2E | DOOR_MOD_PRES |
| .1 | - Test for CTS asserted. |
| .2 | - Test for DSR asserted. |

After the microcomputers have successfuly powered up and/or self-test operation have been finished successfuly, the printer can then display faults/alarms as described below.

## 7-9. FAULT/ALARM MONITOR

Built into the electronics software module are fault/alarm monitors.

## 7-10. FAULT DETECTION

In the event of a fault condition, the IMD will shut down and a fault indication will appear on the SEM. The cause of the shutdown will be displayed on the printer terminal located onthe control panel door. If more than one fault has occurred, the fault finder will store and print the subsequent faults (up to 10 total).

To reset the control after a fault condition, it is necessary to press the reset switch on the HLCB card in the SEM on the reset pushbutton on the control panel door. But first record the fault/alarm/relay status, and I/O states. If the fault condition is clear, the fault light will go out after a door reset, and stay out when the reset is completed.

Consult the system elementary ladder diagram for a complete list of faults. Most of the faults are standard and are defined in the following paragraphs, along with suggested corrective action.

LOSS OF COOLING FAN- This fault indicates a loss of bridge cooling fan(s). In units having redundant fans, this fault occurs only after both the normal and backup fans have failed.

Possible action to be taken:

1. Replace fan motor fuses
2. Replace defective fan motor (and/or centrifugal switch)
3. Replace HRDA boards
4. Replace I/O module

SOURCE, LOAD, MASTER, SLAVE HARDWARE OVERCURRENT - This fault indicates that the source or load side current has exceeded an acceptable level for more than 26 milliseconds. Under normal circumstances, the electronic control can reduce current to zero in less than 26 milliseconds. Actuation of the hardware overcurrent generally indicates a severe problem from which the control cannot recover.

Possible action to be taken:

1. Check converter bridges for shorted cells
2. Replace NLIB card

MICROPROCESSOR STALL - This microprocessor has failed to execute its program in a normal amount of time.

Possible action to be taken:

1. Replace HMPG card
2. Replace HITA card
3. Replace NLCB and NLIB cards

LOAD, SOURCE, MASTER, SLAVE SOFTWARE OVERCURRENT - Source and load current are compared in software to an adjustable overcurrent thresh-

old. If the current exceeds this threshold, both source and load converters will act to reduce the current to zero. If it is not possible to reduce current to zero and regain control, a fault will be declared and the drive will be tripped.

Possible action to be taken:

1. Go into gate-test-mode and look for missing pulse
2. Replace in order NLCB, NLIB, HAIA cards
3. Reset and try again

SOURCE, LOAD OVERVOLTAGE - Indicates an excessive level of voltage on the source or load ac buses.

Possible action to be taken:

1. If source overvoltage, check incoming voltage from switchgear.
2. Replace HLCB, NLIB cards

OVERSPEED - Indicates excessive motor speed has occurred.

Possible action to be taken:

1. Replace NLCB PWB
2. Check attenuators

GROUND FAULT - Indicates a ground exists in the converter bridges, transformer, reactor, or motor. The ground detector will also actuate if the load converter ac line filter circuits are unbalanced due to an open fuse or defective capacitor.

Possible action to be taken:

1. Check AC and DC filter fuses and capacitors in bridges
2. Check for grounds in motor, transformer, reactor (megger)
3. Replace NLCB PWB

For other possible faults and solution patterns see Appendix C.

## 7-11. ALARM DETECTION

Alarms are conditions which are cause for concern but which do not warrant immediate shutdown of the drive but shoud be investigated as soon as possible. A composite alarm signal is generally supplied to operate a light on the control panel door, a remote annunciator supplied by the user, or both.

Some of the alarms will vary from one installation to the next, so the software ladder diagram section of the system elementary must be consulted for a complete list of. The following alarms are standard (if applicable) and so will be defined here.

LOSS OF SPEED REFERENCE - When the drive is controlled from a 4 - 20 milliamp speed reference, a check is made to determine if this signal has gone to zero (Less than 4 mA). If so, an alarm is declared and the drive speed is locked at its present value.

The remaining alarms to be defined deal with unused or unexpected interrupts. In case one of these interrupts occurs, software is present to record data, declare an alarm, and then return to normal processing. The data saved upon execution of one of these interrupts is stored in a stack in RAM memory. There is also a counter for each of these interrupts so that the number of occurrences can be logged. The stack variable and event counter names are given in Table 5-2. The definition of the information contained in the stack is given in Table 5-3. It is not expected that the user would be able to fully utilize this information, but it is given here in the interest of completeness.

Rapid repetition of these interrupts can result in shutdown of the system but it has been shown that the system may recover from these if they are infrequent. The cause of these interrupts is generally due to noise or other unusual conditions stated in the following definitions.

I/O ADDRESSING - Whenever any memory or I/O device is addressed by the microprocessor, it is expected that that device will respond with an acknowledge signal. If an acknowledge is not received, the microprocessor will wait for up to 80 microseconds, generate a bus hog interrupt, and then proceed to the next task. Failure to receive this acknowledge indicates that some memory or I/O device is "hogging" the data bus, and an alarm is declared. Actually, addressing an undefined address with the terminal will invoke a bus hog alarm. The bus hog alarm may be tested in this manner by executing the following command:

DW500:500 CR

Continued receipt of the bus hog alarm usually indicates that a digital card is defective or not plugged in properly, or a wrong address was requested.

The following are combined to form one composite alarm - MICROPROCESSOR ALARM - DUMMY INTERRUPT (DUM INT) - If any of the masked (disabled) interrupts should occur, a dummy interrupt alarm will occur.

DIVISION BY ZERO INTERRUPT (DIV ZERO) - This is a non-maskable interrupt that is invoked whenever the microprocessor attempts a division by zero.

POWER DOWN INTERRUPT (POW DN) - This is a non-maskable interrupt that places the microprocessor in a special mode where all instructions and data must be one byte in length. This interrupt is generated whenever the result of a calculation exceeds the available register size.

SINGLE STEP INTERRUPT (SING STP) - A single step alarm is an alarm declared whenever the non-maskable interrupt used to single step the microprocessor execution is invoked. Single step mode is not used in the LCI control.

SYSTEM TIMING INTERRUPT (SYS TIM) - This is a non-maskable interrupt from an external timer that is ordinarily used to give a "heartbeat" interrupt for overall system timing functions. This interrupt is not used on the LCI.

USART RECEIVE, TRANSMIT (USART R, USART T) - USART receive, transmit is an interrupt that is generated whenever the USART fails to indicate that it is ready to receive, transmit data. These interrupts are not used in the LCI control.

*Table 5-2.*
INTERRUPT ALARM COUNTER AND STACK VARIABLES

| | | |
|---|---|---|
| ALARM | : STACK ARRAY VARIABLE | : TRAP CNTR BH |
| DUM INT | : SAVE STACK DUM | : TRAP CNTR DUM |
| DIV ZERO | : SAVE STACK 0 | : TRAP CNTR 0 |
| POW DN | : SAVE STACK 2 | : TRAP CNTR 2 |
| ONE B | : SAVE STACK 3 | : TRAP CNTR 3 |
| OV FL | : SAVE STACK 4 | : TRAP CNTR 4 |
| SING STP | : SAVE STACK 1 | : TRAP CNTR 1 |
| SYS TIM | : SAVE STACK SYS TIM | : TRAP CNTR SYS TIM |
| USART R | : SAVE STACK US R | : TRAP CNTR US R |
| USART T | : SAVE STACK US T | : TRAP CNTR US T |

NOTE: In order to access information in the stacks, it is necessary to obtain the address of the desired SAVE STACK variable from a set of memory maps. Then the data from SAVE STACK ADDRESS to SAVE STACK ADDRESS + 1AH must be printed. To do this, use the following command structure:

<DW SAVE STACK ADDRESS , SAVE STACK ADDRESS + 1AH CR>

NOTE: See paragraph entitled "Monitor Command Structure" for an explanation of < >.

To obtain the trap counter value, the following must be done:

<DW TRAP CNTR ADDRESS CR>

## 7-12. SECTION B - GATE-TEST-MODE

**7-12.1. Gate-Test-Mode.** The gate test mode allows off-line testing of much of the analog signal conditioning, microprocessor firing control, and gate pulse generation circuitry. In the gate-test-mode, a local test oscillator supplies 3 phase, approximately 60 Hz signals in place of the attenuated bus signals from the source and load bridges. The electronic control can then lock onto these signals and generate signals

*Table 5-3.*
MAP FOR INTERRUPT SAVE STACKS

| STACK OFFSET | REGISTER | DEFINITION |
|---|---|---|
| 0 | BP | |
| 2 | ISR | CALL RETURN (OFFSET) |
| 4 | ISR | CALL RETURN (SEGMENT) |
| 6 | DI | |
| 8 | DI | |
| A | BX | |
| C | DX | |
| E | CX | COUNT REGISTER |
| 10 | AX | 16BIT ACCUMULATOR |
| 12 | DS | DATA SEGMENT |
| 14 | ES | EXTRA SEGMENT |
| 16 | IR | INTERRUPTED ROUTING (OFFSET) |
| 18 | IR | INTERRUPTED ROUTING (SEGMENT) |
| 1A | FLAG | FLAG REGISTER |

to "dry fire" the thyristors in both bridges. Thyristor gate pulses may then be examined for proper magnitude and phasing. Firing may be adjusted between advance and retard limit for each bridge. In the gate-test-mode, the control produces pulse train firing, thus it is necessary to run the converter ventilating fans for this test.

To perform the following tests, a dual channel oscilloscope with voltage probes and a current probe is required.

### 7-12.2. Getting Into Gate-Test-Mode

a. The drive should be stopped, primary switchgear and/or transformer isolation switch (if any), and load isolation device (if any) should be open. Check the software ladder diagram in the system elementary for any other conditions that must be met.
b. Put the NLCB (1E) "TEST" switch (SW1) in the test position.
c. Reset the electronics module by pressing the "RESET" (SW1) switch on the HLCB (2E or 2J).

Note: Always use DC coupling on the oscilloscope channels.

### 7-12.3. Gate-Test-Mode Checkout

a. Connect channel one of the oscilloscope to NLIB (1C) "FCBR" (TP5) and channel two of HLCB (2E) "VA1" (TP12). Trigger the scope on channel one and verify the waveforms look like figure 7B-1.
b. Connect channel one of the oscilloscope to NLIB (1D) "FCBR" (TP5) and channel two to HLCB (2E) (TP13). Trigger the scope on channel one and verify the waveforms look like figure 7B-1.
c. Connect channel one of the oscilloscope to HLCB (2E) "VA1" (TP12) and trigger on

this signal. Adjust the variable time/division scale so that one period equals six major divisions. (i.e, 60 degrees per major division or 12 degrees per minor division).

d. One at a time, connect channel two to SIG1 thru SIG6 on the source side "A" NSFC (2C) (TP1 thru TP6) and verify that the gate pulses are positioned as in figure 7B-2A. Note: The firing angle can be adjusted by varying the speed demand reference (place in "LOCAL" and adjust "LOCAL REF").

e. One at a time, connect channel two to SIG1 thru SIG5 on the source side NSFC (2A) (TP1 thru TP6) and verify that the gate pulses are positioned as in figure 7B-2B. Note: The slave firing lags the master firing by 30°.

f. Connect channel one of the oscilloscope to HLCB (2E) "VA2" (TP13) and trigger on this signal. Adjust the variable time/division scale so that one period equals six major divisions. (i.e., 60 degrees per major division or 12 degrees per minor division).

g. One at a time, connect channel two to SIG1 thru SIG6 on the load side NSFC (2B) (TP1 thru TP6) and verify that the gate pulses are positioned as in Figure 7B-2A.

h. Verify that each of the red gate pulse LEDS on the HPTN cards in the leg cell modules is illuminated indicating that the cell is getting gate pulses.

i. With a current probe, look at the gate pulse of each thyristor while simultaneously monitoring the corresponding gate pulses on the NSFC card. The two signals should line up with each other as shown in Figure 7B-4A.

j. Connect a current probe to the scope and trigger on this channel. Hook the current probe to each of the gate leads (white wire) of the SCRs with the arrow on the current probe pointing away from the gate lead termination point. Verify that the waveform has the same characteristics as Figure 7B-4B. If the pulse fails the criterion, the HPTN card in that leg cell module is most likely the cause, or the NTDA connectio0ns may be reversed.

k. Connect channel one of oscilloscope to NLCB (2E) "VA2" (2E) and channel two to NCCC (1F) "GSH" (TP13). Verify against figure 7B-5A. Note that the top GTO pulse train is for the source angle at approximately 120°. As the angle increases the time is reduced until the bottom GTO train is reached. The GTO doesn't fire when the angle is less than 120°. Also verify that "GSR" (TP14) NCCC (1F) is as shown.

l. Connect channel one of oscilloscope to NCCC (1F) "GSH" (TP13) and with a current probe look at the gate pulse at each of the shunt GTOs (A-D). Verify Figure 7B-5B. The off pulse varys in width and amplitude, increasing as the actual drive current increases. Also check the series GTOs (A-D) (if used).

m. Put the NLCB (2E) "TEST" switch (SW1) in the OFF position and reset the electronics module by pressing the "RESET" (SW1) switch on the HLCB (2E).

*Table 2.*
Master Bridge

| CORRESPONDING TP ON NSFC | GATE LEAD OF EACH SCR |
|---|---|
| SIG1(TP1, 2B) | LTM1 |
| SIG1(TP2, 2B) | LTM2 |
| SIG1(TP3, 2B) | LTM3 |
| SIG1(TP4, 2B) | LTM4 |
| SIG1(TP5, 2B) | LTM5 |
| SIG1(TP6, 2B) | LTM6 |

*Table 2A.*
Master Bridge

| CORRESPONDING TP ON NSFC | GATE LEAD OF EACH SCR |
|---|---|
| SIG1(TP1, 2C) | STM1A |
| SIG1(TP2, 2C) | STM2A |
| SIG1(TP3, 2C) | STM3A |
| SIG1(TP4, 2C) | STM4A |
| SIG1(TP5, 2C) | STM5A |
| SIG1(TP6, 2C) | STM6A |

*Table 2B.*
Slave Bridge

| CORRESPONDING TP ON NSFC | GATE LEAD OF EACH SCR |
|---|---|
| SIG1(TP1, 2A) | STM1B |
| SIG1(TP2, 2A) | STM2B |
| SIG1(TP3, 2A) | STM3B |
| SIG1(TP4, 2A) | STM4B |
| SIG1(TP5, 2A) | STM5B |
| SIG1(TP6, 2A) | STM6B |

## 7-13. SECTION D - SERIAL MONITOR AND HAND TERMINAL (OR OTHER DIAGNOSITIC TERMINAL)

Integrated into the firmware for both source and load electronics is a monitor program which supports manipulation of system memory and I/O. The serial monitor drives an RS-232 port on the front of each microcomputer card. Communicaion with the microcomputer can be established by connecting the handheld terminal suppllied with the drive to RS-232 port. This connection must be made using the HNMB null modem (supplied) with the electronics module) as illustrated in Chapter 5. Observe the plug orientation and Berg jumper positions marked on the null modem.

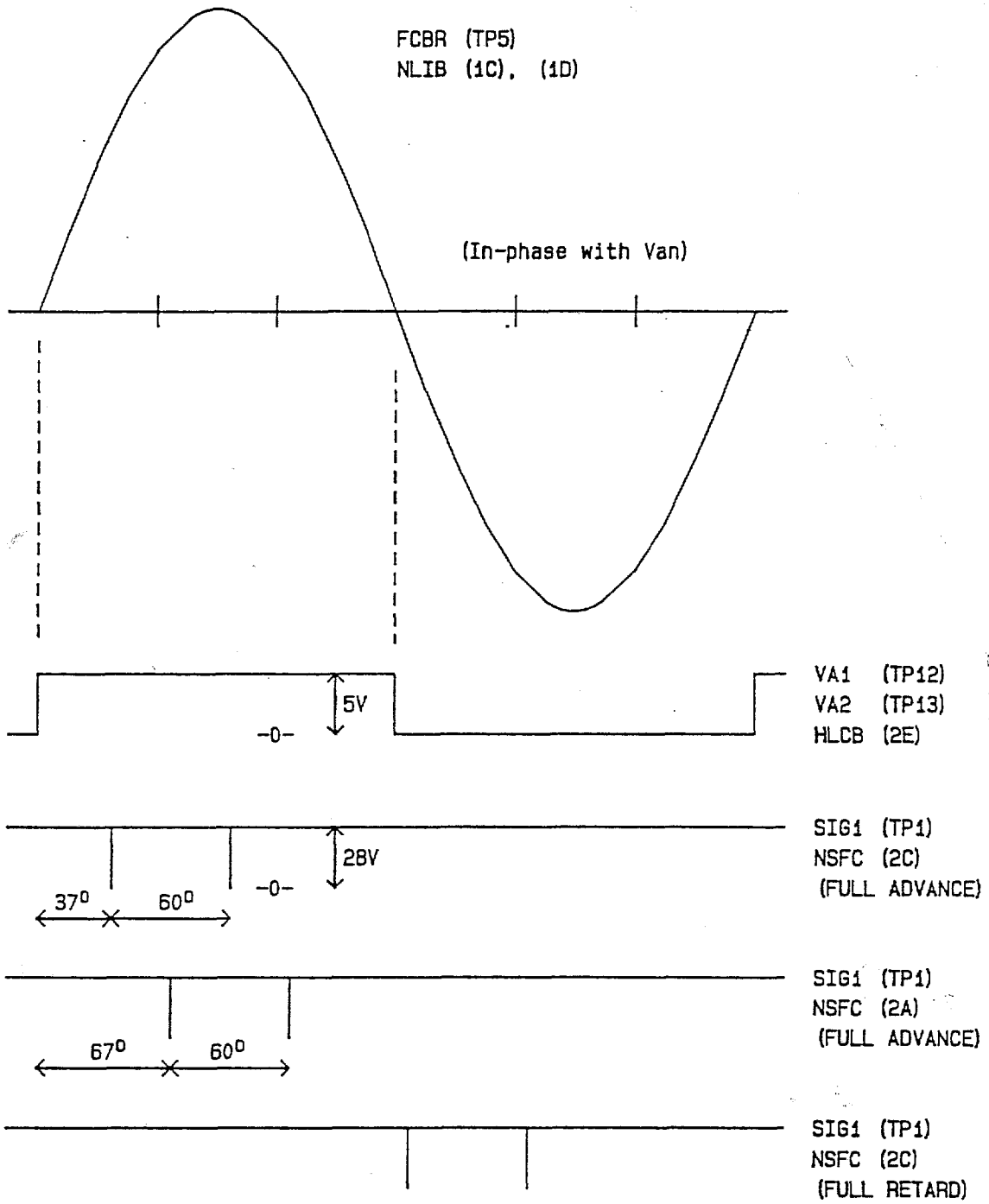
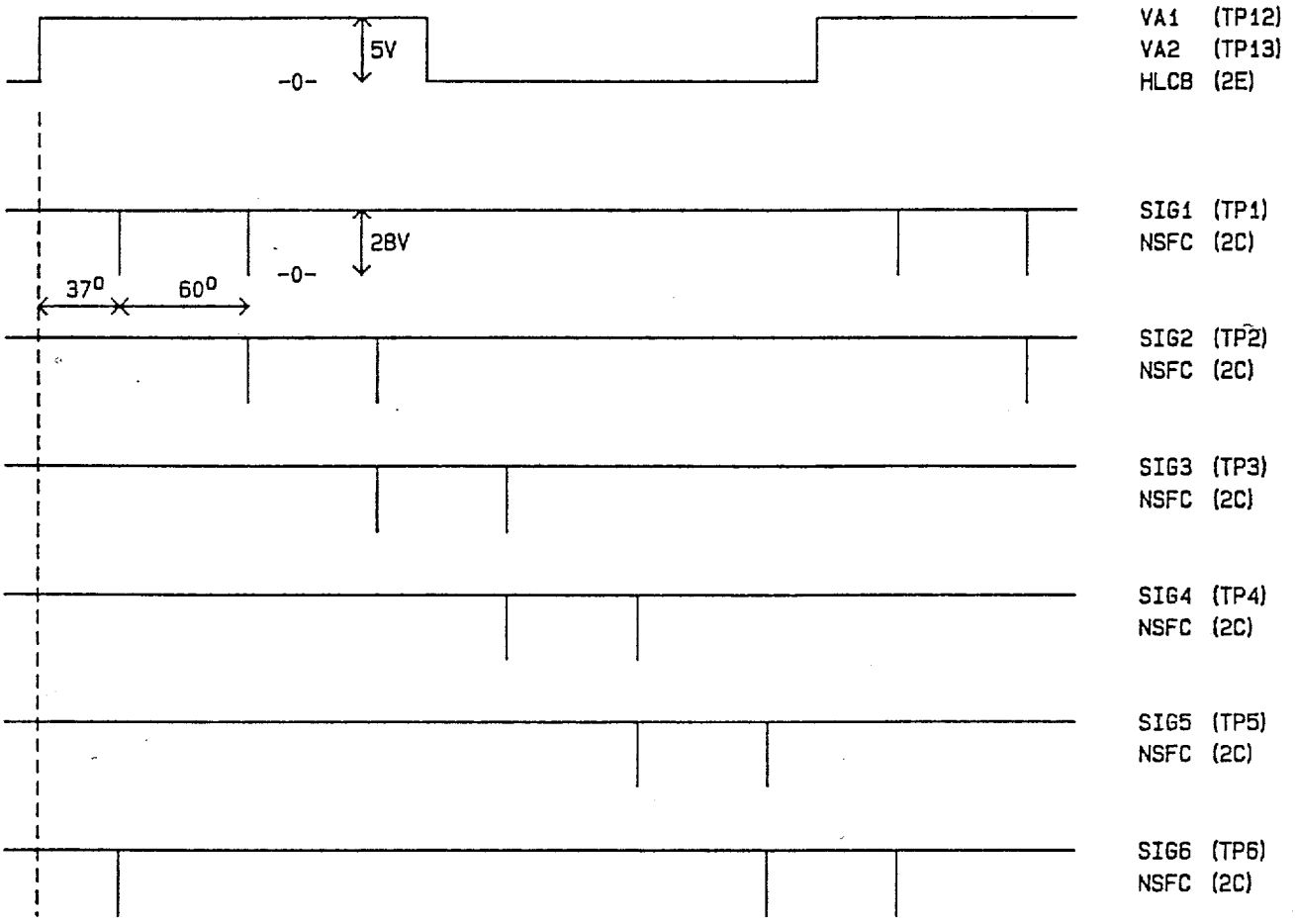
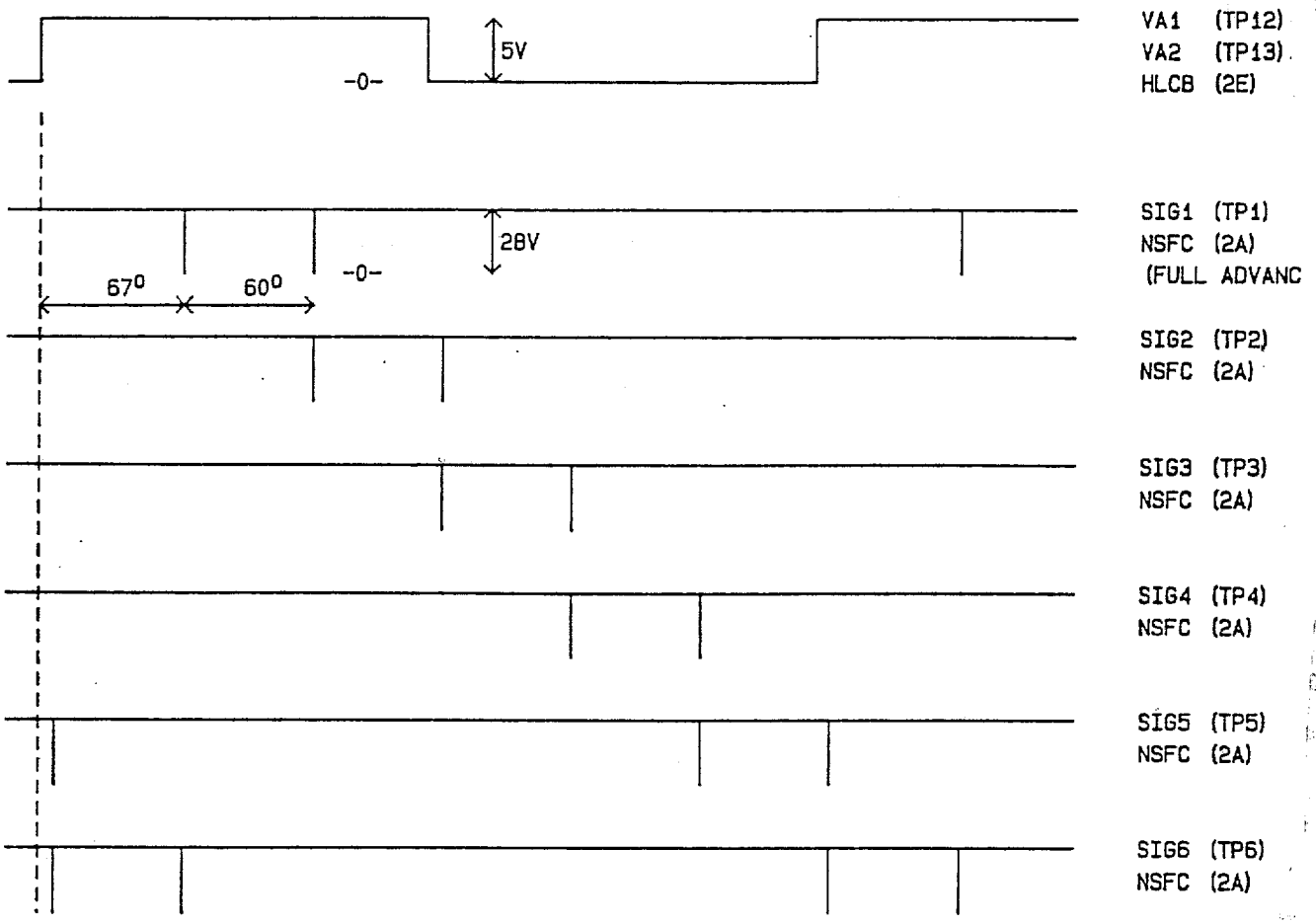

FCBR (TP5)
NLIB (1C), (1D)
(In-phase with Van)
5V
-0-
VA1 (TP12)
VA2 (TP13)
HLCB (2E)
28V
-0-
37°
60°
SIG1 (TP1)
NSFC (2C)
(FULL ADVANCE)
67°
60°
SIG1 (TP1)
NSFC (2A)
(FULL ADVANCE)
SIG1 (TP1)
NSFC (2C)
(FULL RETARD)

*Figure 7B-1.*




SOURCE FIRING (MASTER) - FULL ADVANCE

*Figure 7B-2A.*




Source firing (slave) - full advance

*Figure 7B-2B.*

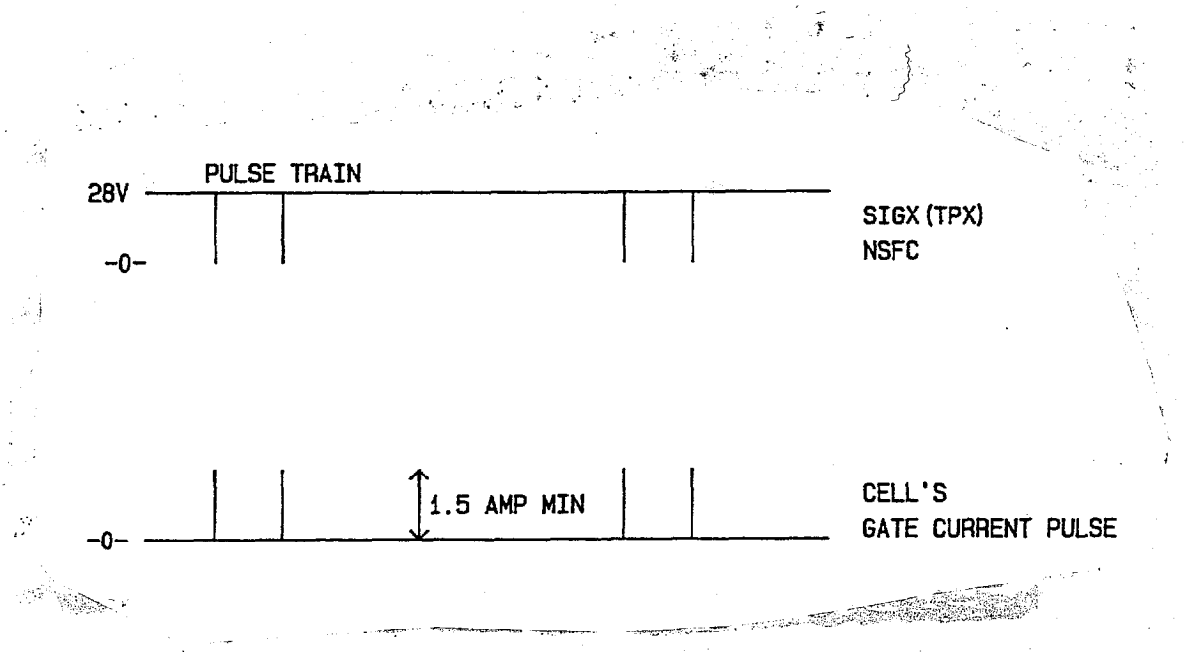

PULSE TRAIN
28V
-0-
SIGX (TPX)
NSFC
1.5 AMP MIN
-0-
CELL'S
GATE CURRENT PULSE

*Figure 7B-4A.*

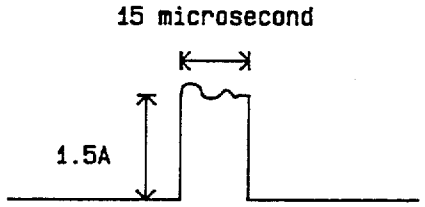



*Figure 7B-4B. Typical firing Pulse*

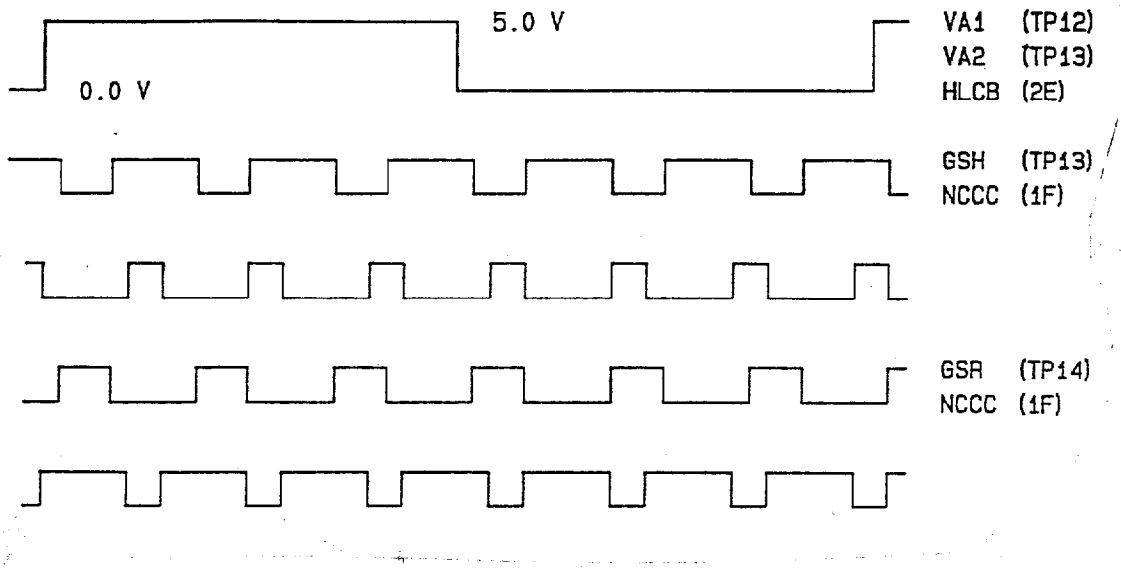

5.0 V
0.0 V
VA1 (TP12)
VA2 (TP13)
HLCB (2E)
GSH (TP13)
NCCC (1F)
GSR (TP14)
NCCC (1F)

*Figure 7B-5A.*

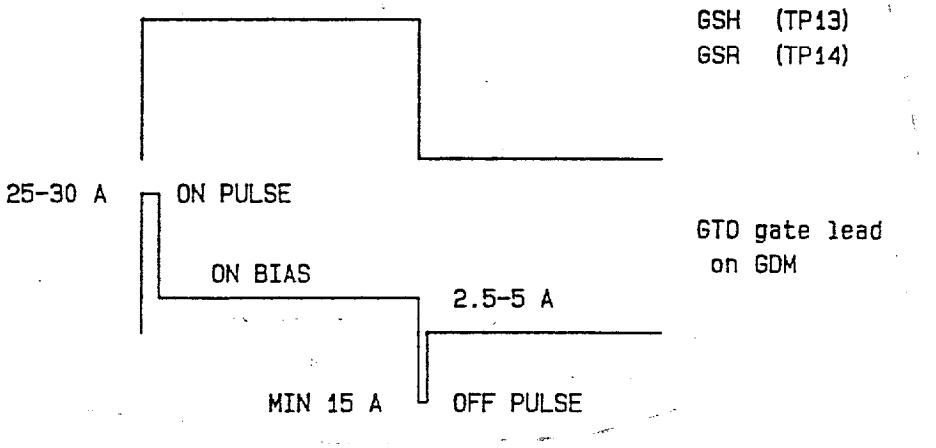

GSH (TP13)
GSR (TP14)
25-30 A
ON PULSE
GTO gate lead
on GDM
ON BIAS
2.5-5 A
MIN 15 A
OFF PULSE

*Figure 7B-5B.*



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